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- Product use and build issues
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- This product should be used only when there is adult supervision present as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and parts, which are sharp. This product contains electrically conductive parts. Use caution with electrically conductive parts near or around power supplies, batteries and powered (live) circuits.
- When the product is turned ON, activated or tested, some parts will move or rotate. To avoid injuries to hands and fingers, keep them away from any moving parts!
- It is possible that an improperly connected or shorted circuit may cause overheating. Should this happen, immediately disconnect the power supply or remove the batteries and do not touch anything until it cools down! When everything is safe and cool, review the product tutorial to identify the cause.
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- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi, micro: bit and Raspberry Pi Pico W.
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

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Contents

Welcome....1
Contents....1
Preface....2
List....3
Metal Case....5
Machinery Parts....5
Acrylic Parts....6
Electronic Parts....7
Wires....10
Tools....10
Required but NOT Contained Parts....11
Chapter 1 Main Components of the Computer Case....12
1.1 Introduction to Freenove Case Adapter for Raspberry Pi....12
1.2 Introduction to Freenove Case GPIO Adapter for Raspberry Pi....14
1.3 Introduction to Raspberry Pi 5 (RPi 5)....15
Chapter 2 Assembly....18
Step 1 Installation of Fans and Speakers....18
Step 2 Installing Rpi 5 and Case Adapter Board....21
Step 3 Connecting Cables....28
Step 4 Installing the Top Cover....33
Step 5 Installing the Side Board....40
Chapter 3 APP Control....41
3.1 Boot Behavior & Environment Settings....41
3.2 About the Case Control Software....50
Chapter 4 Speed Test & PCIe Gen3.0....70
4.1 Disk Speed Test....70
4.2 PCIe Gen3.0....73
What's next?....75

Preface

Welcome to the Freenove Computer Case Kit for Raspberry Pi

This product is exclusively designed for the RPi 5, a popular single-board computer (SBC).Follow this tutorial, you caninstall a sleek, multi-functional computer case kit for your Raspberry Pi 5 (RPi 5).

The RPi 5 delivers highly powerful performance with a CPU frequency of up to 2.4GHz. However, this increased power consumption also leads to significant heat generation. Without an appropriate heatsink, thermal throttling may severely limit the Raspberry Pi 5's performance. Additionally, the removal of the 3.5mm audio jack in the RPi 5 has caused inconvenience for many users.

In this context, our case kit came into being. It not only addresses these issues, but also introduces advanced features.

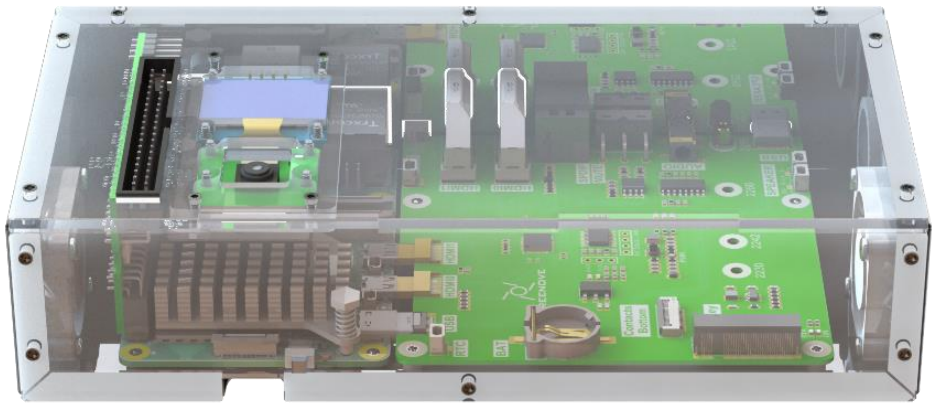
If you encounter any issues, feel free to contact us for prompt and free technical support.

support@freenove.com

List

This product is currently available in four versions: FNK0100A, FNK0100B, FNK0100H, and FNK0100K.
Among them,
FNK0100A is the basic model;
FNK0100H & FNK0100K: Include a 4.3-inch IPS screen (with a DSI interface)
FNK0100B & FNK0100K: Come with an NVMe SSD.
Below are their respective images:

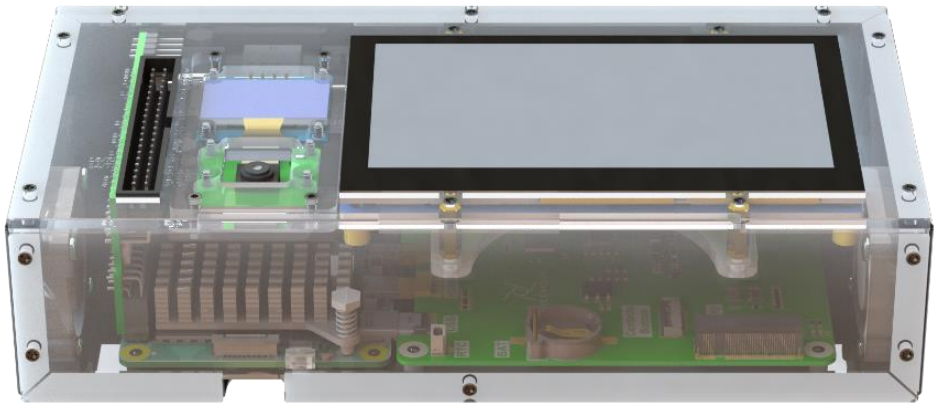
FNK0100A:



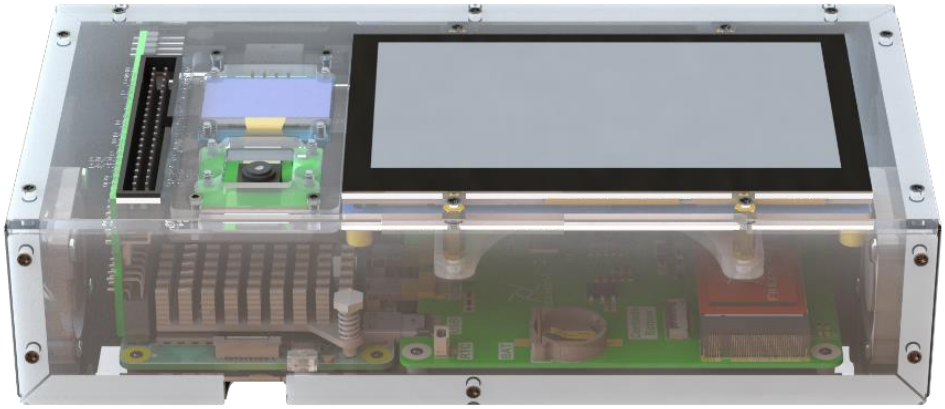
FNK0100B:



FNK0100H:



FNK0100K:

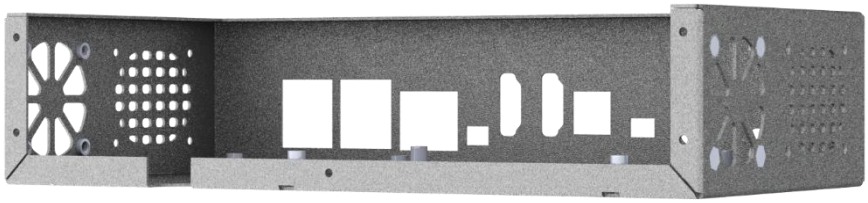


Before getting started, please check the part list. If any component is missing from your kit, please email our support team at

support@freenove.com

Metal Case

Metal case x 1



Machinery Parts

All fasteners come in a larger bag, please open it and check whether they are complete.







Note: Your M2 screws may come in one of the following two quantity combinations due to different product versions.





(Only for FNK0100H and FNK0100K)



(Only for FNK0100A and FNK0100B)



(Only for FNK0100H and FNK0100K)

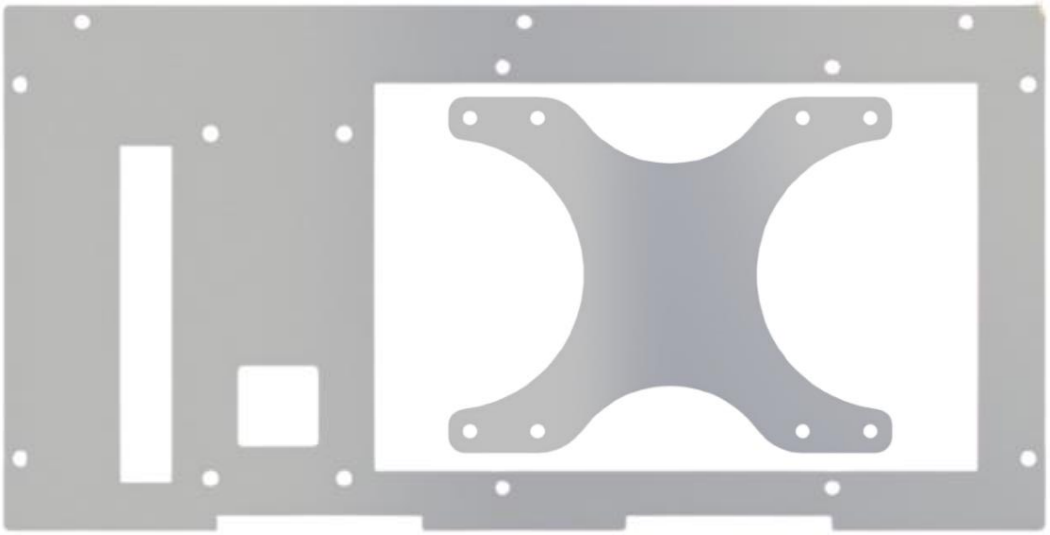


Acrylic Parts

Top Cover x 1 (Only for FNK0100A and FNK0100B)



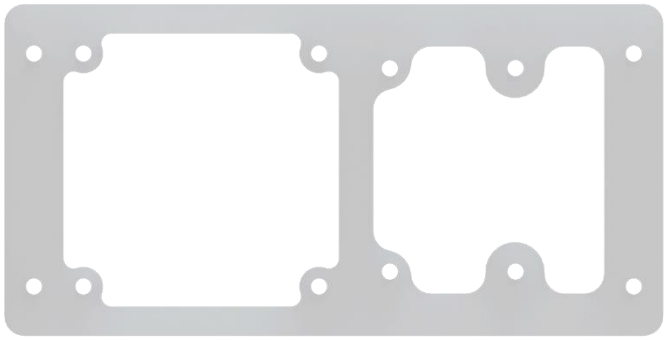
Top Cover X 1, Support Plate for Screen X 1 (Only for FNK0100H and FNK0100K)



Side Plate x 1



Acrylic Part for OLED Screen and Camera x 1

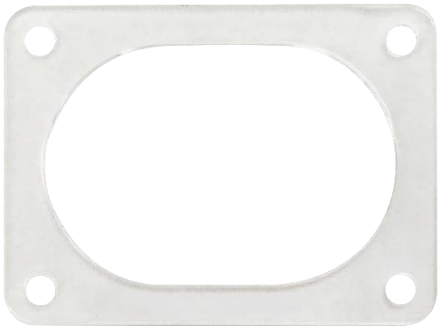


Power Button Caps x 6



Note: The physical power button caps are very small, so be careful not to lose them.

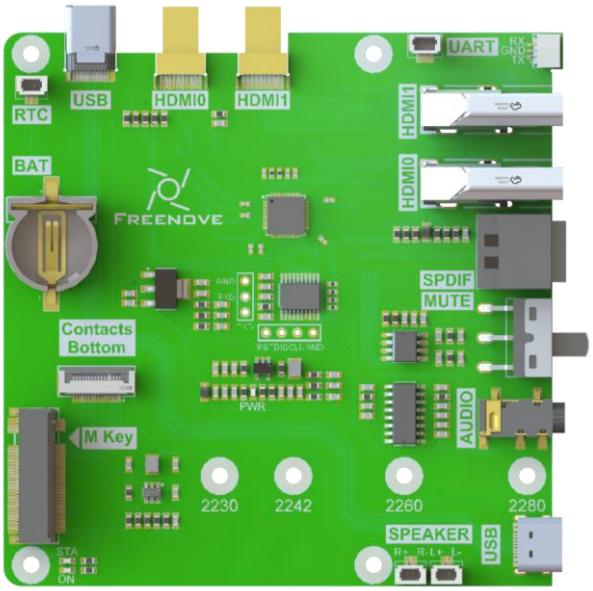
Speaker acrylic pad x 2



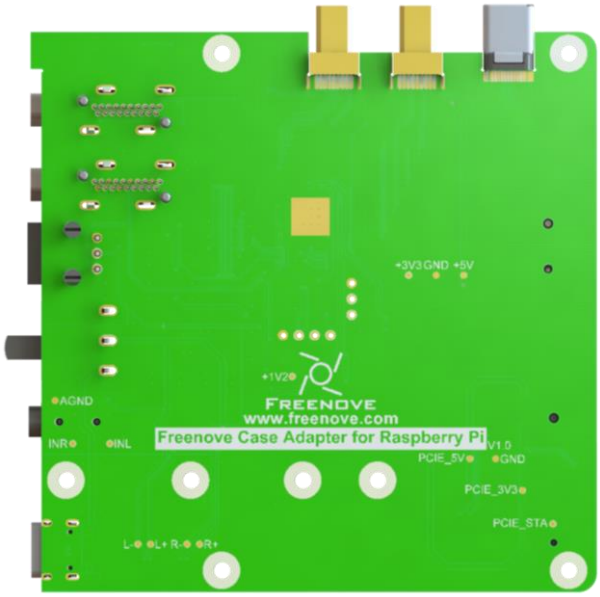
Electronic Parts

Freenove Case Adapter for Raspberry Pi

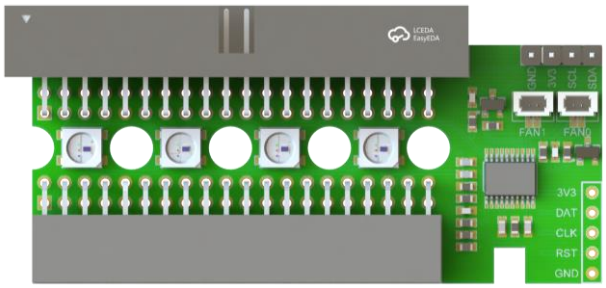
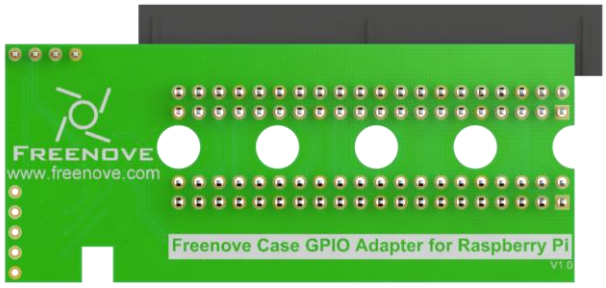
Top



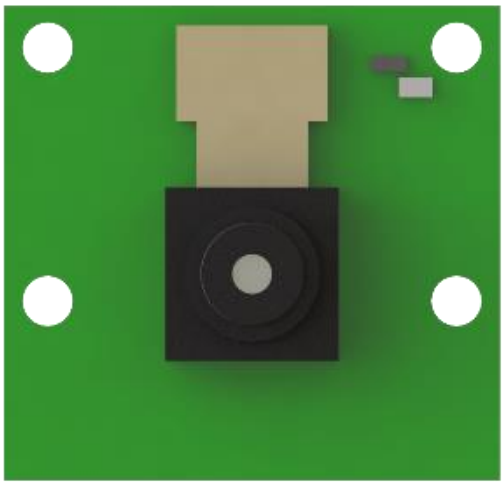
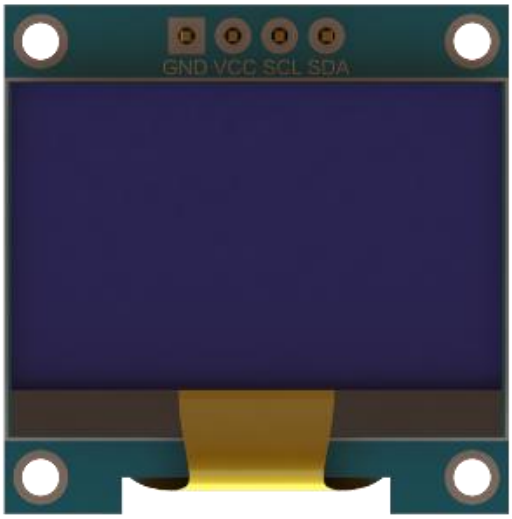
Bottom



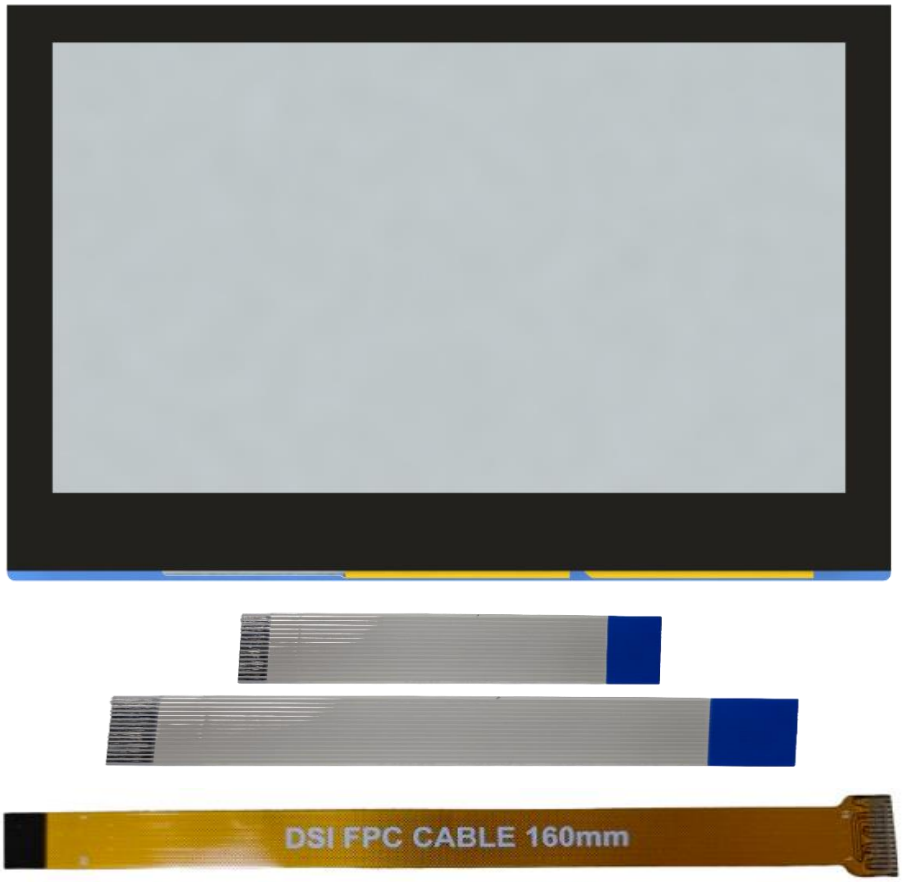
Freenove Case GPIO Adapter for Raspberry Pi

Top	Bottom
	

Electronic Modules

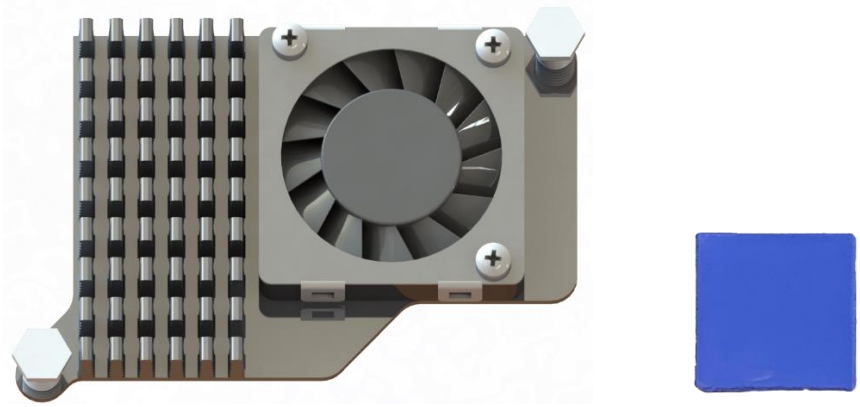
Camera Module x 1	OLED Module x 1
	
3010 Fan x 2	3040 Speaker x 2
	

4.3-inch Screen (with 3 FPC lines) x 1 (Only for FNK0100H and FNK0100K)



Caution: Although the Raspberry Pi camera cable and screen cable may look identical in appearance, they are NOT interchangeable.

Pi5 CPU Active Cooler x 1, Heat Dissipation Silicone Sheet x3



NVMe SSD x 1(Only for FNK0100B and FNK0100K)



Wires

F-F Jumper Wire -4P-20cm x 1



NVMe SSD FPC Line-0.5mm-16P-12cm (same direction) x 1



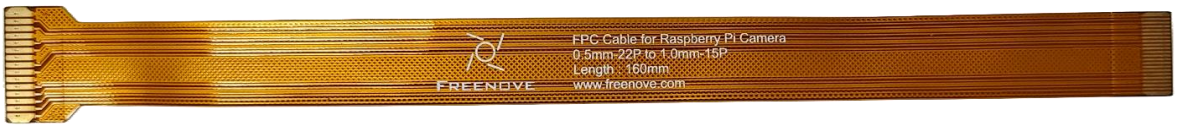
SH-1.0mm-2P-5cm cable (same direction) x 1



SH-1.0mm-3P-5cm cable (same direction) x 1



Camera FPC line 0.5mm-22P to 1.0mm-15P-16cm line (same direction) x 1



Caution: Although the Raspberry Pi camera cable and screen cable may look identical in appearance, they are NOT interchangeable.

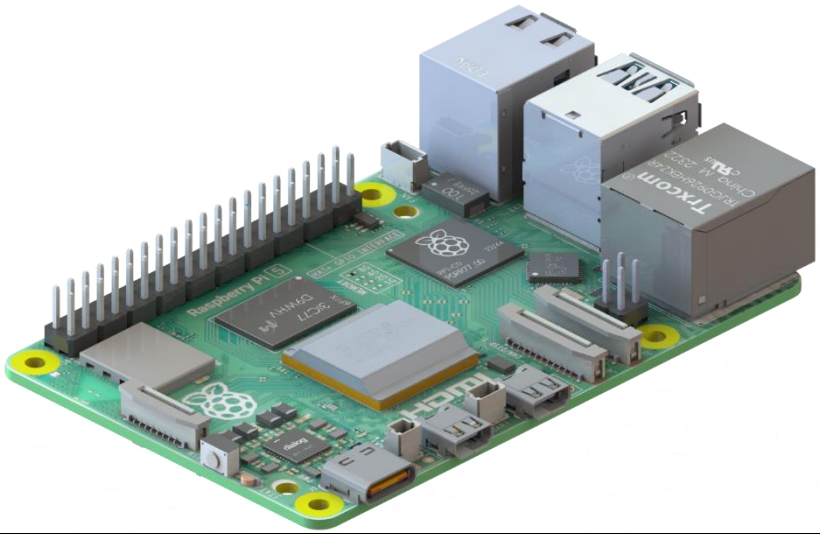
Tools

Cross screwdriver (3mm) x 1



Required but NOT Contained Parts

Raspberry Pi 5 x 1



27W Power Adapter x 1 (or a power adapter compatible with Raspberry Pi official one that can output 5.1V/5A)



Micro SD Card (TF Card), Card Reader x 1

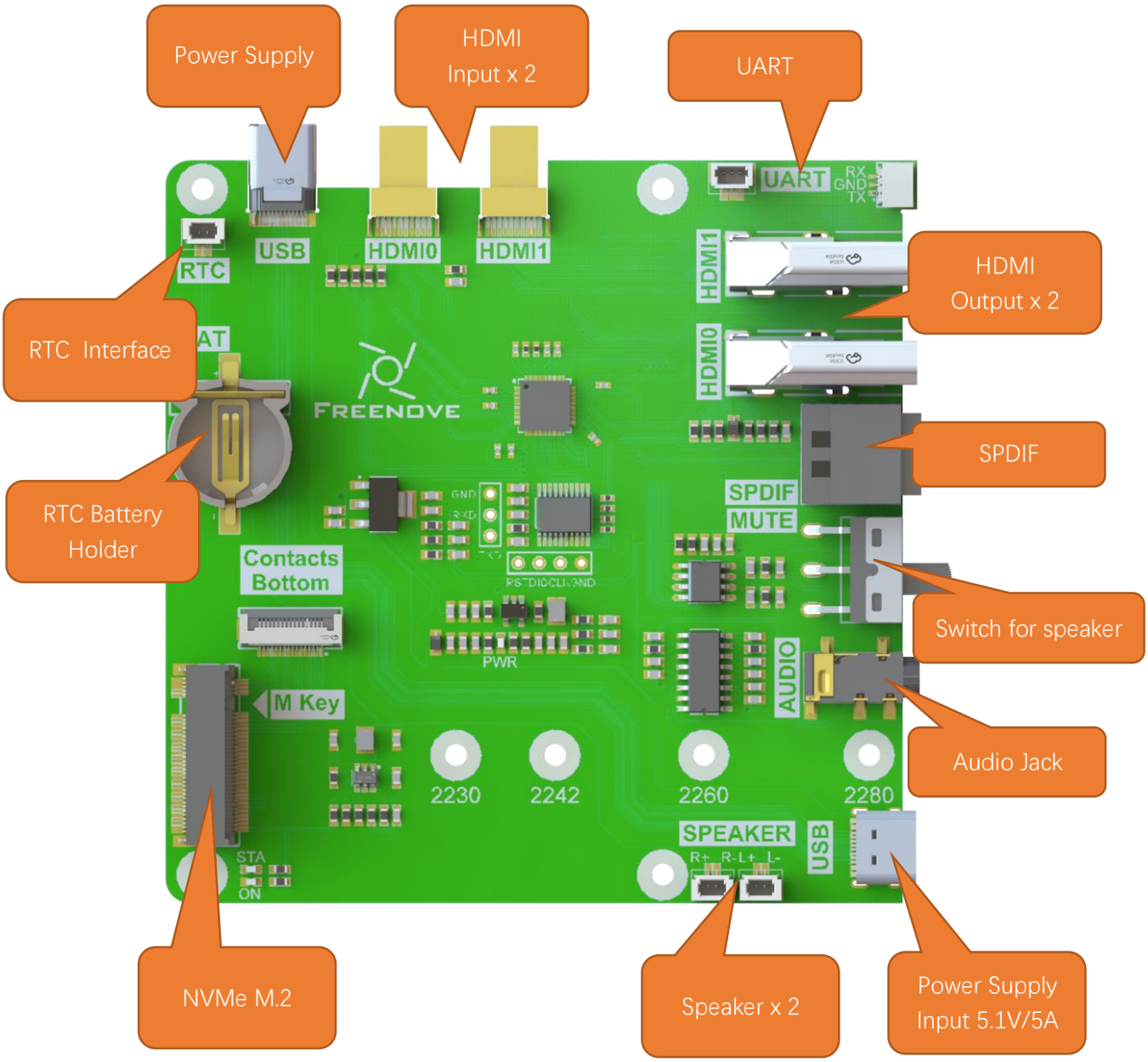


Chapter 1 Main Components of the Computer Case

In this chapter, we will mainly introduce the main components of this case and their functionalities.

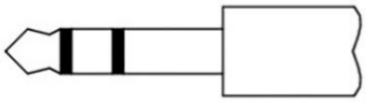
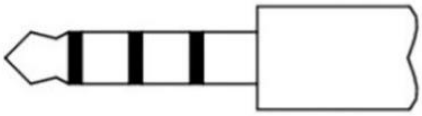
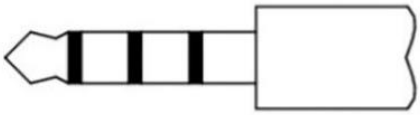
1.1 Introduction to Freenove Case Adapter for Raspberry Pi

Hereinafter we will refer to this component as Case Adapter Board.



The Case Adapter Board is specifically designed for the Raspberry Pi 5 interface, with the goal of transforming the Rpi 5 into a fully functional home computer. This board incorporates a variety of essential interfaces, including a Type-C interface, two HDMI interfaces, an audio output interface, a speaker interface, an M.2 M-Key interface, and an RTC interface.

In the RPi 5 design, the traditional 3.5mm audio output interface has been removed, and audio signals are now solely output through HDMI. To address this, we have developed an audio separation circuit on the Case Adapter Board. This circuit enables audio to be output via the 3.5mm audio interface on the board, You can use the external 3.5mm TRS headphone jack on the Raspberry Pi 5 to listen to audio output. For proper functionality, ensure your headphone connector is compatible. Below is an explanation of common jack types:

<table>
<tr><td colspan="2">TRS JACK
 
Tip Ring Sleeve</td><td>Tip</td><td>Left Channel</td></tr>
<tr><td colspan="2"></td><td>Ring</td><td>Right Channel</td></tr>
<tr><td colspan="2"></td><td>Sleeve</td><td>GND</td></tr>
<tr><td rowspan="4">OMTP</td><td rowspan="4">TRRS JACK
 
Tip Ring 1 Ring 2 Sleeve</td><td>Tip</td><td>Left Channel</td></tr>
<tr><td>Ring1</td><td>Right Channel</td></tr>
<tr><td>Ring2</td><td>MIC</td></tr>
<tr><td>Sleeve</td><td>GND</td></tr>
<tr><td rowspan="4">CTIA</td><td rowspan="4">TRRS JACK
 
Tip Ring 1 Ring 2 Sleeve</td><td>Tip</td><td>Left Channel</td></tr>
<tr><td>Ring1</td><td>Right Channel</td></tr>
<tr><td>Ring2</td><td>GND</td></tr>
<tr><td>Sleeve</td><td>MIC</td></tr>
</table>

Additionally, we have designed an audio amplification circuit. With this circuit, you can control two speakers, each with an impedance of 4Ω and a power rating of 3W. If you prefer not to utilize the speakers within the computer case, you can easily switch them on or off using the MUTE button. For those who have high-fidelity demands, we also offer an SPDIF interface. This allows you to connect your own device for audio playback, ensuring superior sound quality.

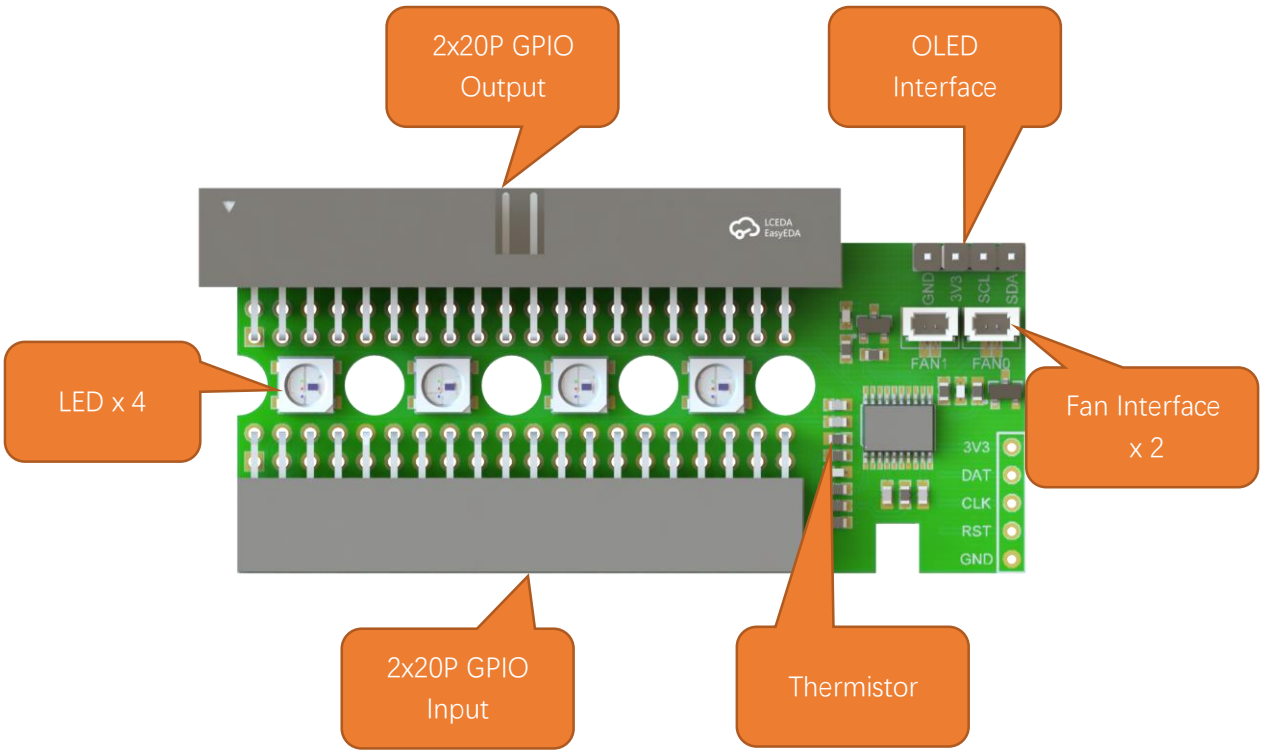
The Case Adapter Board also features a conversion function that transforms the RPi 5's PCIe interface into an M.2 M-Key interface. This conversion provides great convenience when it comes to using NVME SSDs. It supports four different sizes of SSDs, namely 2230, 2242, 2260, and 2280. Important Note: Only the FNK0100B and FNK0100K models include an NVMe SSD. If you purchase any other model, you will need to buy a compatible SSD separately.

To enhance the accuracy of the Raspberry Pi's clock, it is advisable to add a battery to the RTC. For this purpose, we have installed a battery holder on the board. If you decide to add a battery, we recommend either purchasing a rechargeable lithium-manganese button-cell battery (with dimensions of 1220 and a voltage of 3V) or an official battery.
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#real-time-clock-rtc。

Due to the addition of the audio separation function, there is an increase in power consumption. Only the official 5.1V/5A power adapter (or compatible versions) can be used.

1.2 Introduction to Freenove Case GPIO Adapter for Raspberry Pi

Hereinafter we will refer to this component as GPIO Board.



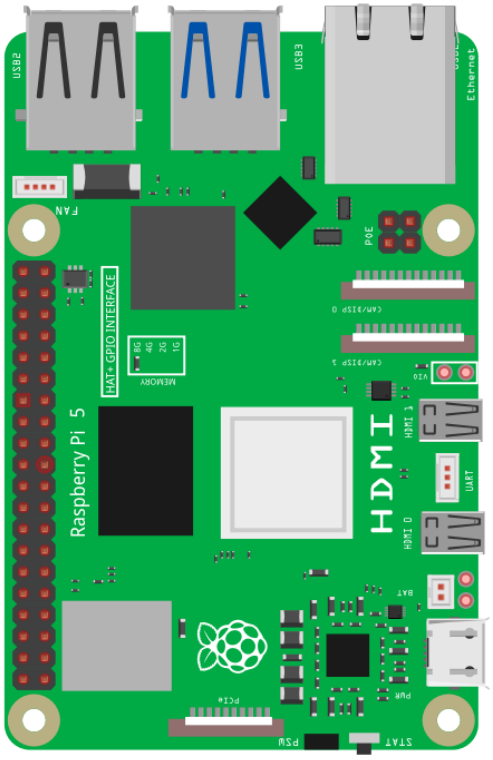
The GPIO board extends the GPIO pins of the Raspberry Pi 5 to the outside of the case. Additionally, it integrates color lights, case fans, and OLED interface, which can be controlled via the IIC of the RPi 5.

The following table shows the power consumption of the GPIO Board.

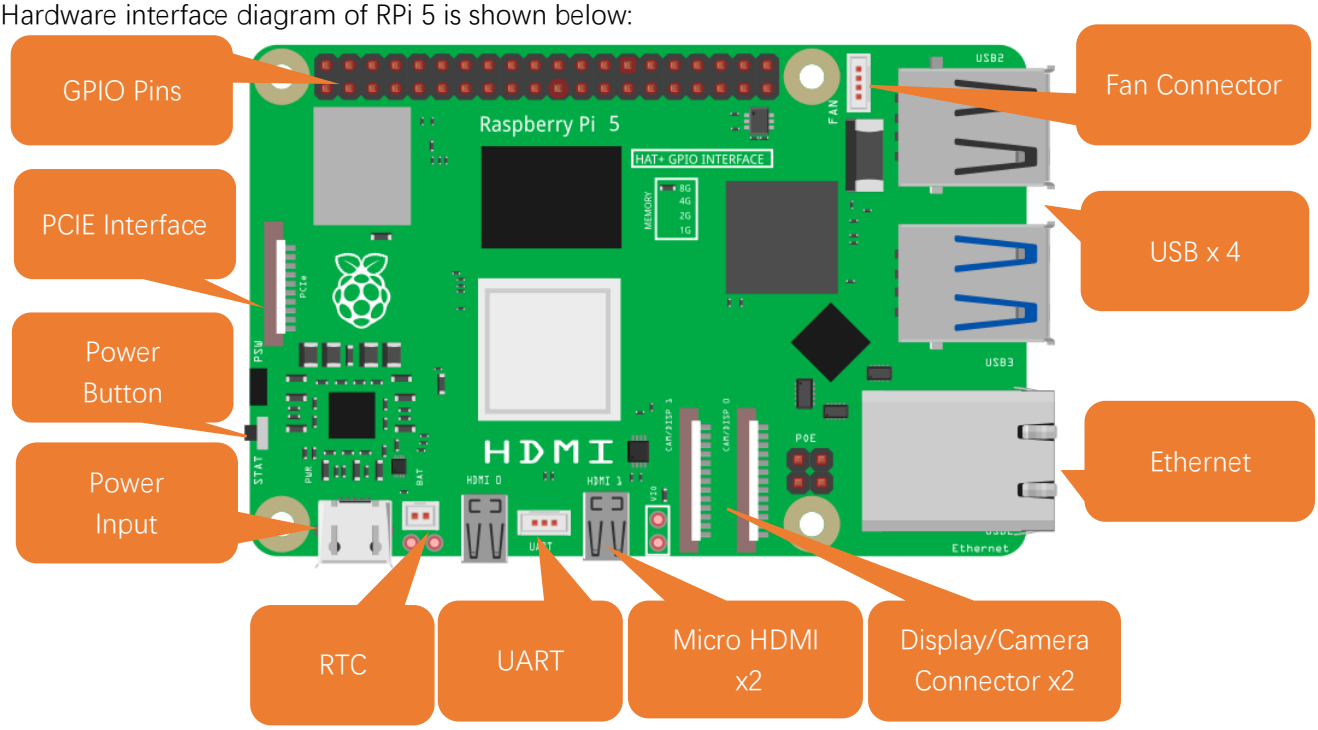
Function	Supply Voltage	Electric current
LED	5V	40*4mA
Fan	5V	130mA
OLED	3.3V	22mA

1.3 Introduction to Raspberry Pi 5 (RPi 5)

At the time of this writing, this product only supports RPi5. The following shows the physical and model figures of an RPi 5.

Practicality picture of Raspberry Pi 5:	Model diagram of Raspberry Pi 5:
	

Hardware interface diagram of RPi 5 is shown below:

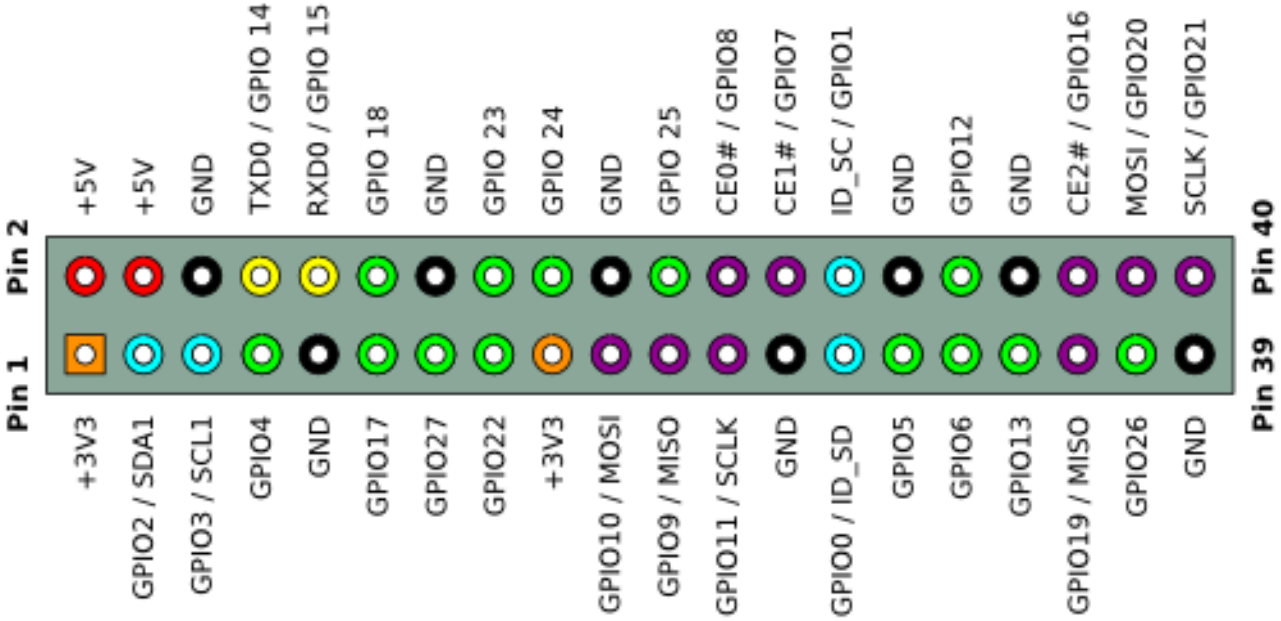


GPIO

GPIO: General purpose input/output. We will introduce the specific feature of the pins on the Raspberry Pi and how you can utilize them in all sorts of ways in your projects. Most RPi Module pins can be used as either an input or output, depending on your program and its functions. When programming the GPIO pins, there are three different ways to reference them: GPIO numbering, physical numbering, WiringPi GPIO Numbering.

BCM GPIO Numbering

The Raspberry Pi CPU uses Broadcom (BCM) processing chips BCM2835, BCM2836 or BCM2837. GPIO pin numbers are assigned by the processing chip manufacturer and are how the computer recognizes each pin. The pin numbers themselves do not make sense or have meaning, as they are only a form of identification. Since their numeric values and physical locations have no specific order, there is no way to remember them, so you will need to have a printed reference or a reference board that fits over the pins.
Each pin is defined as below:



For more details about pin definition of GPIO, please refer to https://pinout.xyz/

Power requirements of various versions of Raspberry Pi are shown in following table:

Product	Recommended PSU current capacity	Maximum total USB peripheral current draw	Typical bare-board active current consumption
Raspberry Pi Model A	700mA	500mA	200mA
Raspberry Pi Model B	1.2A	500mA	500mA
Raspberry Pi Model A+	700mA	500mA	180mA
Raspberry Pi Model B+	1.8A	600mA/1.2A (switchable)	330mA
Raspberry Pi 2 Model B	1.8A	600mA/1.2A (switchable)	350mA
Raspberry Pi 3 Model B	2.5A	1.2A	400mA
Raspberry Pi 3 Model A+	2.5A	Limited by PSU, board, and connector ratings only.	350mA
Raspberry Pi 3 Model B+	2.5A	1.2A	500mA
Raspberry Pi 4 Model B	3.0A	1.2A	600mA
Raspberry Pi 5 Model B	5.0A	1.6A (600mA if using a 3A power supply)	800mA
Raspberry Pi Zero W	1.2A	Limited by PSU, board, and connector ratings only.	150mA
Raspberry Pi Zero	1.2A	Limited by PSU, board, and connector ratings only	100mA

For more details, please refer to
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#power-supply

In this product, the Raspberry Pi 5 is used and it must be powered by a 5.1V/5A power supply. Insufficient power may cause various functions to operate abnormally, or even permanently damage your Raspberry Pi 5. Therefore, we strongly recommend using a 5.1V/5A power supply to ensure optimal performance and avoid potential hardware failure

Chapter 2 Assembly

Video: https://youtu.be/wyHVcwiXlbg

If you need any help, please feel free to contact us at support@freenove.com

It is recommended to assemble and use the Freenove Computer Case for Raspberry Pi according to this tutorial. Otherwise, it may lead to incorrect device installation or damage. Please check all the parts again. If there are any incorrect or missing parts in your kit, please contact us in time.

Please assemble the product when it is powered off.

Before installation, try using the cross-head screwdriver inclueded in the kit to pick up the small screws.
If the magnetic tip of your screwdriver is too weak to pick up the screws, you can magnetize it by gently touching it to the speaker's magnet for a few moments. (Be careful not to damage the speaker diaphragm.)

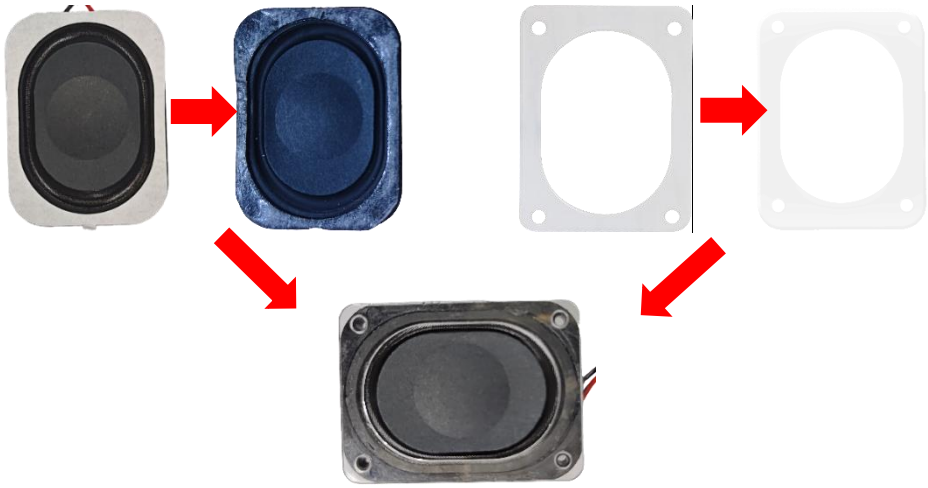


Step 1 Installation of Fans and Speakers

Installing the Speakers

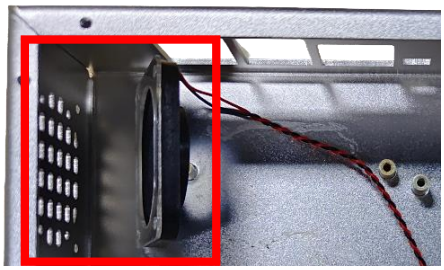
When attaching the speakers:
If there are 20 × M2x4 countersunk head screws and 6 × M2x6 ones available, use M2x4.
If there are 10 × M2x4 and 14 × M2x6 screws available, use M2x6.

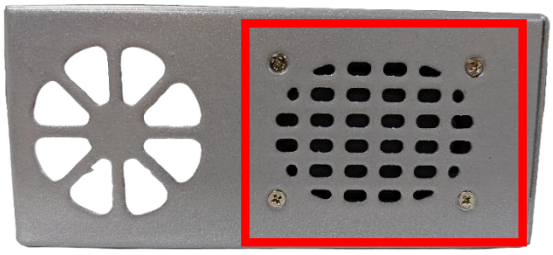
Step 1: Peel off the protective films from both the speaker and the acrylic pad, then align and adhere them together. (Note: Do not attach the speaker directly to the case, as this may make it difficult to remove later. If the speakers come stick to the bottom of the case, please click here to process.)



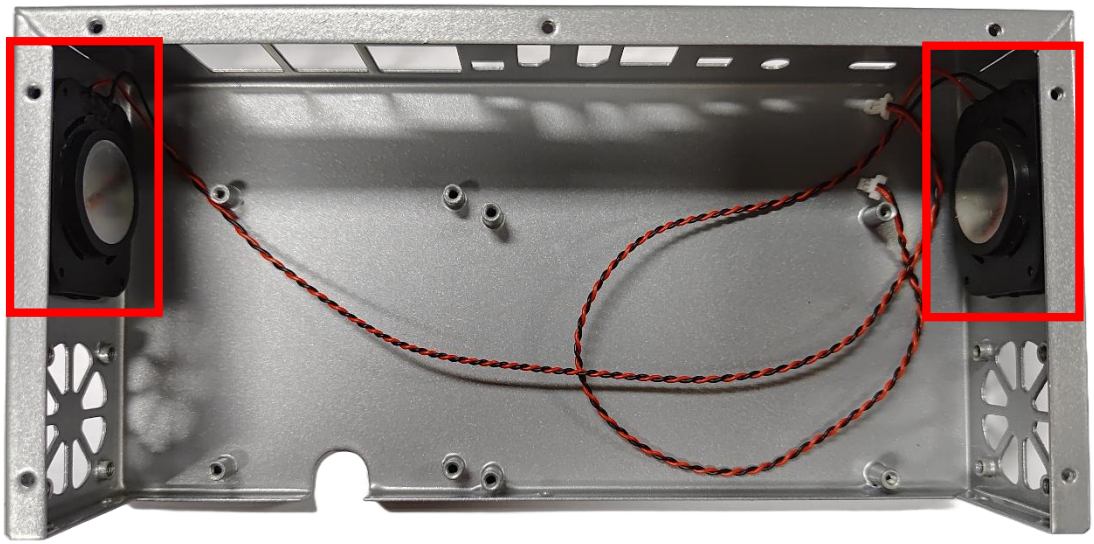
Step 2: Mount the speakers

Fix the assembled unit to the case using four screws.



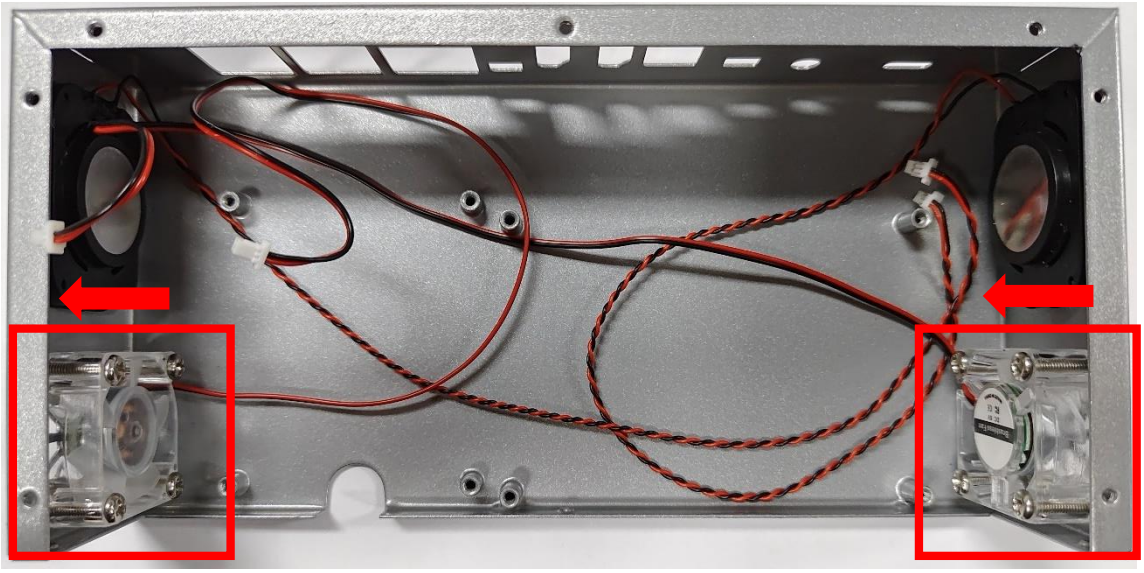


Installation finished as shown below:



Installing the Fans

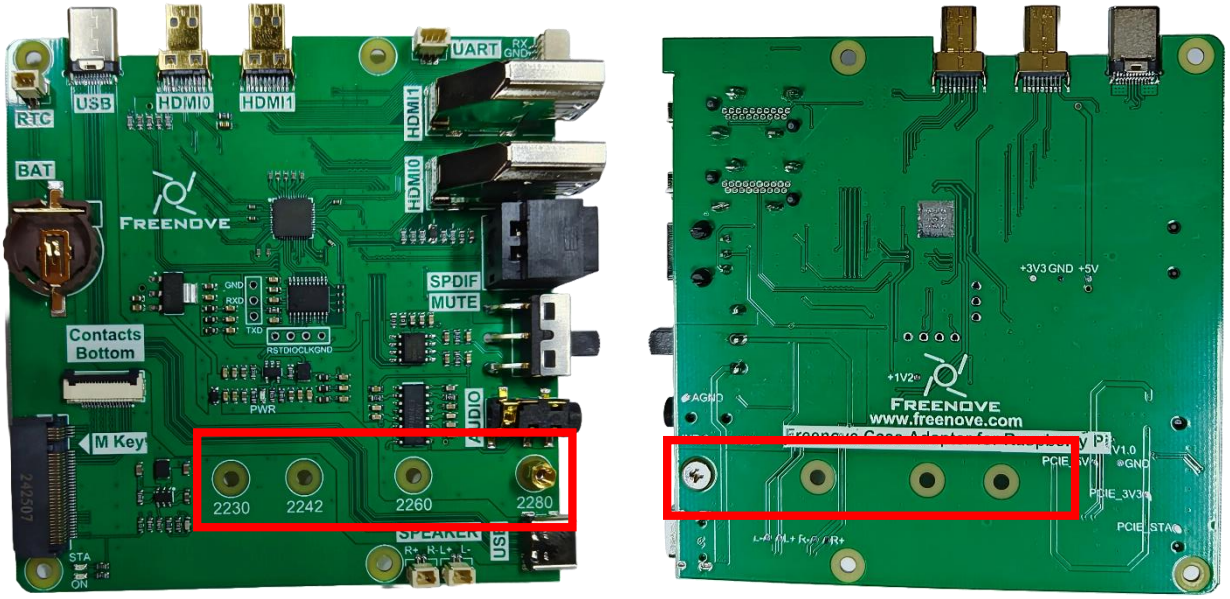
Use the M3 x 12 screws to secure the fans to the inside of the computer case. Pay close attention to the assembly orientation of the fan as they should form an air duct (indicated by the red arrows), which shows the direction of airflow. The fan on the left side is responsible for blowing the hot air out of the computer case's interior, while the fan on the right side draws in cool air.



Step 2 Installing Rpi 5 and Case Adapter Board

Installing the NVMe SSD Securing Standoffs

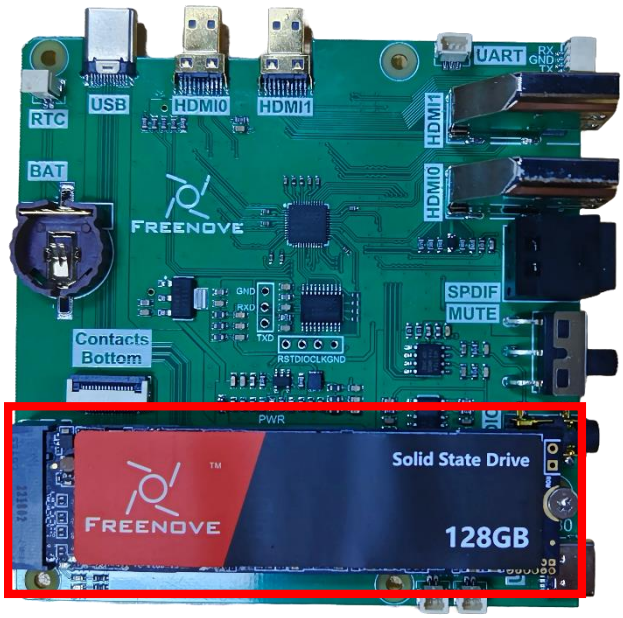
Install the brass standoffs for securing the NVMe SSD on the Case Adapter Board as illustrated below. Utilize M2.5x3 Screws to affix the M2.5x5 Brass Standoffs to the 2280 mounting holes. Depending on the length of your NVMe SSD, please install them on the appropriate holes.



Installing NVMe SSD

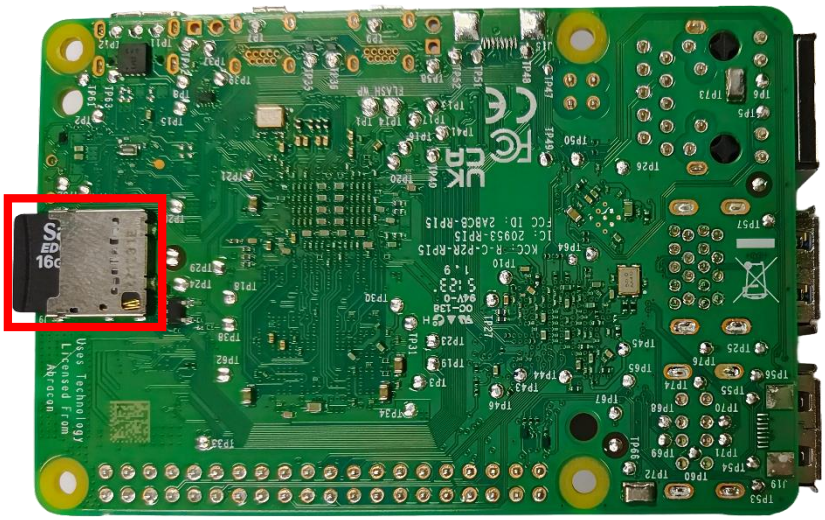
If you do not have an SSD, please skip to the next step.

Insert the NVMe SSD diagonally into the M.2 interface, and use the M2.5x3 Screws to fix it onto the brass standoffs for the NVMe SSD.

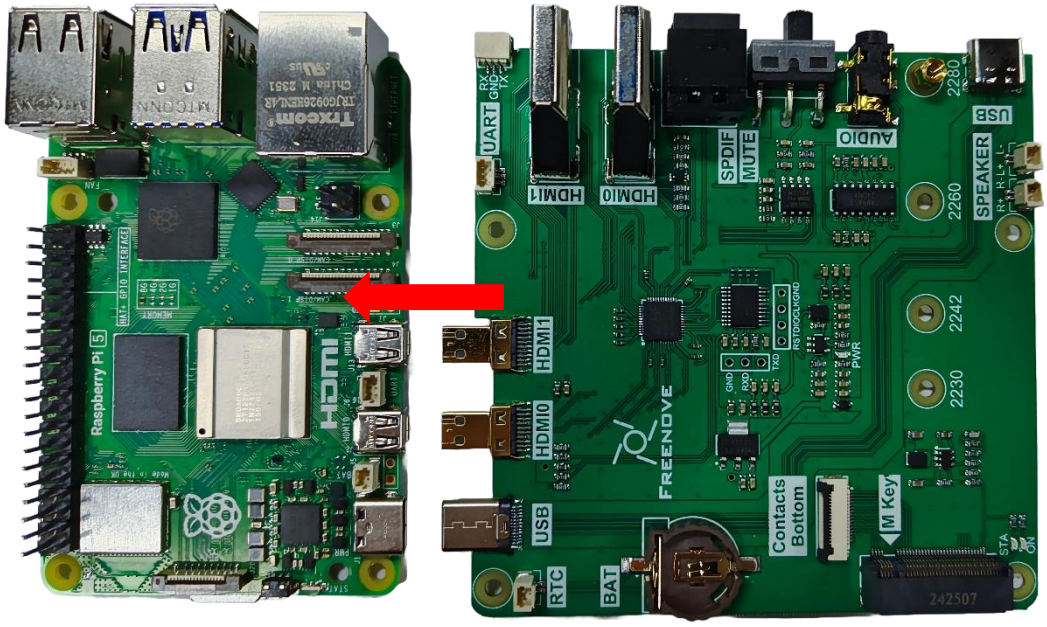


Installing RPi 5 and Case Adapter Board

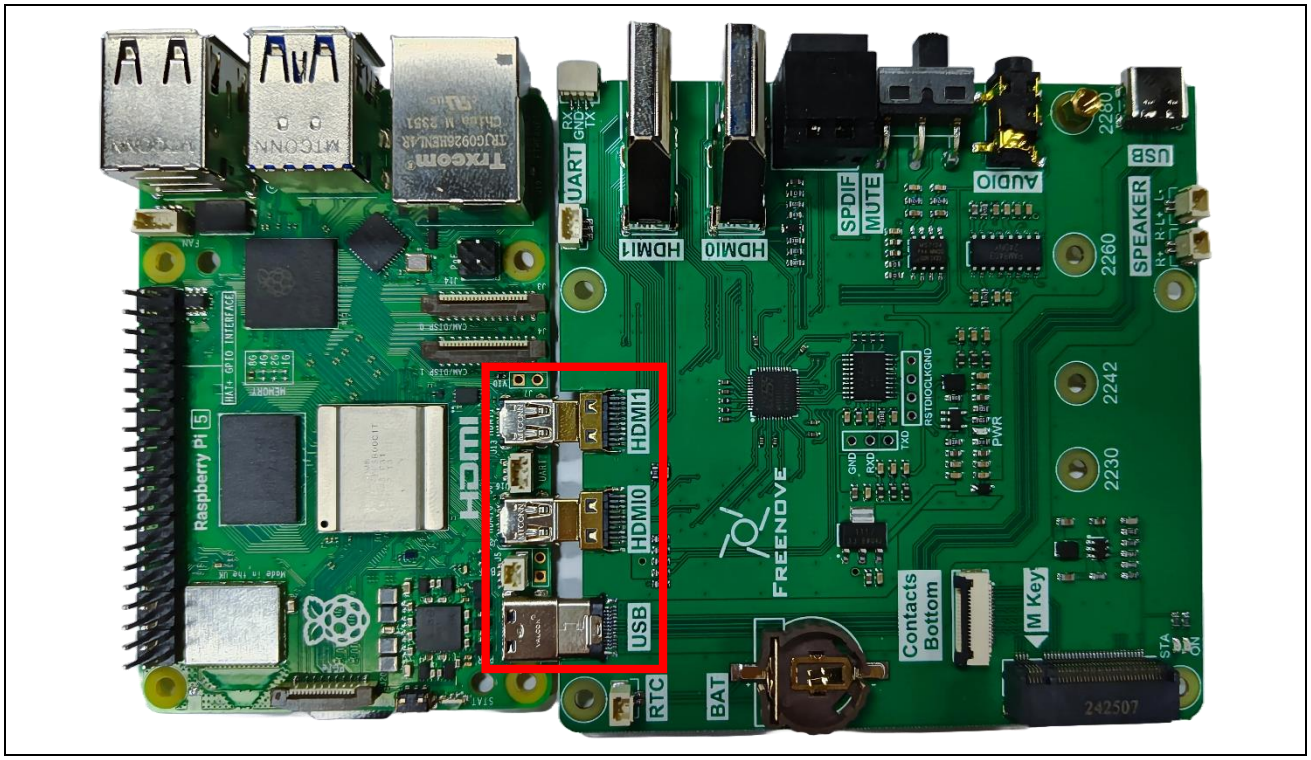
Inserting the SD card to the SD card slot on the back of RPi 5.



Align the Type-C interface, HDMI0 interface, and HDMI1 interface of the RPi 5 precisely with the corresponding interfaces on the Case Adapter Board. Then, connect the two boards together. Please follow the arrow directions for assembly.

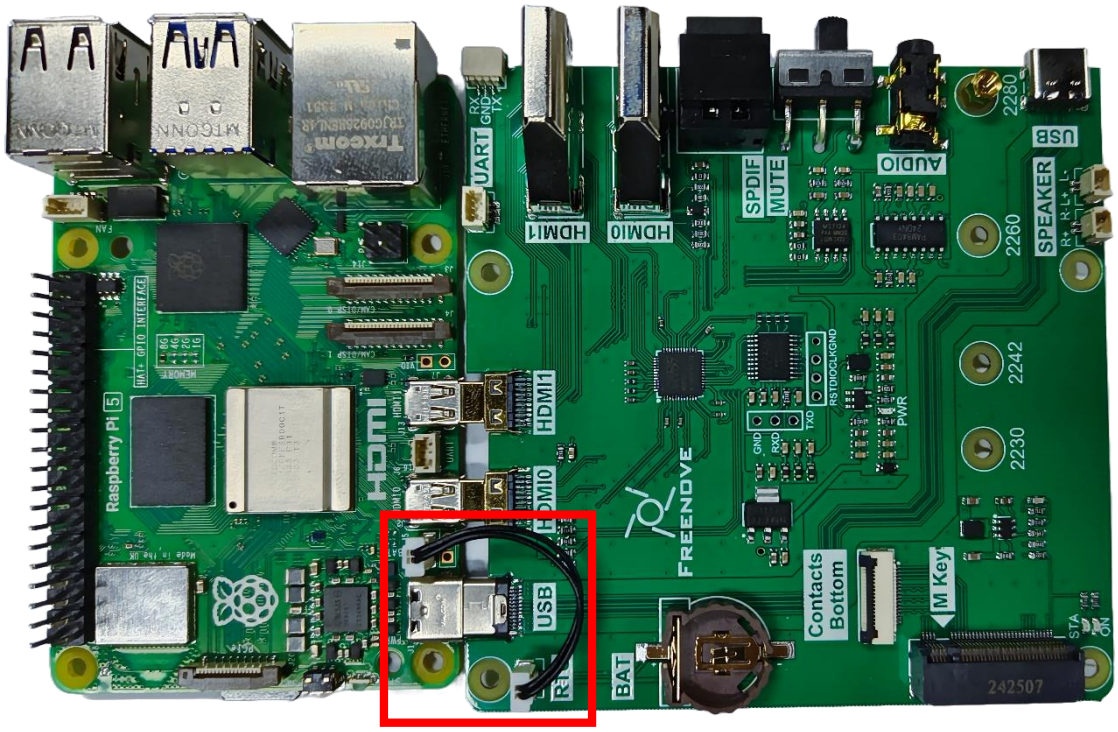


The following image shows the two boards after connection.

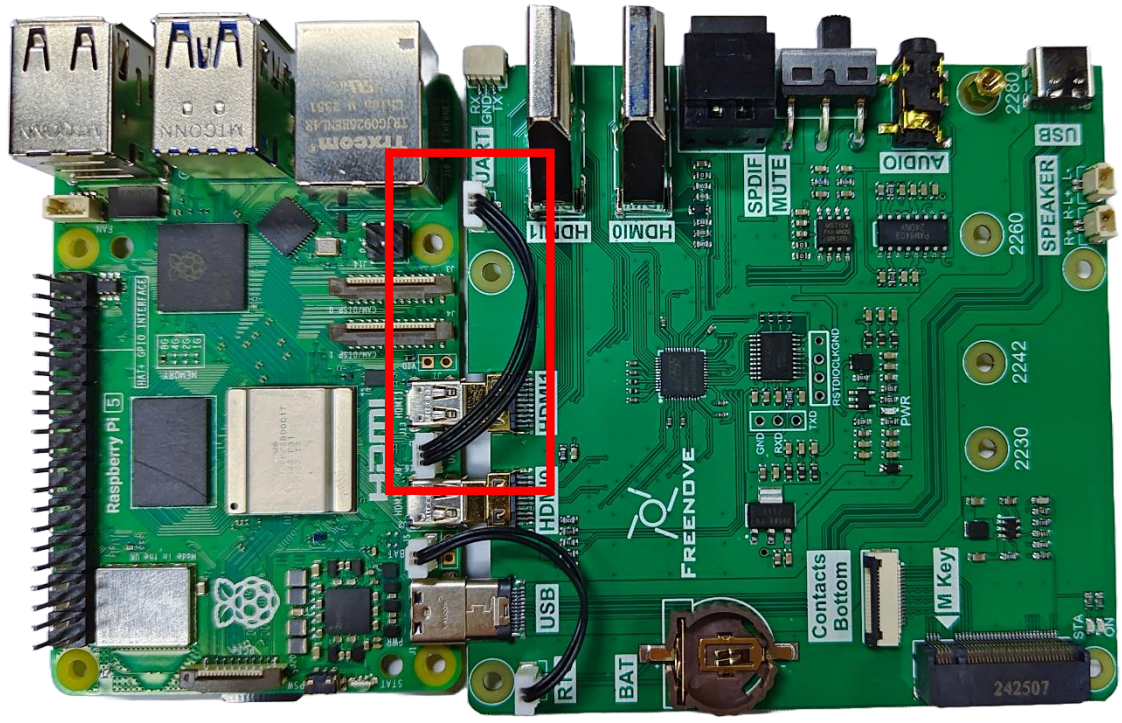

HDMI1
HDMI0
USB
RTC
BAT
UART
SPDIF
MUTE
AUDIO
USB
SPEAKER
Contacts Bottom
M Key
FREENOVE
Raspberry Pi 5

Connecting Cables of RTC, UART and the Speakers

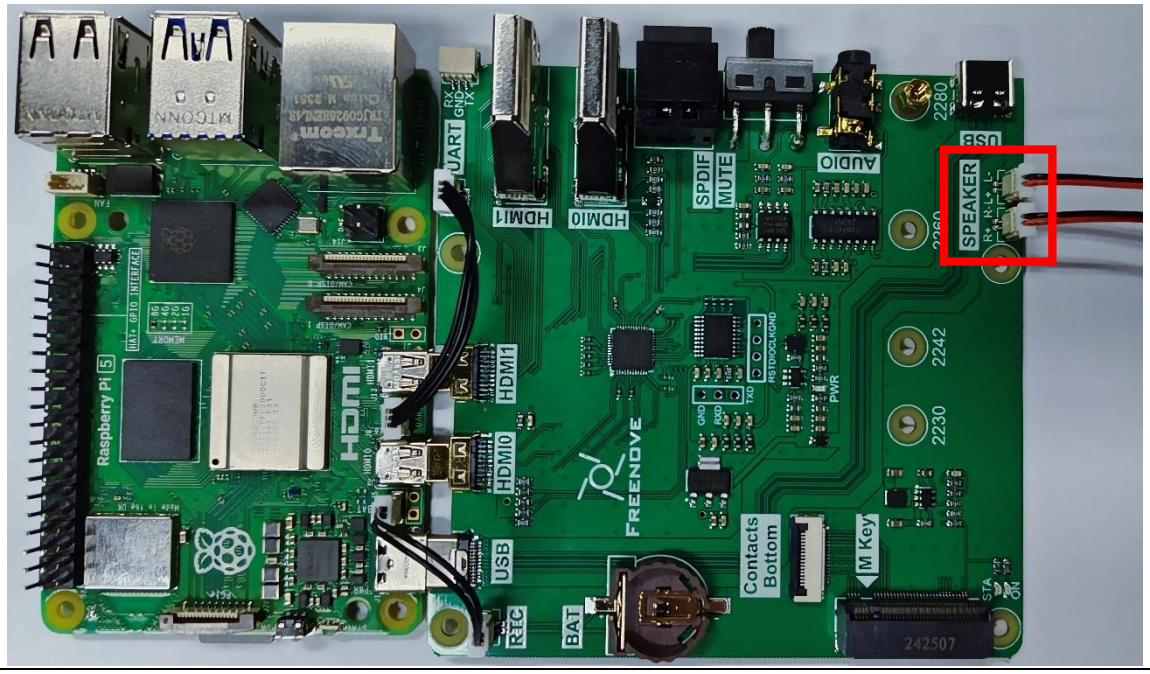
Connect the RPi 5's BAT connector to the RTC connector of the Case Adapter Board with the SH-1.0mm-2P cable.



Connect the RPi 5's UART connector to the UART connector of the Case Adapter Board with the SH-1.0mm-3P cable.

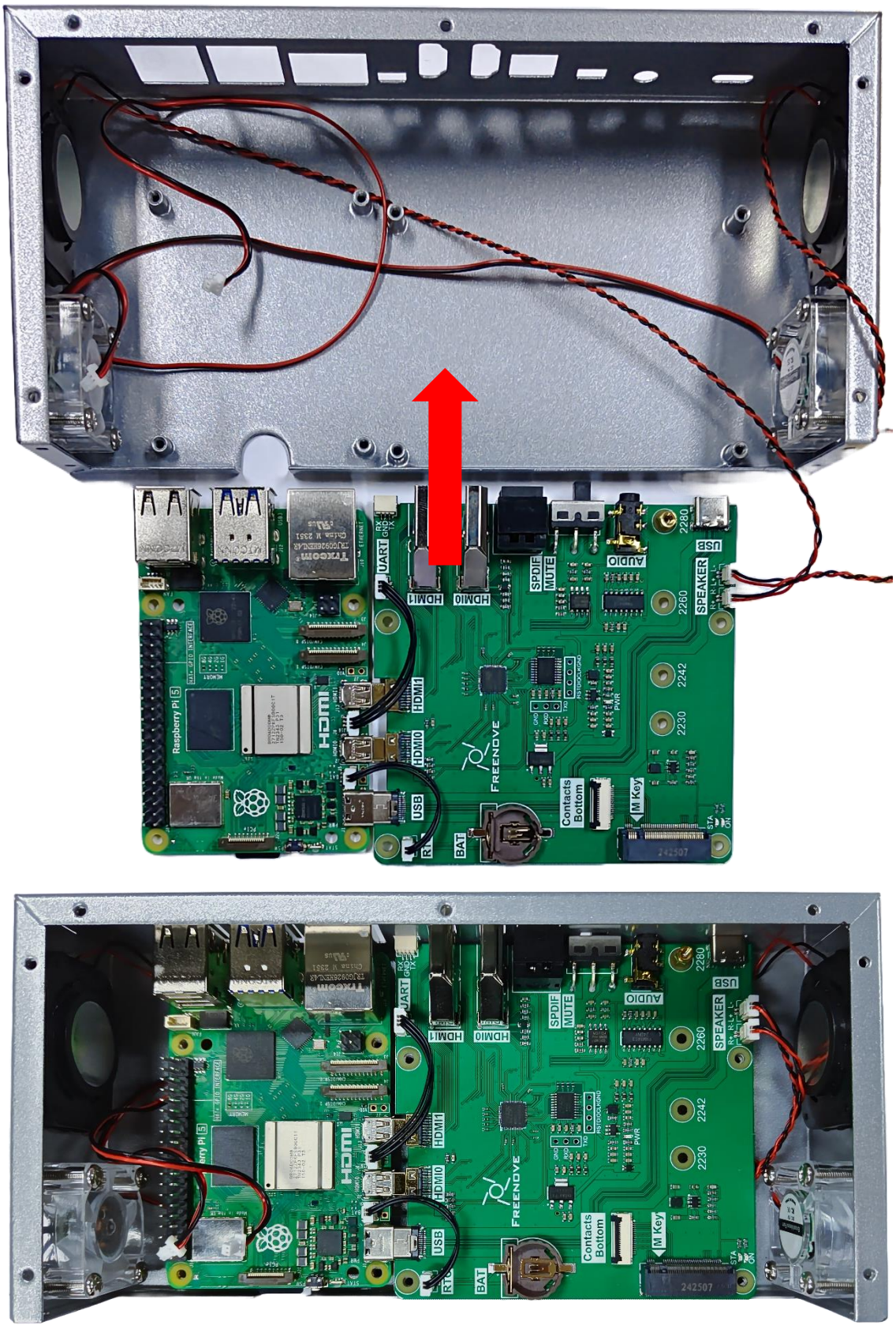


Connect the cables of the two speakers to the SPEAKER interface on the Case Adapter Board.

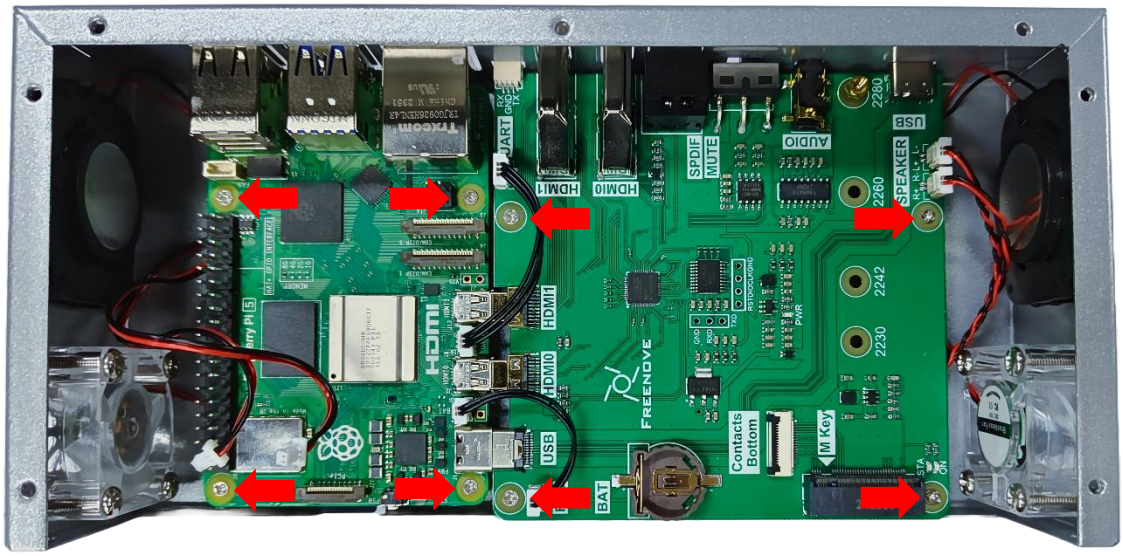


Assembly with the Case

Place the combined Raspberry Pi 5 and Case Adapter Board into the case.



Fix them to the case with M2.5 x 5 Countersunk Head Screws.



Step 3 Connecting Cables

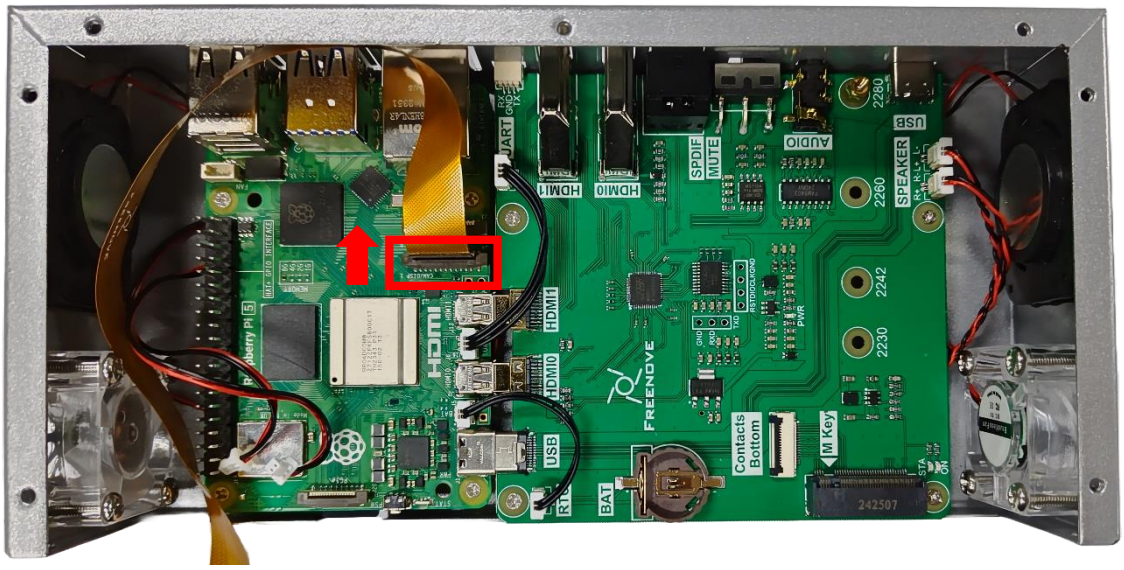
Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.

Connecting Cables

1. Connecting the camera cable

   Connect the shorter end of the camera to the CAM/DISP 1 interface on the RPi 5.

   Pay attention to the orientation of the contacts. (The red arrow below indicates the orientation of the contacts).

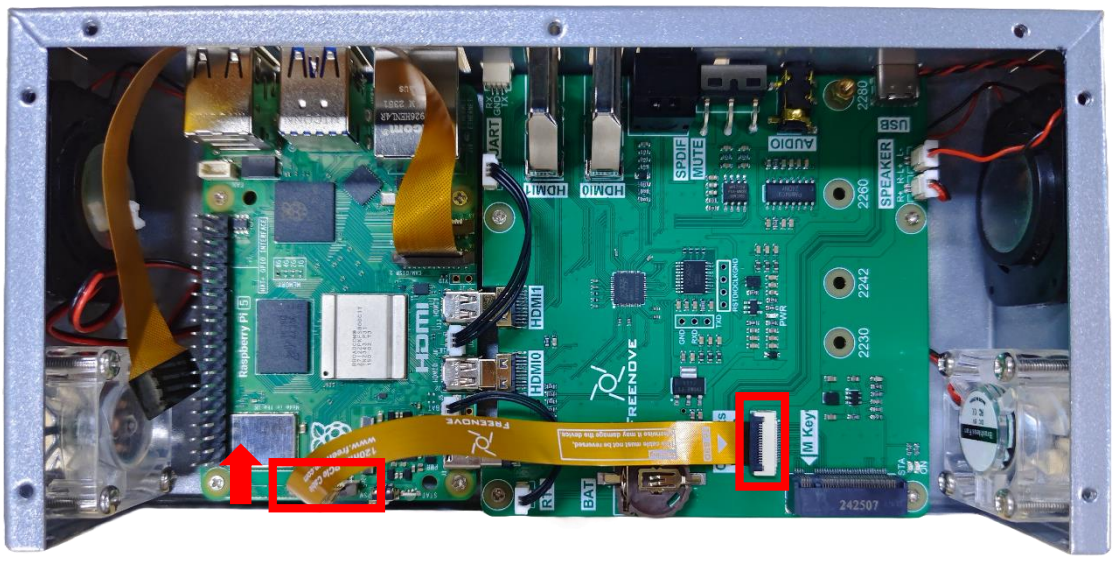


Caution: Although the Raspberry Pi camera cable and screen cable may look identical in appearance, they are NOT interchangeable.

2. Connect the NVMe SSD cable

Connect the cable for NVME SSD to the PCIe interface on the RPi 5. Pay attention to the orientation of the contacts (The red arrow below indicates the contacts orientation.)

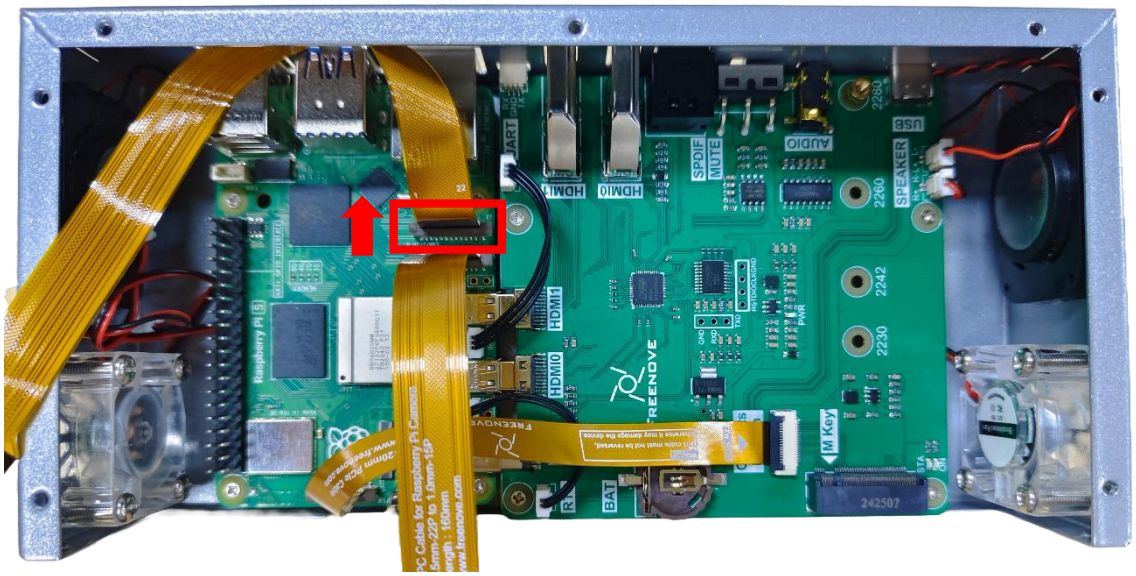
Connect the other end of the cable to the FPC connector of the Case Adapter Board.



3. Connect the screen cable

   If your purchase is not an FNK0100H or FNK0100K, please skip this.

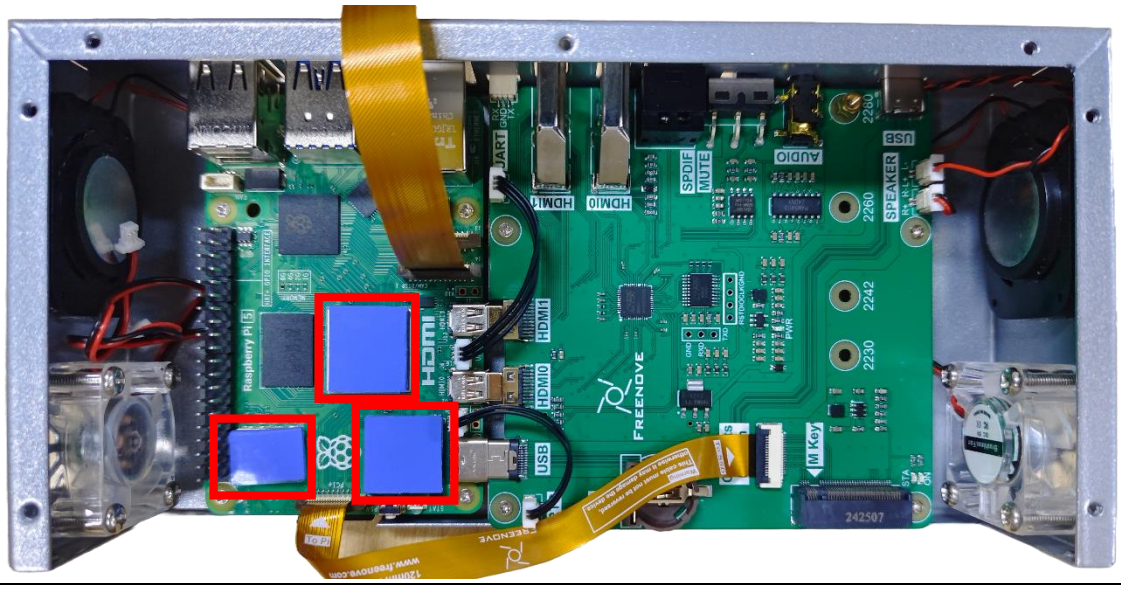
   Connect the shorter end of the cable to the CAM/DISP 0 inferface of RPi 5. Pay attention to the contacts orientation (the red arrow below indicates the contacts orientation).



Caution: Although the Raspberry Pi camera cable and screen cable may look identical in appearance, they are NOT interchangeable.

Installing the RPi 5 CPU cooler

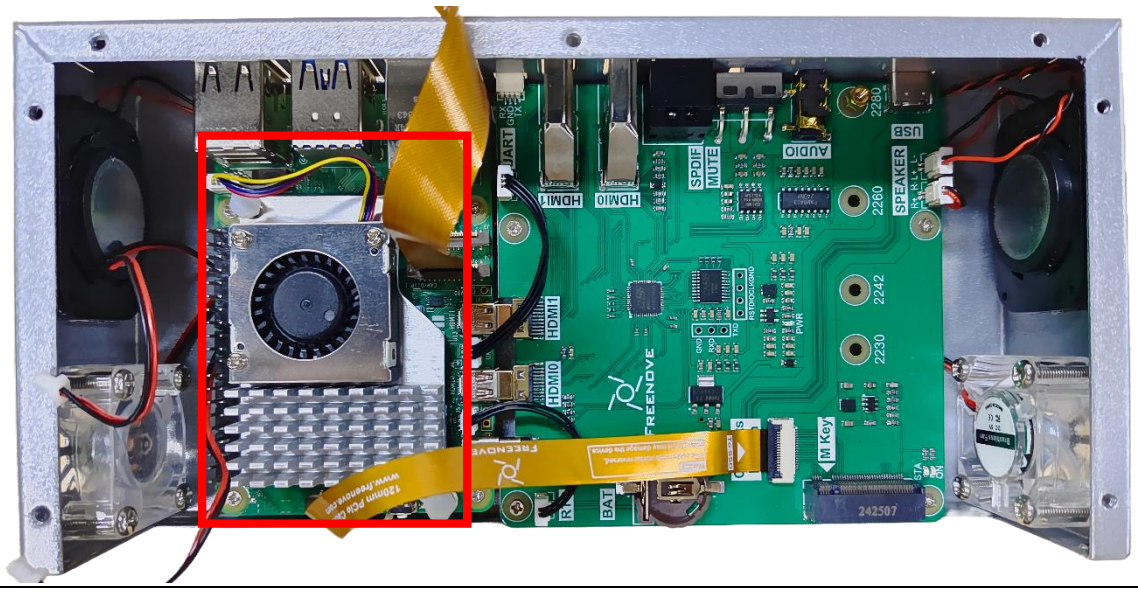
Attach the heat dissipation silicone pads to the power management chip, CPU, and WIFI module of the Raspberry Pi 5 respectively. (Note: Peel off the adhesive tape on the heat sink pad.)



Connect the cable of the active cooler to the fan interface on the RPi 5.

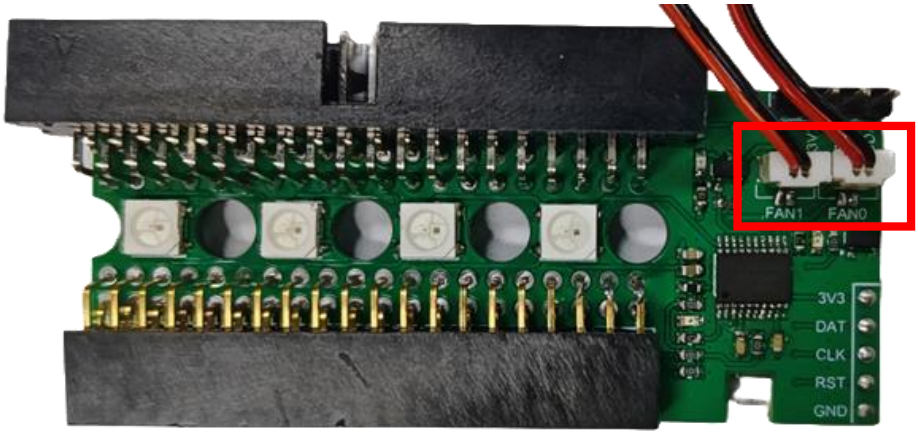


Fix the radiator on the Raspberry Pi 5 with nylon fixing pins.

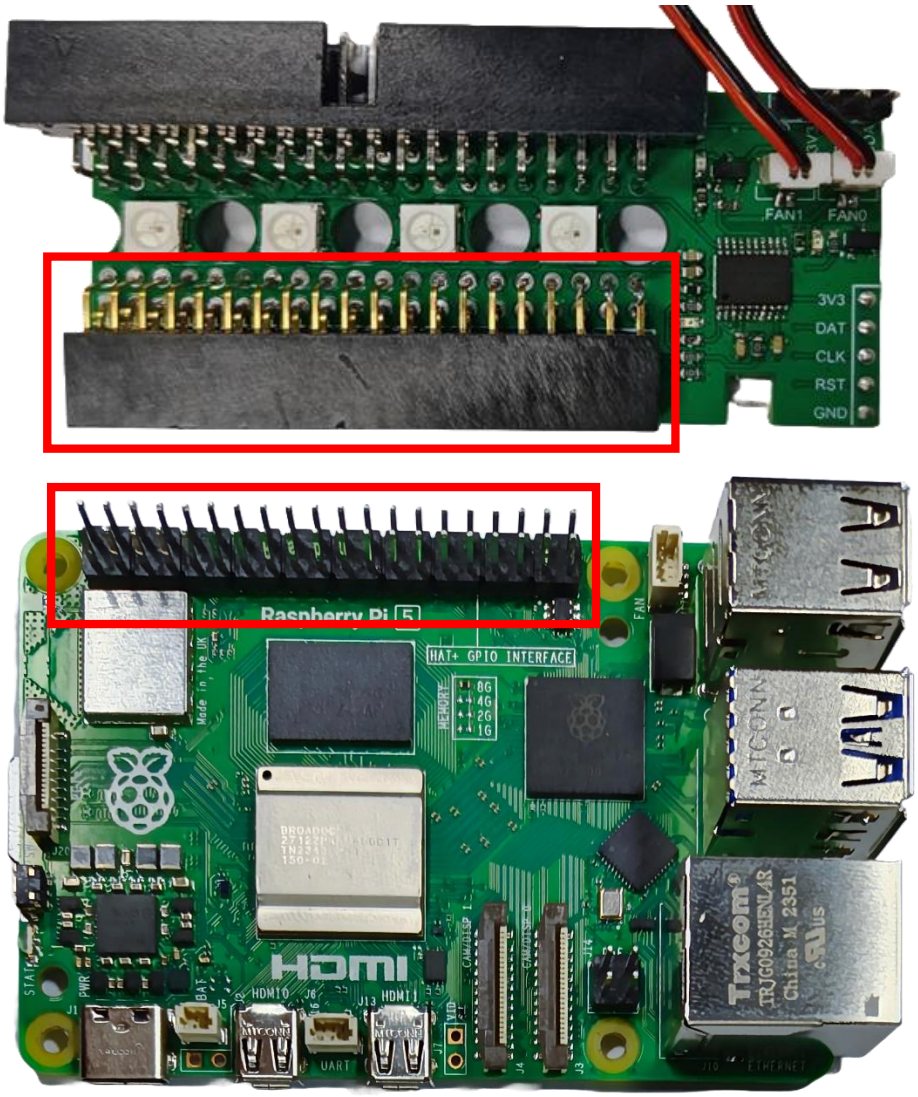


Installing GPIO Board

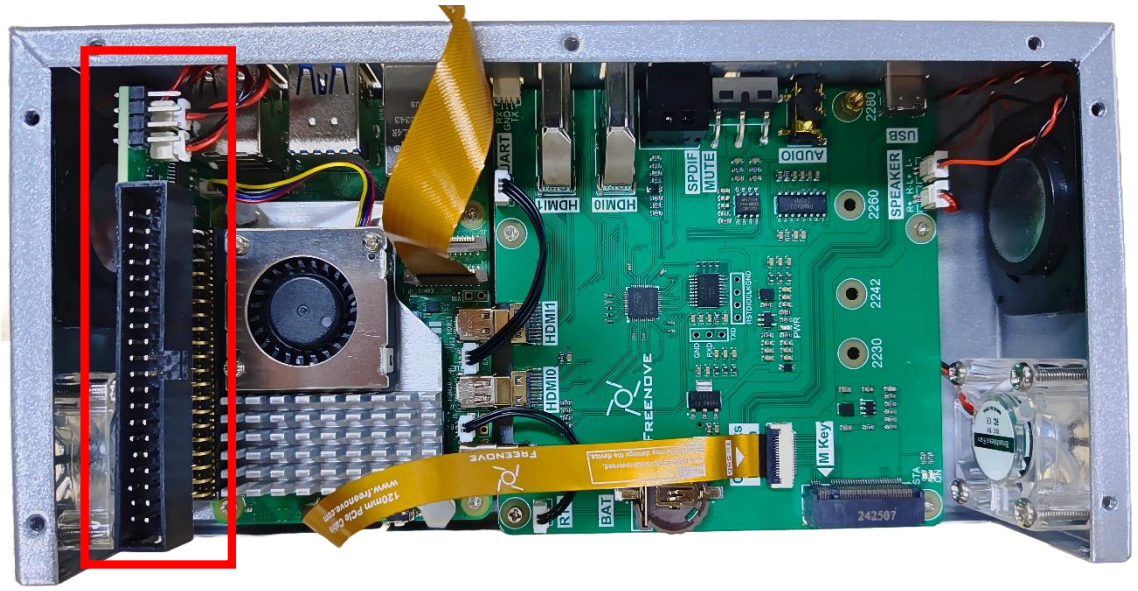
Connect the two fans on the computer case to the FAN0 and FAN1 interfaces on the GPIO Board respectively. You may receive a 2-wire or 4-wire fan.



Install the GPIO Board on the GPIO interface of the Raspberry Pi. Please pay attention to the assembly direction and avoid misalignment during installation.



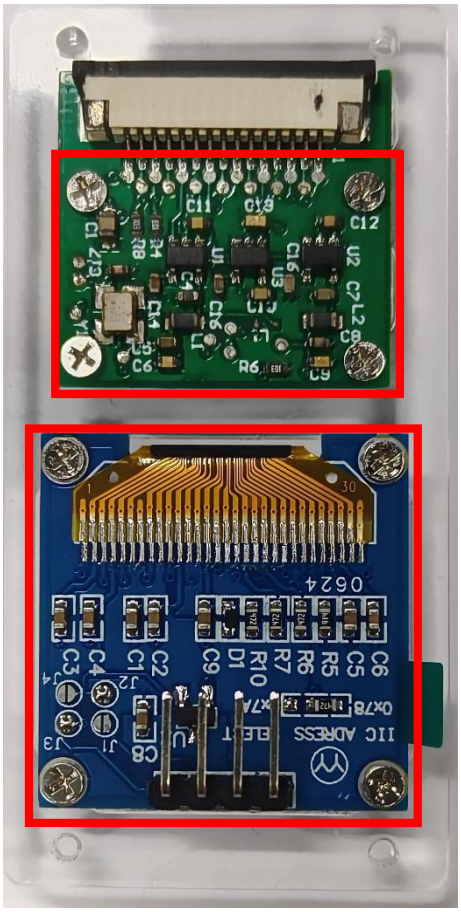
After installation, it should look like the figure below.



Step 4 Installing the Top Cover

Installing the Camera and the OLED Display

Fix the OLED display and the camera module on the acrylic part respectively with M2 x 4 Countersunk Head Screws.



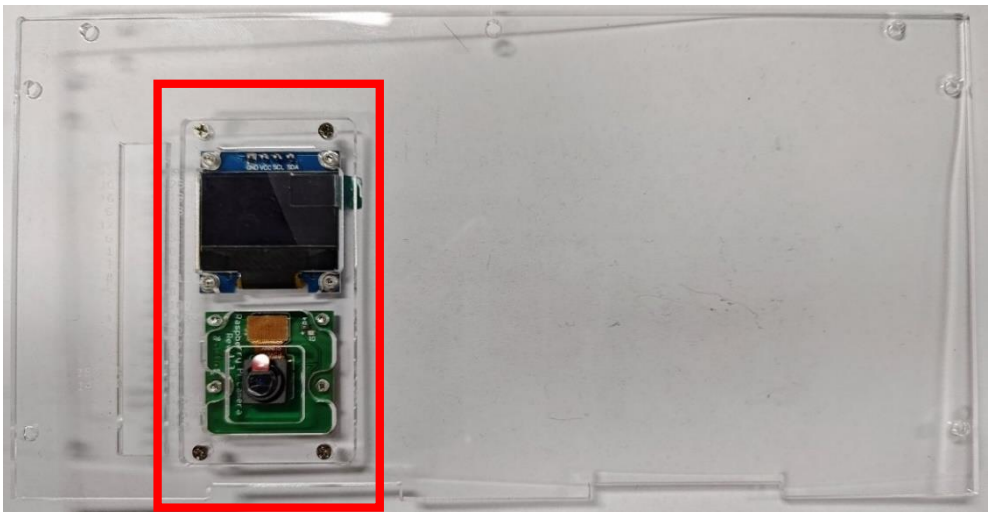


Assembly of FNK0100A or FNK0100B Top Cover

If your purchase is FNK0100H or FNK0100K, please skip to the next step.

Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.

Attach the assembled OLED screen and camera module beneath the top cover. Secure them from above using M2 x 6 Countersunk Head Screws, positioning them specifically at the left side of the cover.

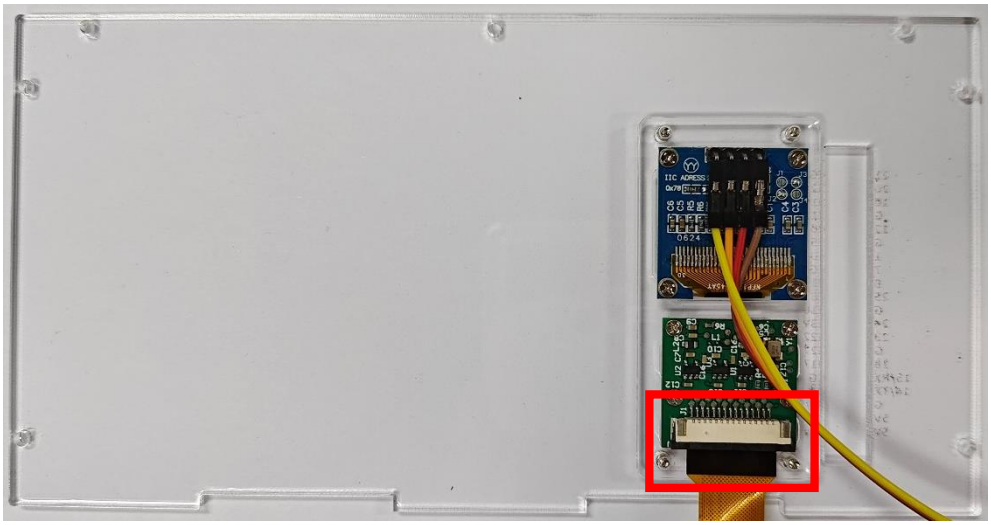


Use the F-F jumper cable to connect the interface of the OLED display to the 1x4P pin header interface on the GPIO Board. Please note the pin numbering carefully. Incorrect connection may result in damage to the components.



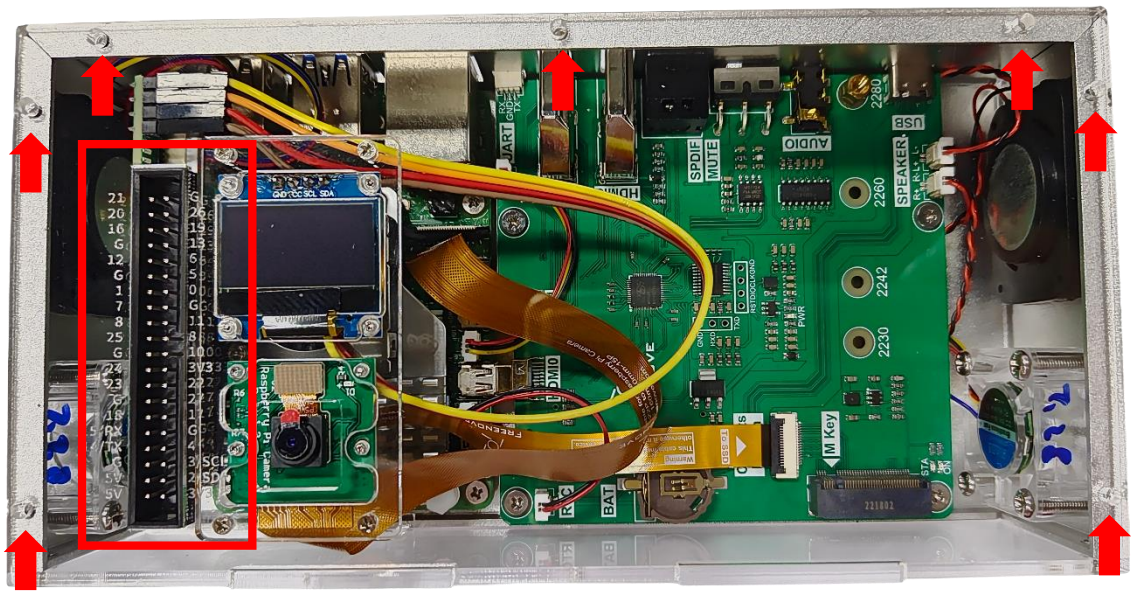


Connect the other end of the camera cable to the camera module, with the contacts facing downward.

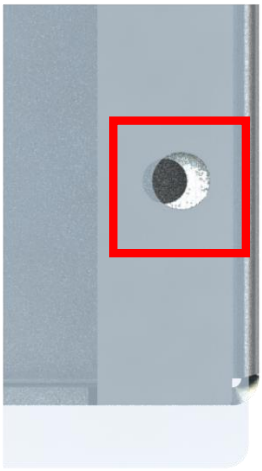


Carefully install the top cover onto the case, ensuring that none of the cables come into contact with the CPU cooler. Failure to do so may result in damage to the CPU cooler.

Align the holes on the left side of the cover with the horn seats of the GPIO Board. Then, use M2.5 x 5 Countersunk Head Screws to secure the top cover firmly in place.



Note: Owing to the machining precision limitations of sheet metal parts, if you notice that certain screw holes do not align properly, this is not an indication of an error. In such cases, you can apply force to bend and align them as needed.

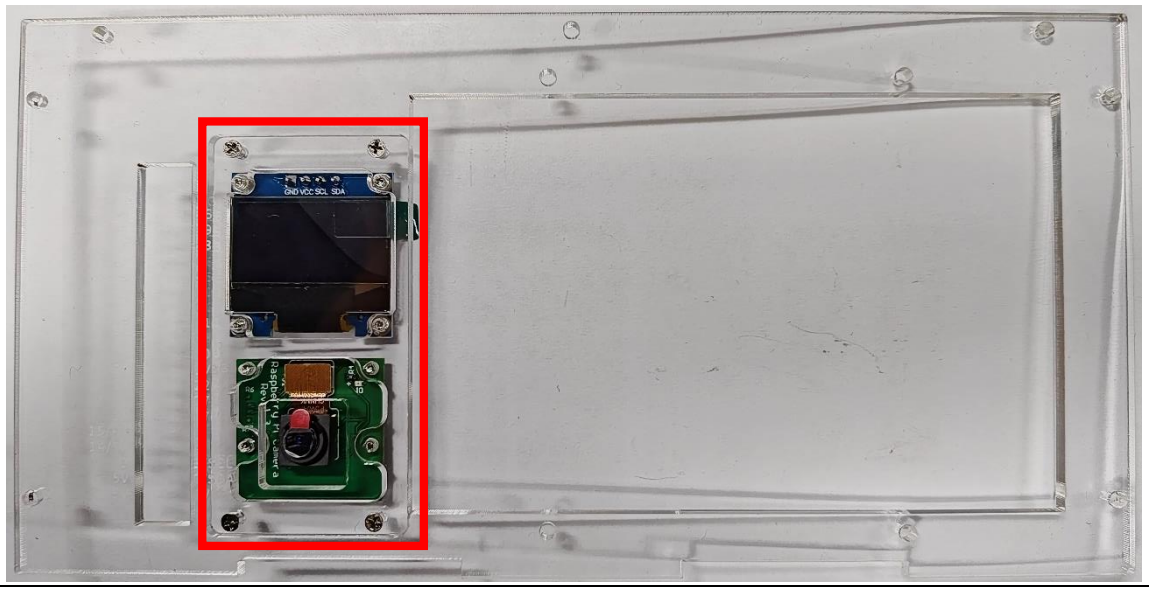


Assembly of FNK0100H or FNK0100K Top Cover

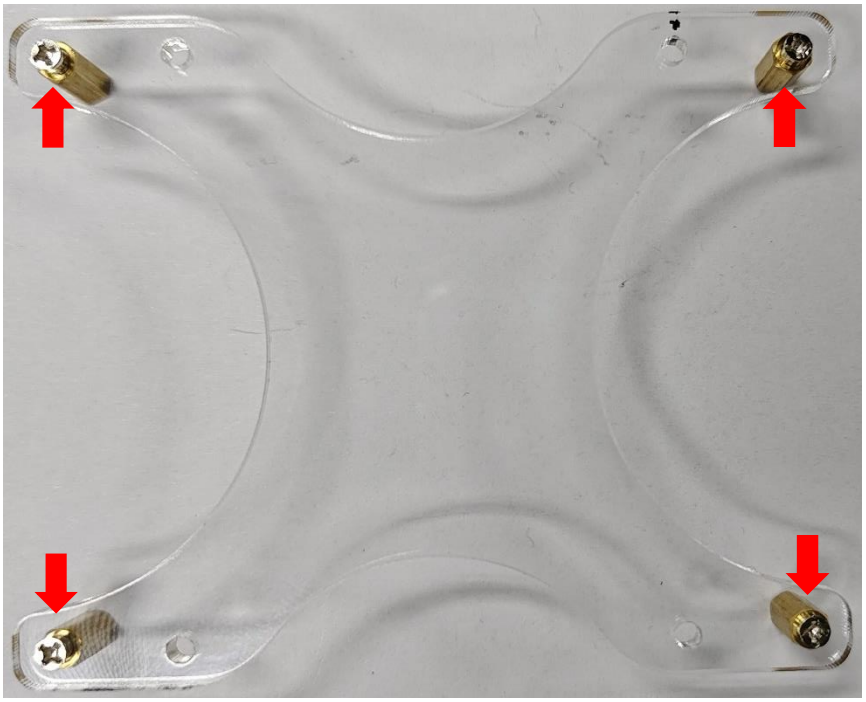
If your purchase is an FNK0100A or FNK0100B and you have assembled to top cover, you can skip to the next step.

Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.

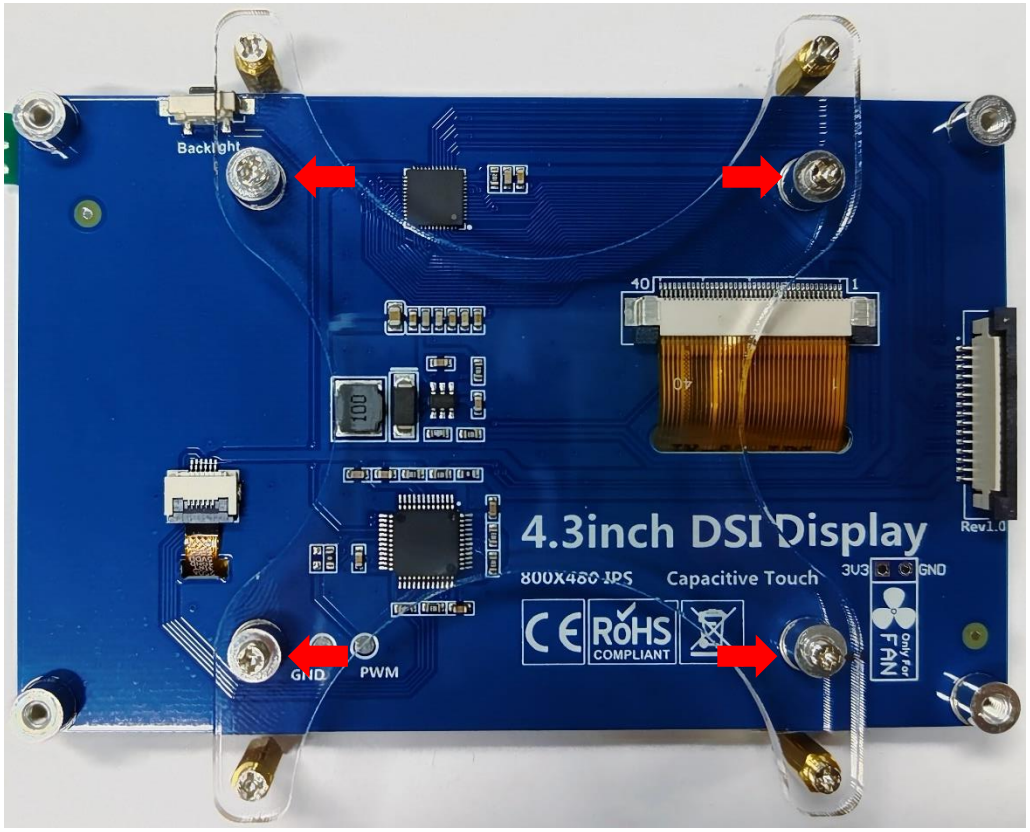
Attach the assembled OLED screen and camera module beneath the top cover. Secure them from above using M2 x 6 Countersunk Head Screws, positioning them specifically at the left side of the cover.



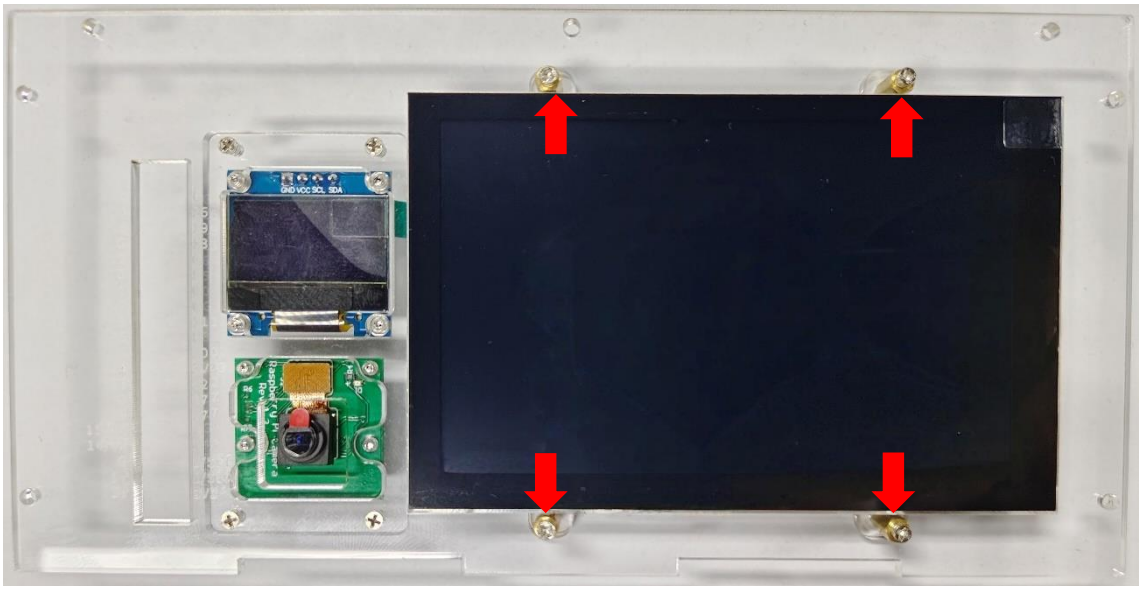
Use a combination of M2.5 x 5 Countersunk Head Screws and M2.5 x 9 Brass Standoffs in the outer holes at the four corners of the screen support plate.



Install the screen on the screen support plate and fix it in the inner holes at the four corners with M2.5 x 5 Countersunk Head Screws.



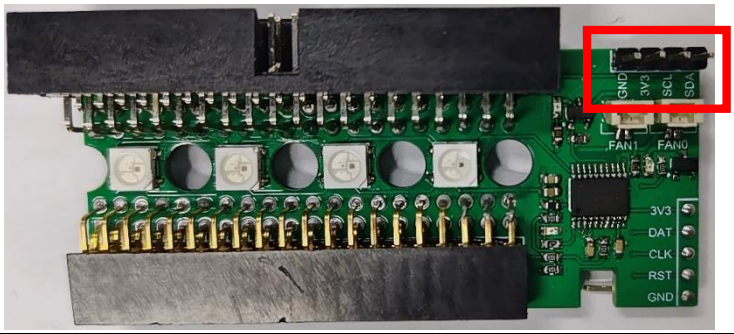
Carefully insert the assembled screen and screen support plate from the underside into the rectangular groove located on the right side of the top cover. Once properly positioned, secure them firmly from above using M2.5 x 5 Countersunk Head Screws.



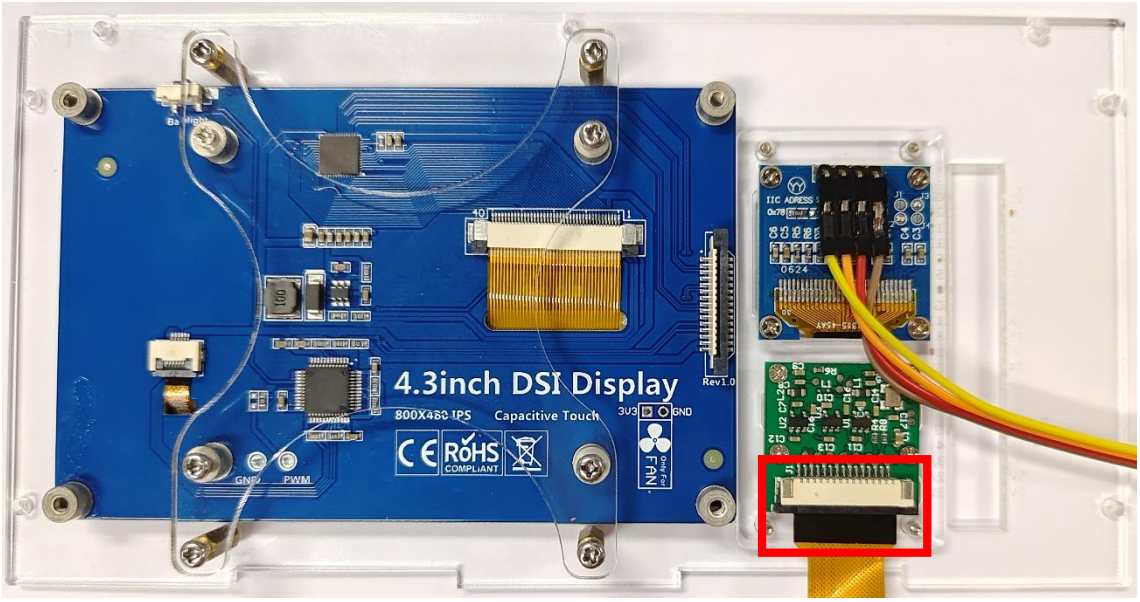
Connect the interface of the OLED display and the 1x4P pin header interface on the GPIO Board with the F-F jumper cable.

Pay close attention to the pin numbering. Incorrect connection may cause damage to the components.

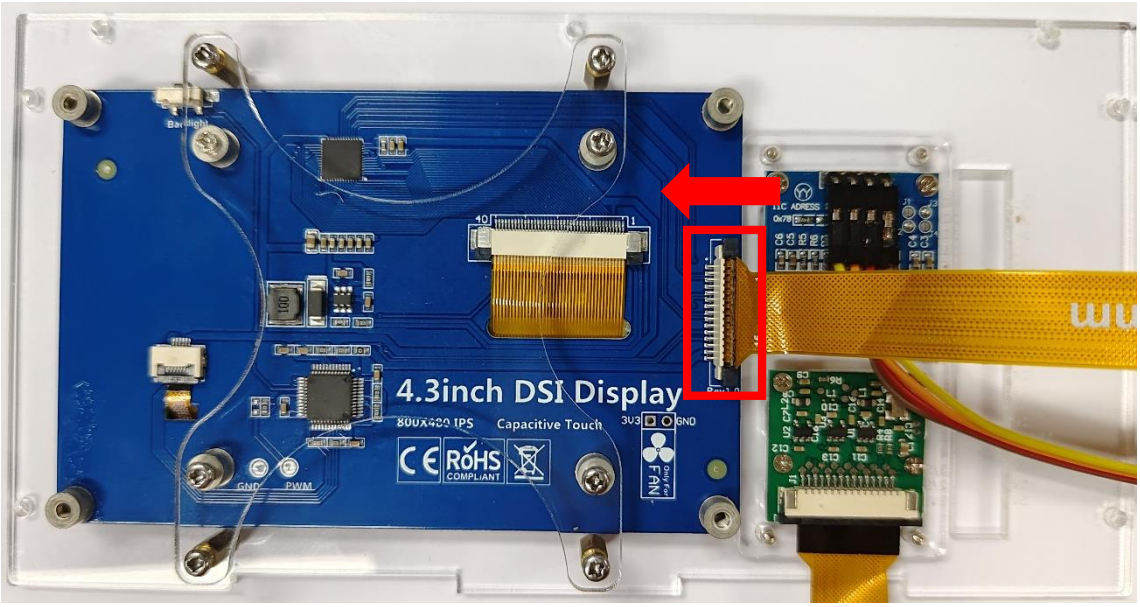




Connect the other end of the camera cable to the camera module, with the contacts facing downward.

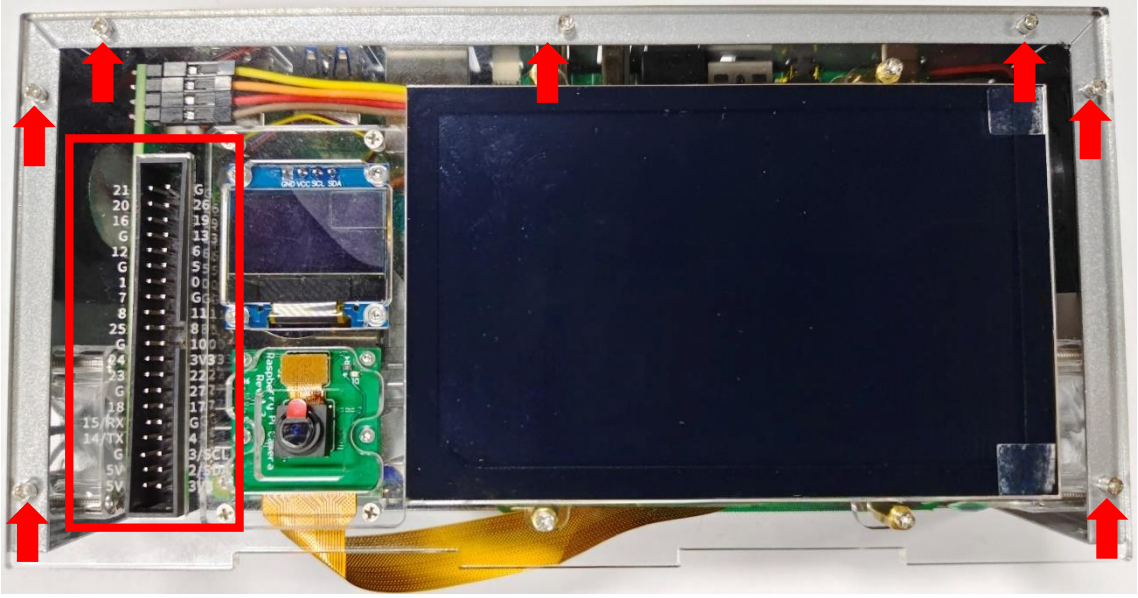


Connect the other end of the screen wire to the screen, paying attention to the orientation of the contacts (the red arrow indicates the direction of the contacts).

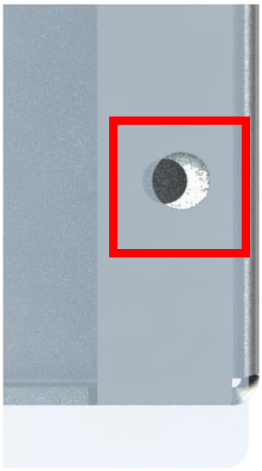


Carefully install the top cover onto the case, ensuring that none of the cables come into contact with the CPU cooler. Failure to do so may result in damage to the CPU cooler.

Align the holes on the left side of the cover with the horn seats of the GPIO Board. Then, use M2.5 x 5 Countersunk Head Screws to secure the top cover firmly in place.

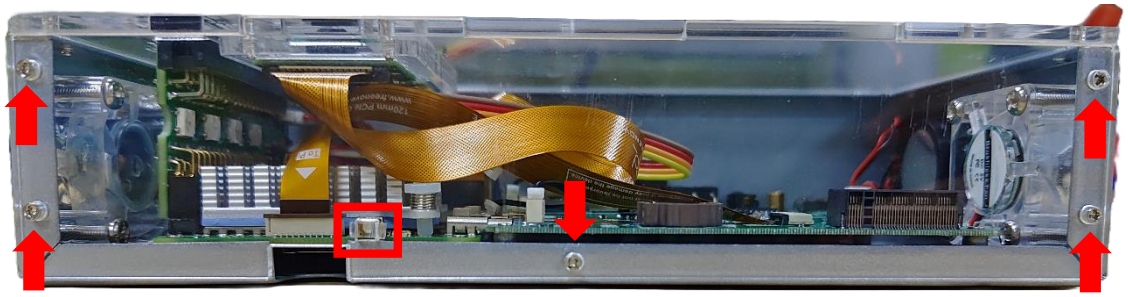


Note: Owing to the machining precision limitations of sheet metal parts, if you notice that certain screw holes do not align properly, this is not an indication of an error. In such cases, you can apply force to bend and align them as needed.



Step 5 Installing the Side Board

Insert the power button cap into the square hole on the side panel, and use the M2.5 x 5 Countersunk Head Screw to fix the side panel to the computer case.



Note: The Raspberry Pi's power button may be slightly recessed to prevent accidental shutdowns.

Following the above assembly steps, you have now successfully assembled your computer case. If you encounter any issues during the assembly process, please feel free to contact us at any time.
support@freenove.com

Chapter 3 APP Control

Before powering on the Freenove Computer Case Kit for Raspberry Pi, please make sure that all cable connections are correct.
Due to its multiple functions, this case requires an adequate power supply. We highly recommend using the official Raspberry Pi 5.1V / 5A power adapter (https://www.raspberrypi.com/products/27w-power-supply). Failure to do so may result in the Freenove Computer Case Kit for Raspberry Pi being unusable or causing damage to components.

Installing Raspberry Pi OS

If you have not yet installed the Raspberry Pi OS, please refer to the documentation "Installing Raspberry Pi OS.pdf" under the directory **Freenove Computer Case Kit for Raspberry Pi** to install the OS to your SD card or SSD.

If you have a display ready, please continue reading the tutorial below.

3.1 Boot Behavior & Environment Settings

3.1.1 What to Expect on First Startup

When you first power on the assembled chassis, the Raspberry Pi does not talk to the GPIO adapter board. This causes the board to operate on its own in a default mode, and you will observe the following.



1. The case RGB lights will operate in rainbow mode.
2. During initial startup, the fan will run at maximum speed for 3 seconds. After this period, the system switches to temperature-controlled mode, where fan speed is regulated by the thermal sensor on the GPIO adapter board

3. The screen will remain off during this phase - this is normal and expected.

Troubleshooting Guide:

If the RGB lights fail to illuminate, or if the fan doesn't perform the 3-second full-speed initialization, please:
Immediately power off the system

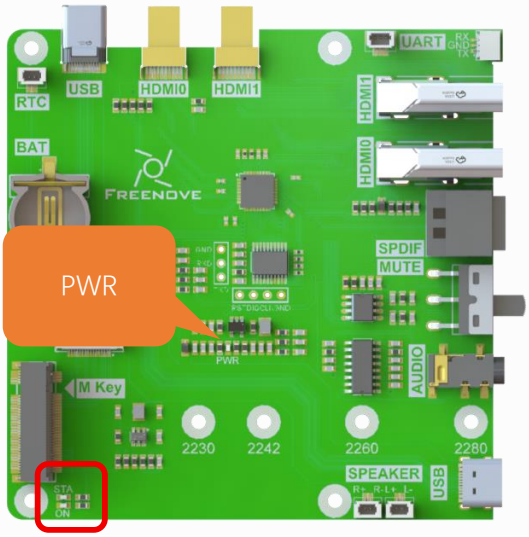
- Verify proper alignment and connection between the GPIO adapter board and Raspberry Pi
- Check all relevant power and data connections

If you have any questions of the above, please contact us at support@freenove.com

4. RPi 5 Status LED: The green STAT LED will remain steadily illuminated. If this LED is not lit or displays any pattern other than a steady green light, it indicates that the Raspberry Pi operating system has not booted successfully. In this case, please check your Raspberry Pi hardware and OS installation separately.



5. The case adapter board's onboard PWR LEDs remain solid, while the STA LED flashes in sync with SSD activity.



Troubleshooting Guide:

If any indicator shows abnormal behavior, immediately power off the system.

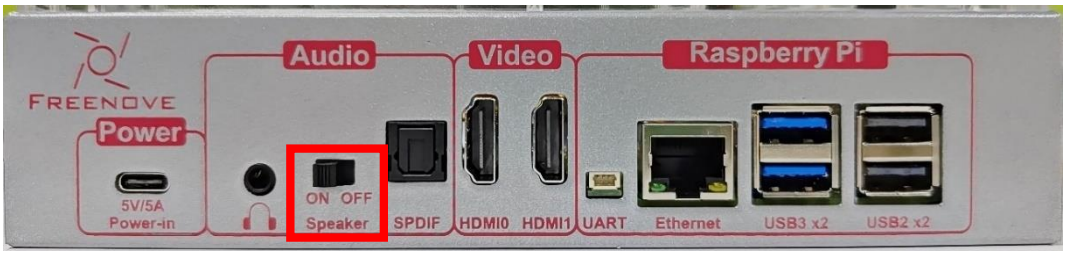
- If the PWR LED is off: Check the USB port and power cable connections.
- If the ON LED is off: Verify the FPC cable connection.
- If the STA LED remains solid (with the SSD connected): Inspect all SSD-related connections.

If you have any questions of the above, please contact us at support@freenove.com

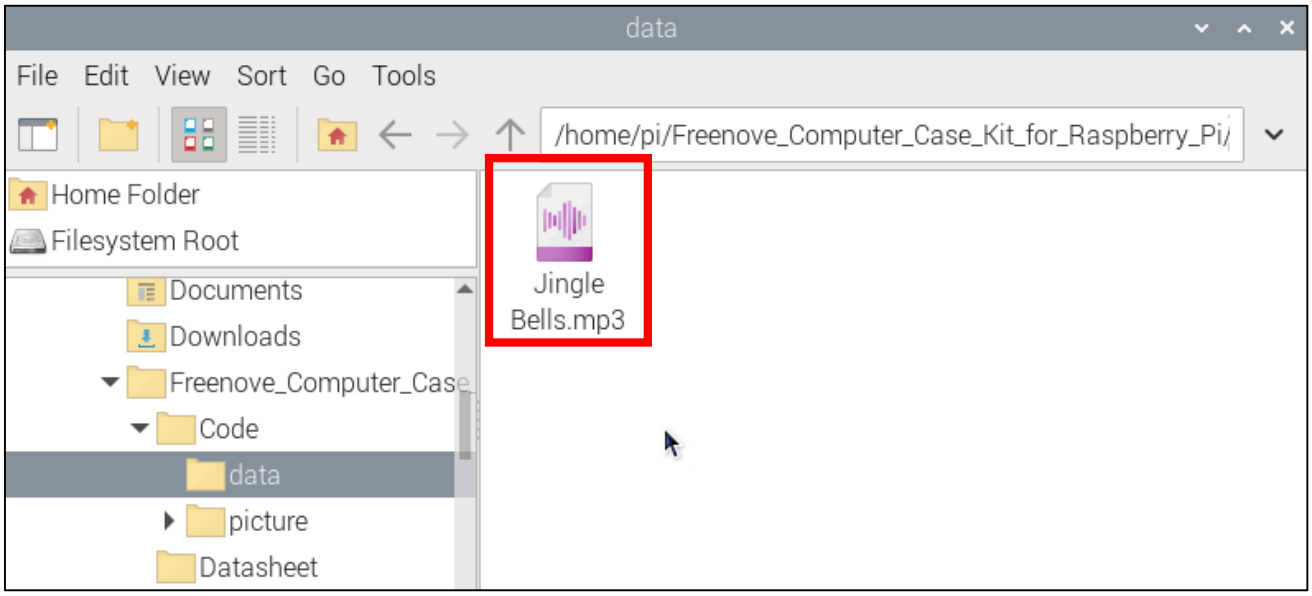
If all components behaves as expected, then your computer case is correctly assembled and functioning well. In this case, you can connect a screen to your Raspberry Pi or access it via VNC viewer.

Speaker Test

Before playing music, please check whether the speak switch is ON.

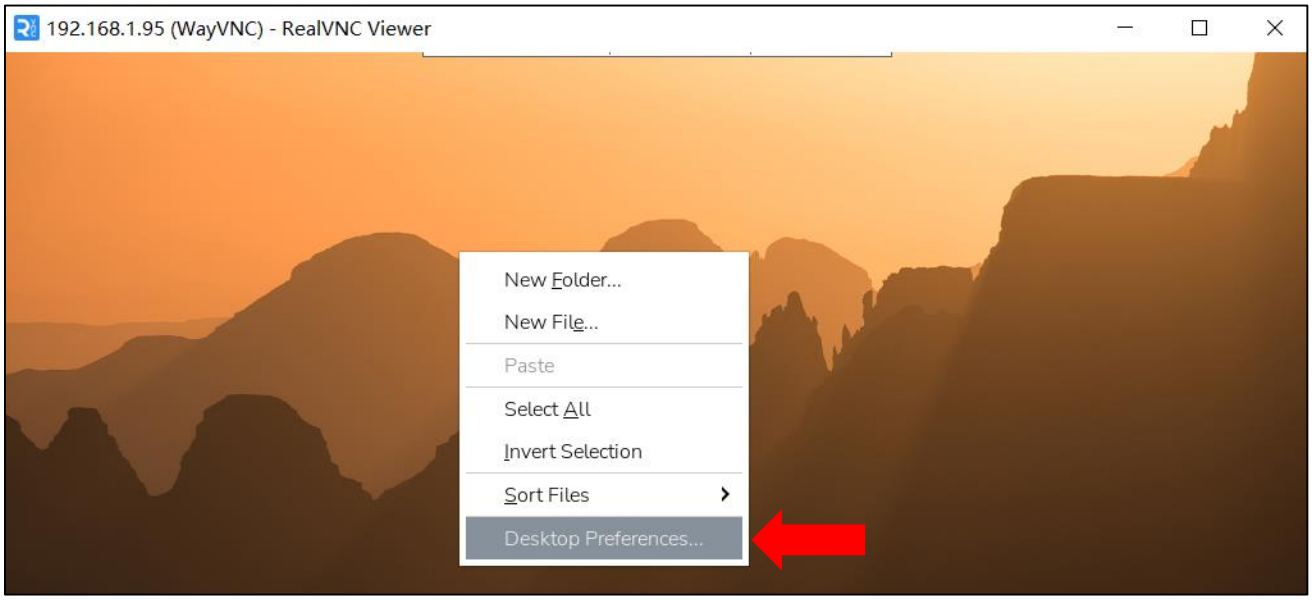


Open the data folder under the Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code/data directory, and click the mp3 file to display.

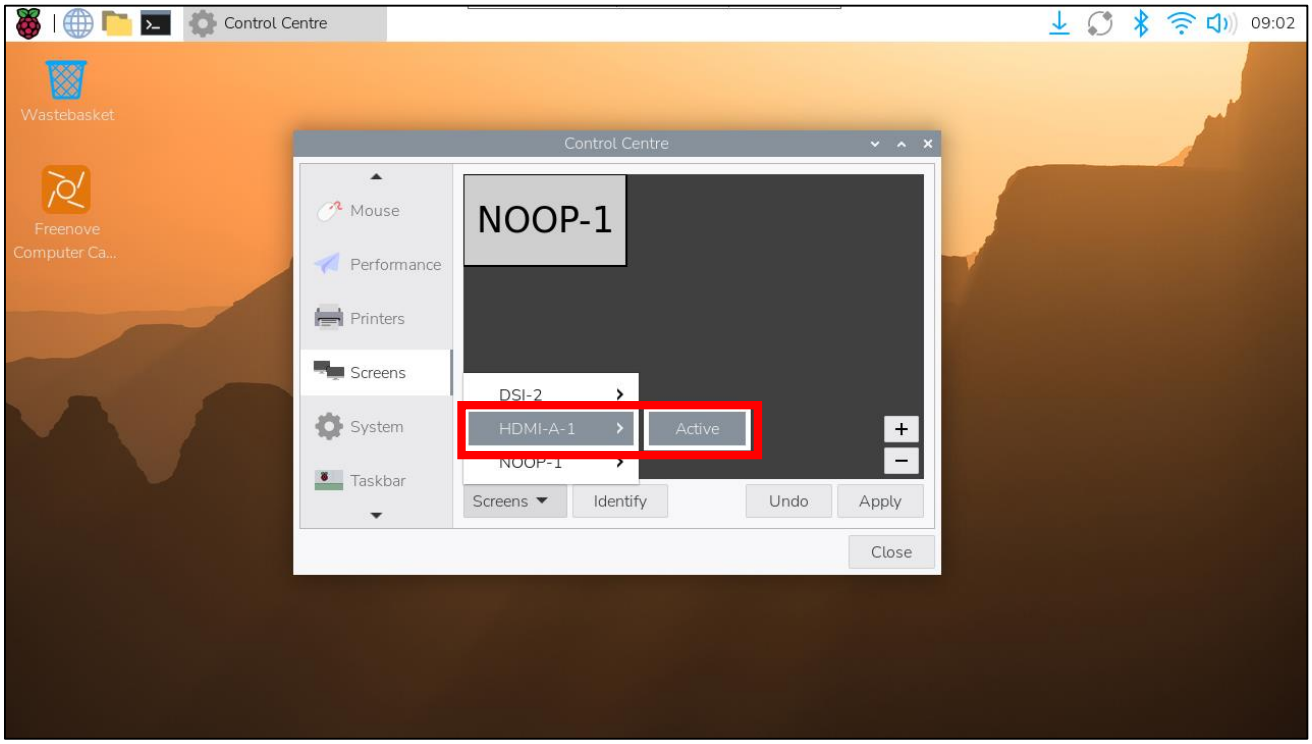


If you have enabled Advanced Options -> A7 Wayland -> W1 X11 in **raspi-config** and experience abnormal audio output, please re-enable HDMI to restore normal audio function.

Right-click in any blank area upon the desktop and choose "**Desktop Preferences...**".



In the pop-up Control Centre window, select "**Screens**" from the left pane. Select Screens → HDMI-A-1 → Active to reactivate HDMI。

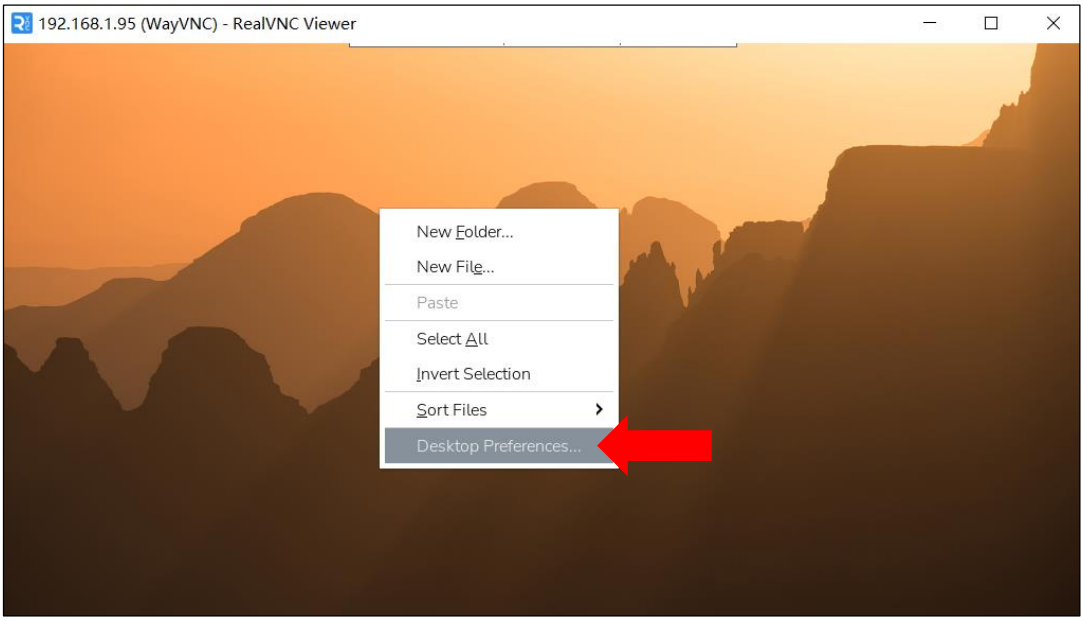


3.1.2 Switching Taskbar Display

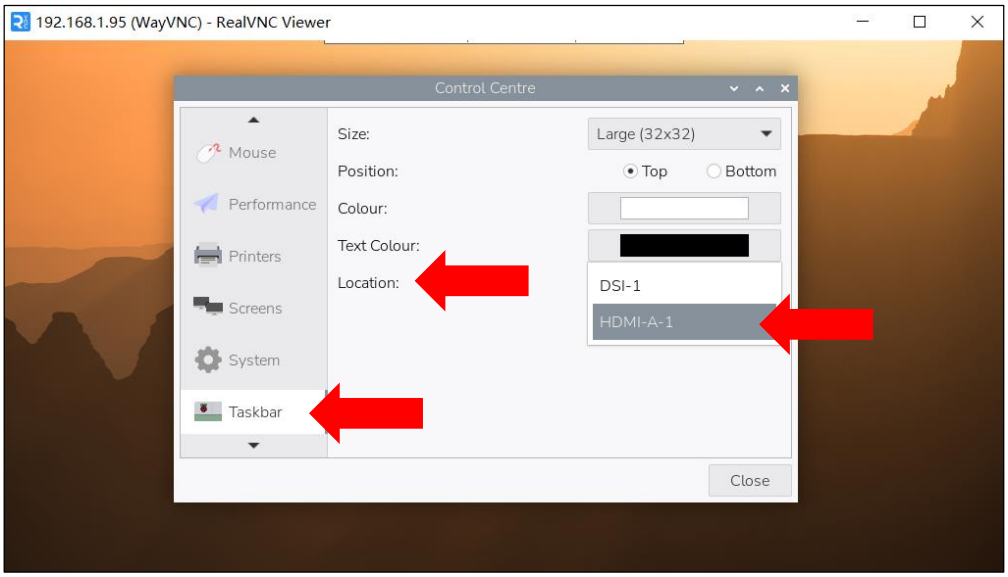
If your kit is a non-display verion (**Model numbers: FNK0100A/B**), you can skip this section by clicking here.

For versions equipped with a 4.3-inch IPS screen (**Model numbers: FNK0100H/K**), after booting the RPi 5, the system will create two independent display areas: DSI-1 and HDMI-A-1. When connecting via VNC, you are viewing a mirrored display of HDMI-A-1. Since the system status bar is prioritized for display on the DSI screen, it is not included by default in the VNC interface. Please follow the instructions below to display the status bar in the HDMI-A-1 area.

Right-click in any blank area upon the desktop and choose "**Desktop Preferences...**".



In the pop-up Control Centre window, select "**Taskbar**" from the left pane. Find the "**Location**" setting and change it from DSI-1 to HDMI-A-1.



3.1.3 Software Setup

Software Packages Update

Run the following command on the terminal to update your Raspberry Pi's package list to the latest version. In the latest Raspberry Pi 13 (Trixie), passwordless sudo is officially disabled. You will need to enter your user password every time you run a sudo command. Please enter your user password when prompted. (Note: The default password is "**raspberry**", please replace it with your user password.)

```
sudo apt update
```

```
pi@raspberrypi:~ $ sudo apt update
[sudo] password for pi:
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://archive.raspberrypi.com/debian trixie InRelease
Hit:4 http://deb.debian.org/debian-security trixie-security InRelease
127 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Code downloading

Open the Raspberry Pi Terminal, type the following two commands one by one to download the code for the case.

```
cd
git clone
https://github.com/Freenove/Freenove_Computer_Case_Kit_for_Raspberry_Pi.git
```

```
pi@raspberrypi:~ $ git clone https://github.com/Freenove/Freenove_Computer_Case_
Kit_for_Raspberry_Pi.git
Cloning into 'Freenove_Computer_Case_Kit_for_Raspberry_Pi'...
```

OLED Library Installation (Important)

Run the following command to install the OLED library. Fail to do so will result in software malfunction.

```
sudo apt install python3-luma.oled
```

```
pi@raspberrypi:~ $ sudo apt install python3-luma.oled
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  python3-luma.oled
0 upgraded, 1 newly installed, 0 to remove and 322 not upgraded.
Need to get 29.5 kB of archives.
After this operation, 168 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main arm64 python3-luma.oled arm64 3.10.0-1 [29.5 kB]
Fetched 29.5 kB in 0s (120 kB/s)
Selecting previously unselected package python3-luma.oled.
(Reading database ... 134691 files and directories currently installed.)
Preparing to unpack .../python3-luma.oled_3.10.0-1_arm64.deb ...
Unpacking python3-luma.oled (3.10.0-1) ...
Setting up python3-luma.oled (3.10.0-1) ...
```

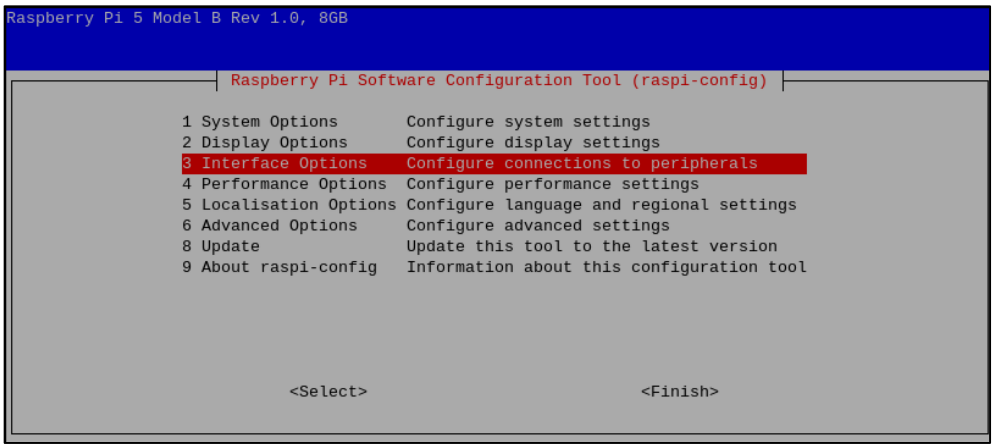
Enable I2C (Important)

The I2C function must be enabled, as it is required for communication between the Raspberry Pi, the OLED display, and the GPIO adapter board. Without it, the chassis software will fail to operate.

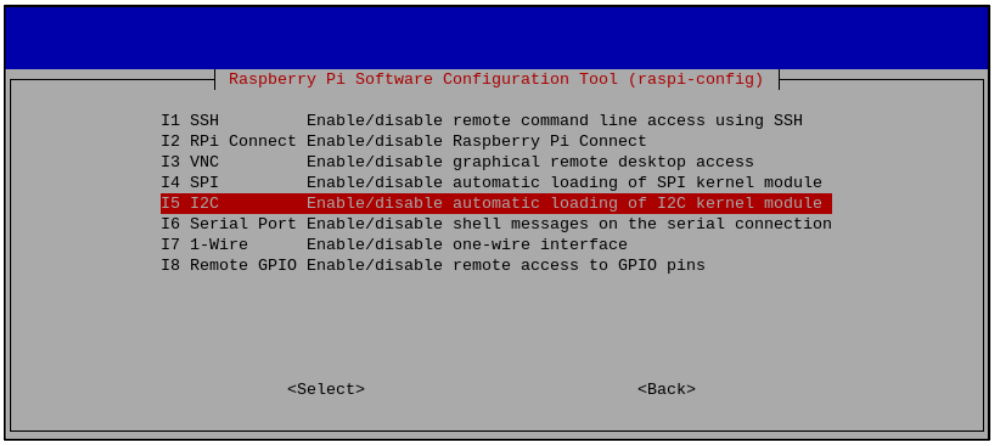
Run the following command to open the Raspberry Pi configuration interface.

```
sudo raspi-config
```

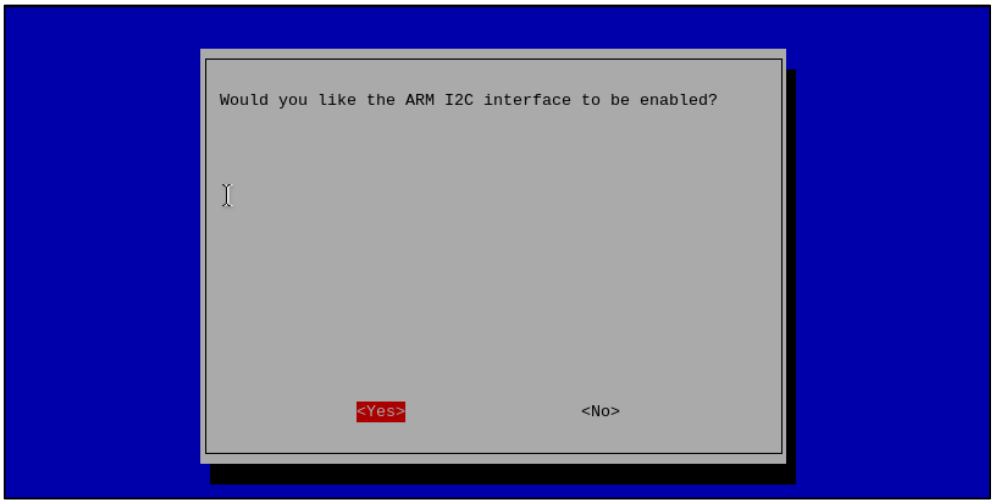
Navigate to Interface Options and press Enter.



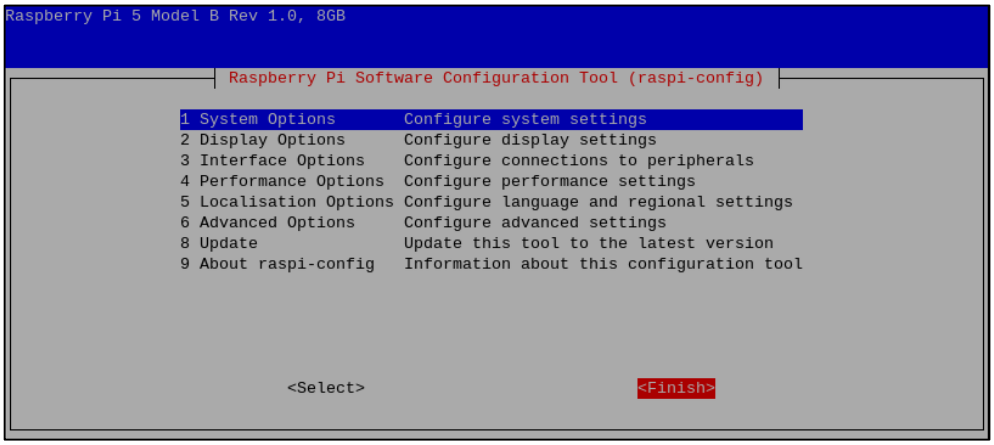
Select I2C and press Enter to enable it.



Select Yes, and press Enter.



Select Finish to save the change and exit the configuration inferface.



Creating Desktop Shortcut

Run the following two commands one by one in the Raspberry Pi terminal to create a desktop shortcut for the case control software.

```
cd Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code/
sudo python create_desktop_shortcut.py
```

```
pi@raspberrypi:~ $ cd Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code/
pi@raspberrypi:~/Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code $ sudo python
create_desktop_shortcut.py
Current directory: /home/pi/Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code
Deleted existing desktop file: /home/pi/Desktop/Freenove.desktop
Created new desktop file directly on desktop
Desktop file and script set as executable
Desktop shortcut created successfully!
```

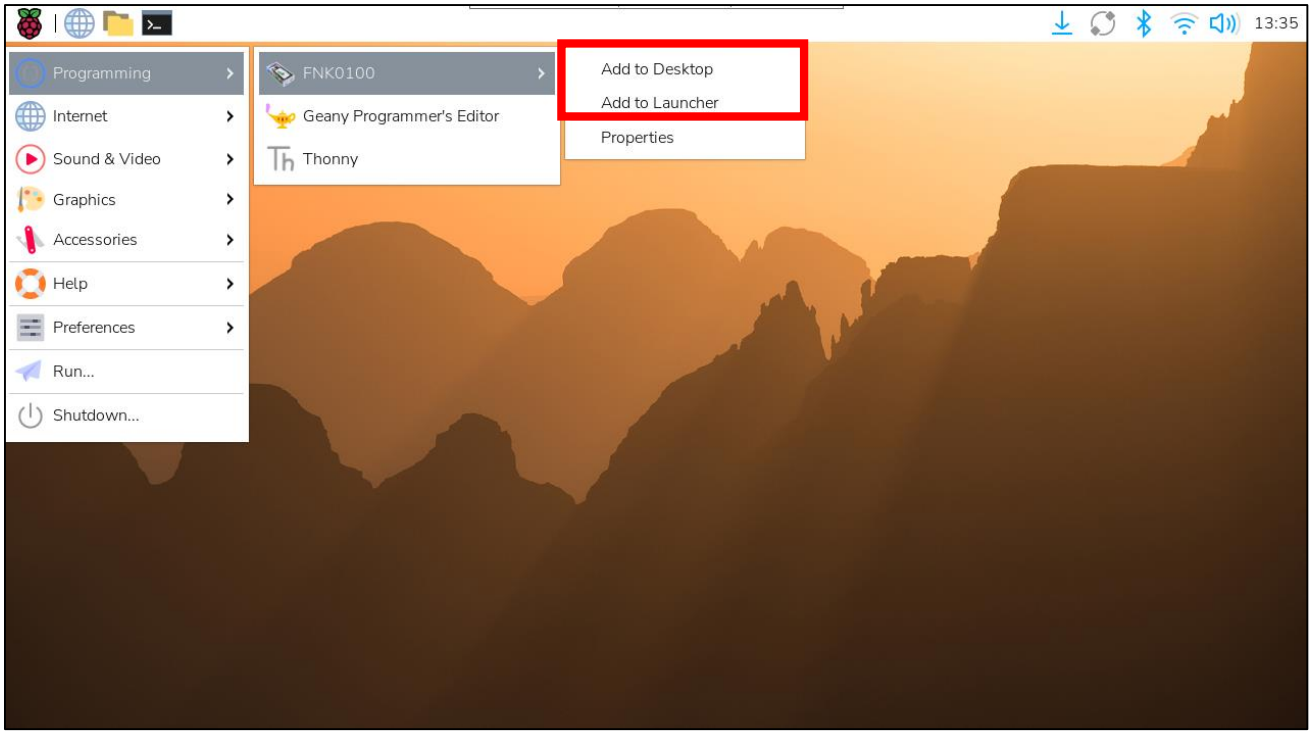


The icon above corresponds to the command shown below.

```
cd Freenove_Computer_Case_Kit_for_Raspberry_Pi
python app_ui.py
```

```
pi@raspberrypi:~ $ cd Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code/
pi@raspberrypi:~/Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code $ python app_ui.py
```

After completing this step, you will find the application in the "**Programming**" section of the menu. If you accidentally delete the desktop shortcut or wish to add the app to the top launcher, simply right-click "**FNK0100**" and select "**Add to Desktop**"or "**Add to Launcher**".

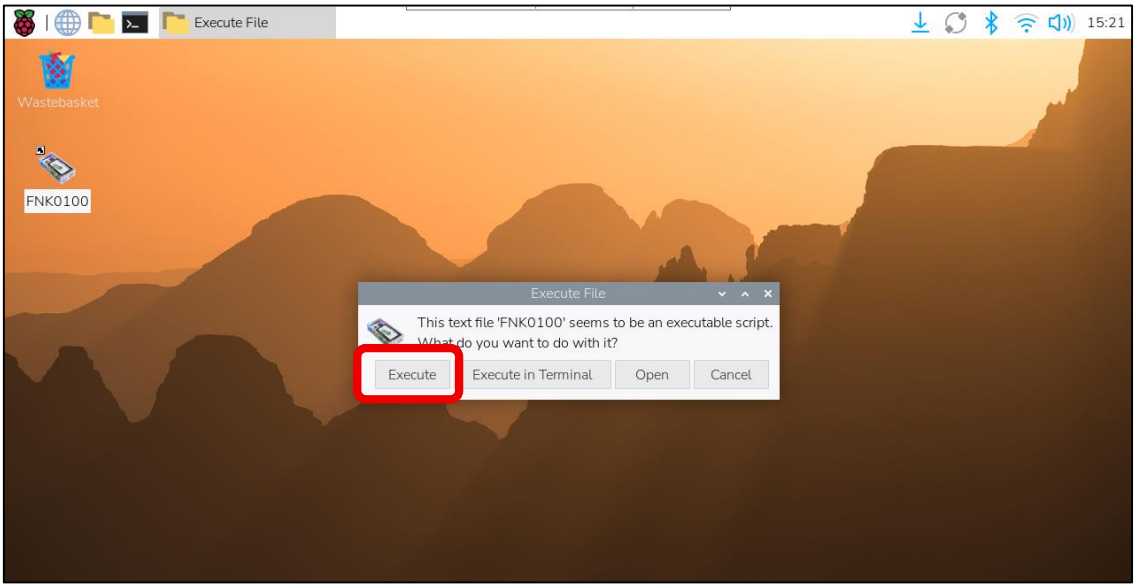


If you are interested in the code implementation, you can explore the files in the Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code directory.
Should you wish to modify the code, please ensure you back it up first to avoid potential software malfunctions caused by unintended changes.

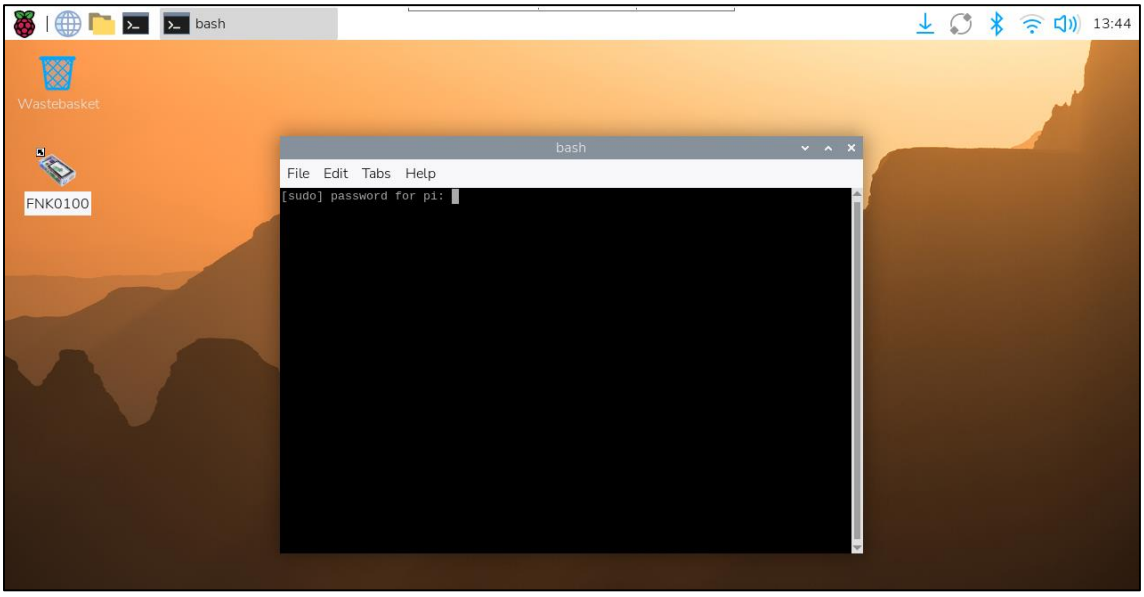
3.2 About the Case Control Software

With the environment configured from the previous chapter, the accompanying host software can now be used to manage case functions including ARGB lighting, the OLED display, and fan control.
This chapter provides a detailed guide on the software's usage. For insight into the interface design, the app_ui.py file is available in the Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code directory.

Double click the software with Freenove logo on RPi's desktop, and a window will pop up. Click "**Execute**" to run the program.



Since passwordless sudo is disabled in Trixie, you will also be asked for your password again after reboot. Enter your user password in the pop-up terminal window.

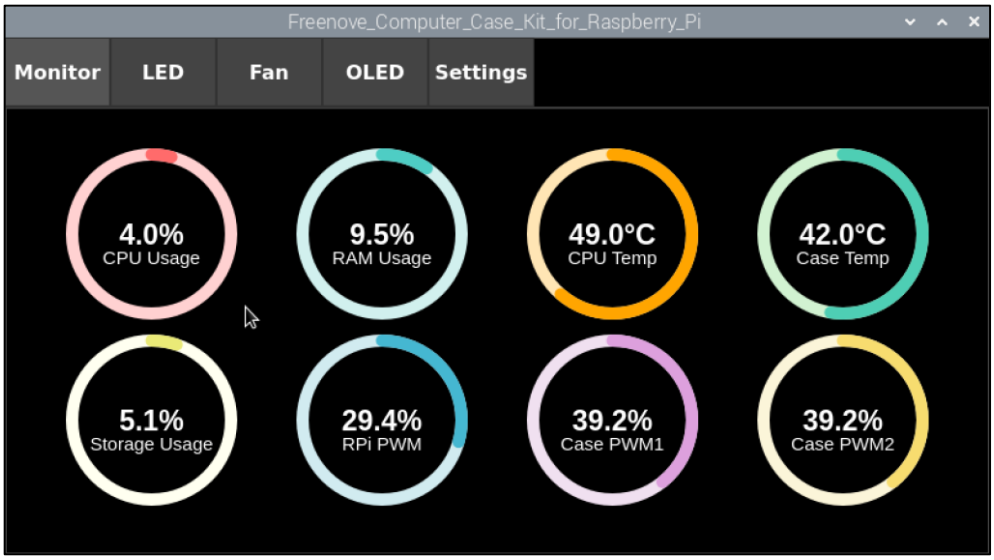


The software interface is shown below. The left window is the main software interface, and the right window displays debugging information.



3.2.1 Dashboard Monitoring

The dashboard provides live monitoring of key RPi 5 and case component stats, giving you an at-a-glance view of the system status.



Below is a detailed explanation of the parameters displayed on the monitoring interface:

- **CPU Usage:** Current processor utilization percentage of the Raspberry Pi 5
- **RAM Usage:** Current system memory utilization percentage of the Raspberry Pi 5
- **CPU TEMP:** The internal temperature of the Raspberry Pi 5's SoC (System on Chip)
- **Case Temp:** The temperature inside the chassis, provided by the temperature sensor on the GPIO Board
- **Storage Usage:** The total storage space utilization. This value reflects the usage of the SD card if one is used. If single or multiple SSDs are installed, it calculates the aggregate usage across all drives.
- **RPi PWM:** PWM duty cycle for the Raspberry Pi's active cooler fan.
- **Case PWM1:** PWM duty cycle for case fan 1.
- **Case PWM2:** PWM duty cycle for case fan 2.

Data Source & Troubleshooting:

CPU Usage, RAM Usage, CPU Temp, Storage Usage, and RPi PWM are retrieved directly from the Raspberry Pi. If any of these values are missing, check the Raspberry Pi for faults.

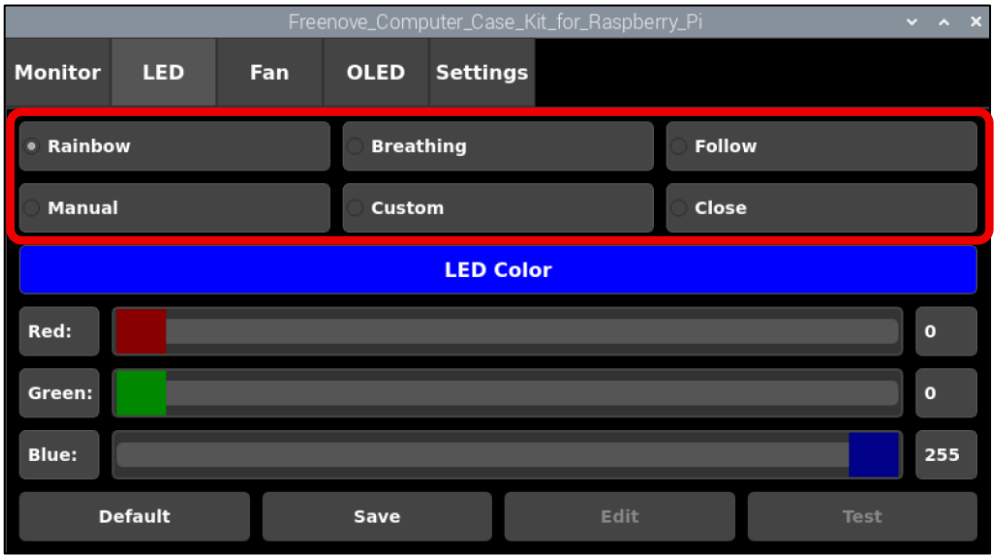
Case Temp, Case PWM1, and Case PWM2 are read from the GPIO Board via I2C communication with Raspberry Pi 5.

If these data cannot be obtained consistently, please contact us by email: support@freenove.com

For those interested in the interface implementation, please refer to the files api_systemInfo.py and api_expansion.py in the Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code directory.

3.2.2 LED Control Interface

This is the control interface for the case ARGB lights. You can select different modes to display various lighting effects.

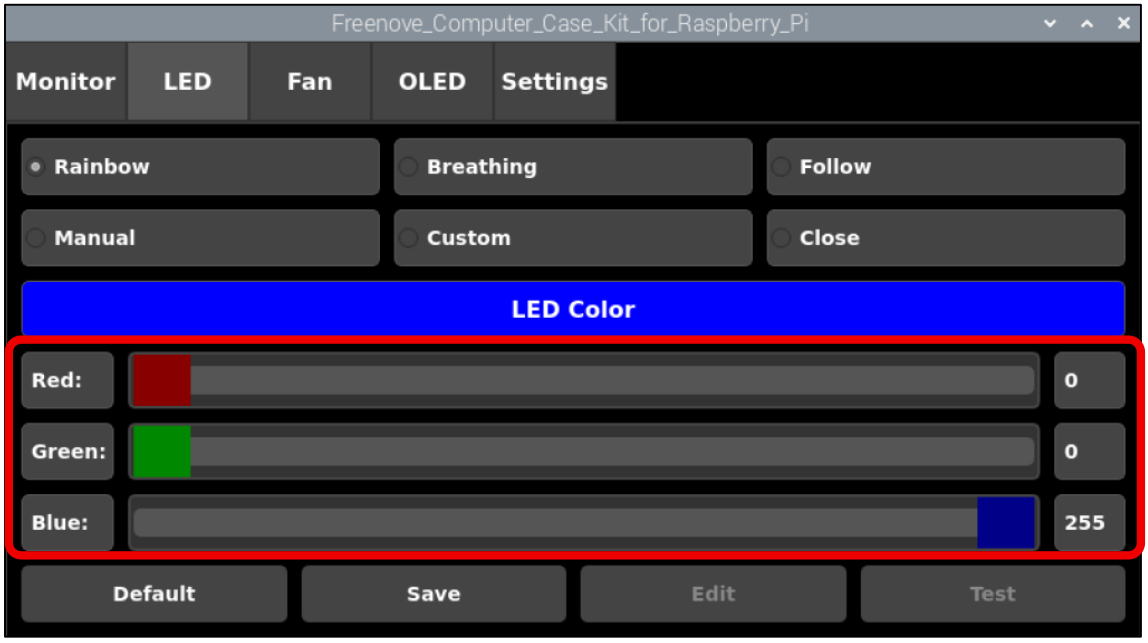


It is set to rainbow mode by default, as shown below.



There are four preset lighting modes available: **Rainbow**、 **Breathing**、 **Follow**、 **Manual**. You can select the corresponding option to switch among the modes.

Note: Only in **Breathing**, **Follow**, and **Manual** modes can the color of the RGB lights be controlled using the slider below. In other modes, the color of the RGB lights cannot be adjusted.



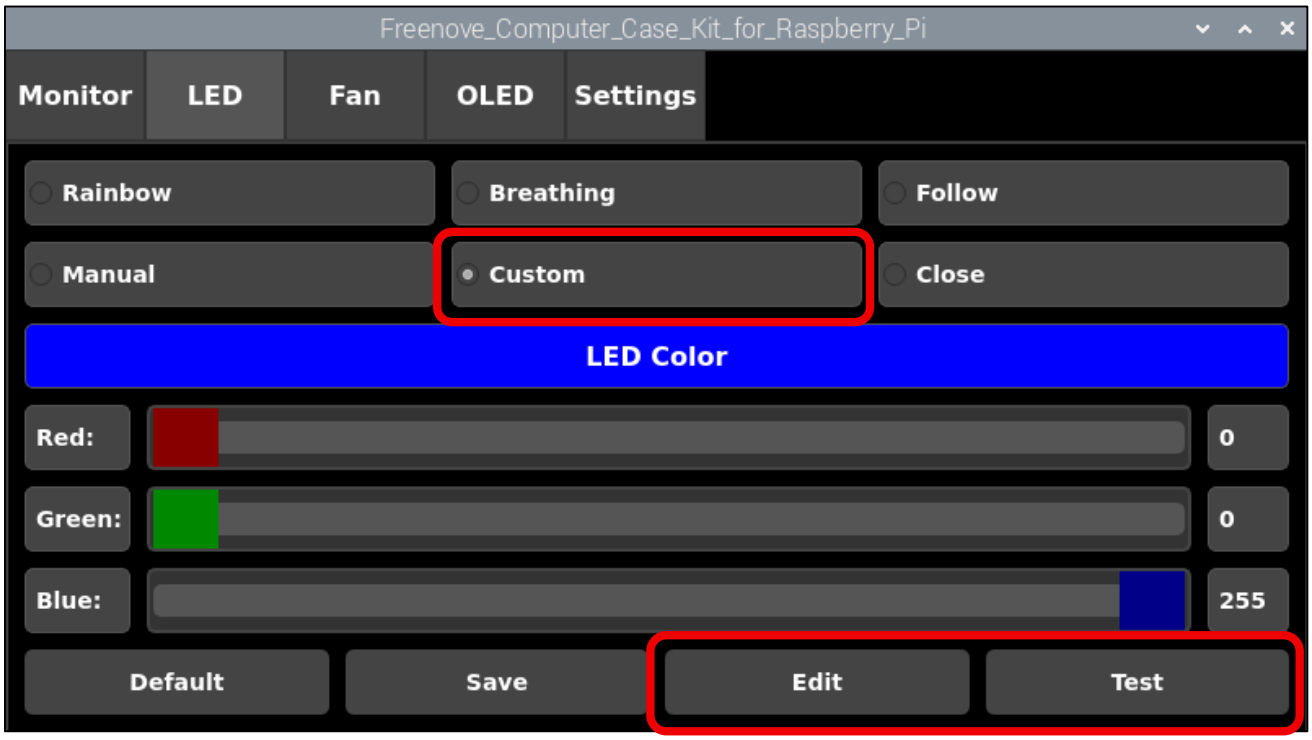
If you do not turn ON the RGB lights, you can select the Close mode to turn off them.

Please note: Any lighting mode you select is temporary. The case will revert to its default mode after a shutdown and restart.
To set a new default mode, select your desired mode and click the "**Save**" button.



To restore the RGB lights and sliders to their default settings, click the "**Default**" button once.

If the four preset lighting modes do not meet your needs, you can use the "Custom" mode for personalized settings.



Click "**Edit**" will open the "task_led.py" file. You can modify the code in the editor.

Click "**Test**" will create a temporary thread and run task_led.py, allowing you to quickly debug and observe the LED light effects.

You can also view the source code directly:

Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code/task_led.py

For more about LED control interface, you can refer to the source file:

Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code/api_expansion.py

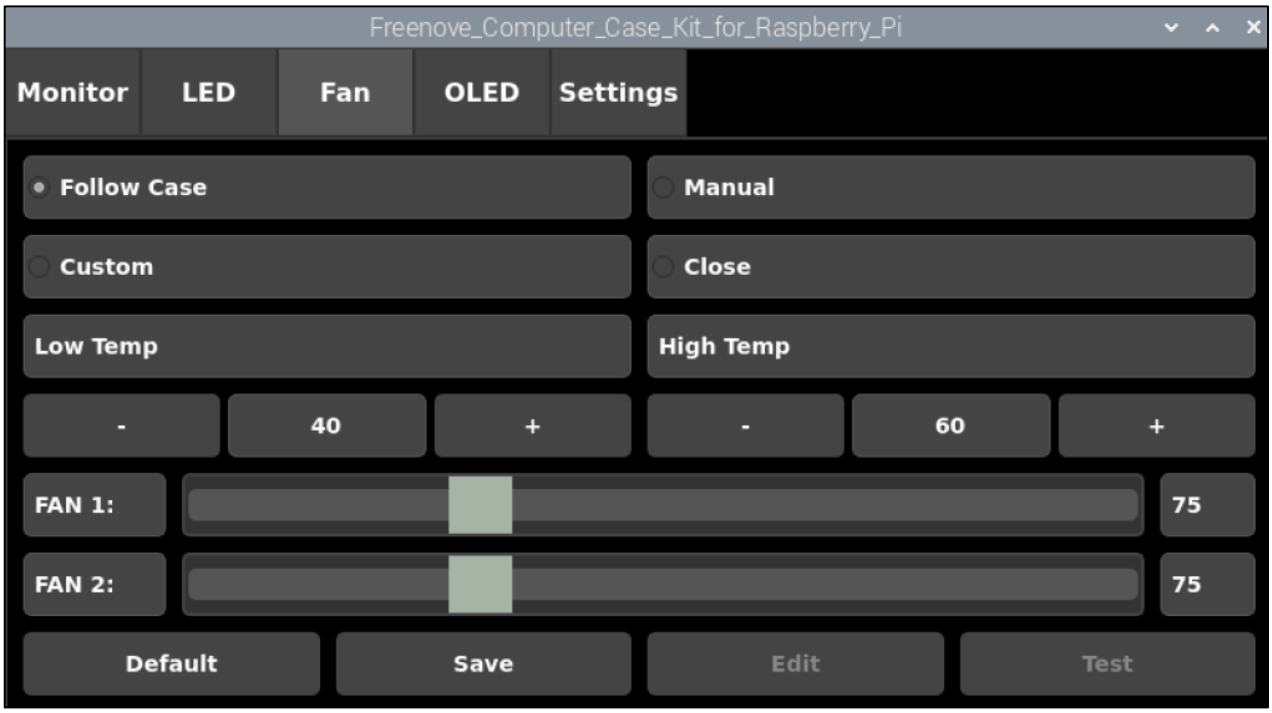
Important Notes:

1. Back up the source code before making any modifications.
2. Custom mode is an advanced feature. Modifying the code in task_led.py requires a basic understanding of Python programming. If you are not familiar with programming, we do not recommend editing this file.
3. The "Test" button only runs the code in the current state. The code will not automatically run after rebooting the Raspberry Pi.
4. If the LED lights show no response after you modify the code and click "Test," it indicates an error in your code preventing it from running properly. Please review your code.
5. If you want the Custom mode code to execute automatically on every startup, please refer to the instructions in the "3.2.5 Settings Interface" Section.

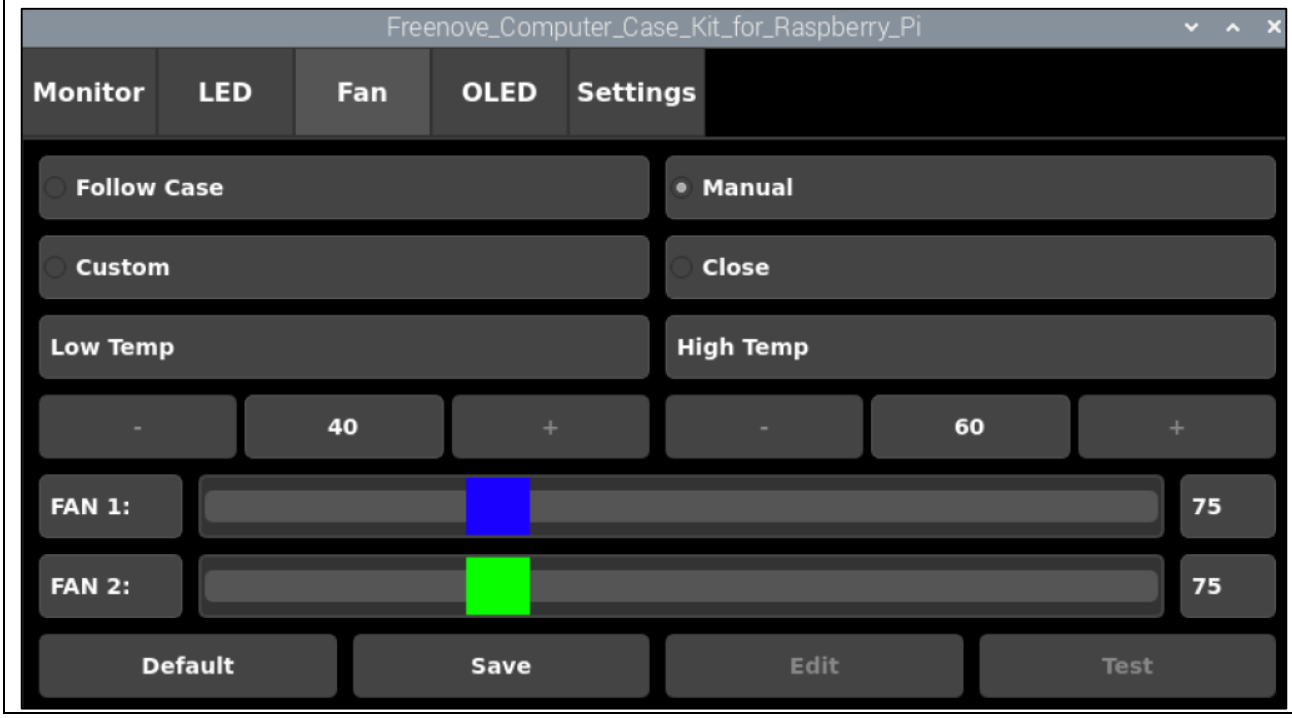
3.2.3 Fan Control Interface

This is the case fan control interface for convenient fan management. The interface will change when you select different modes: Follow Case or Manual mode.

Follow Case Mode

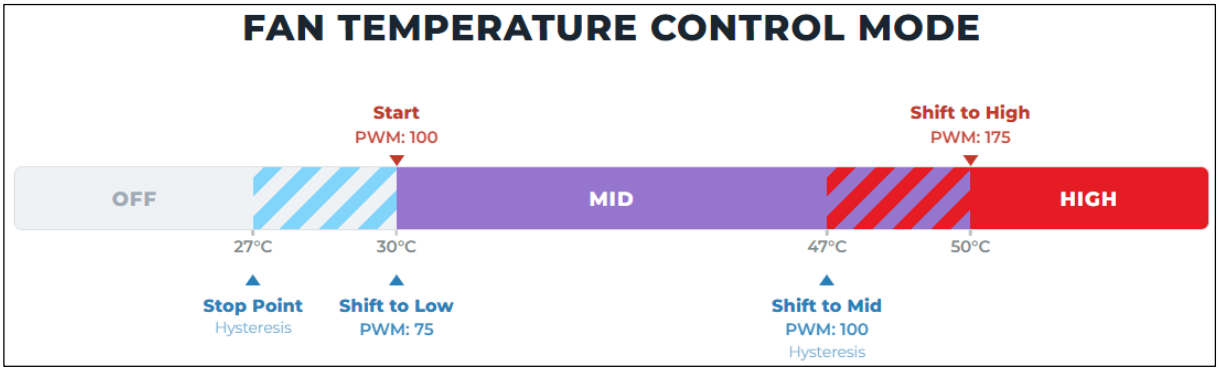


Manual Mode



Follow Case

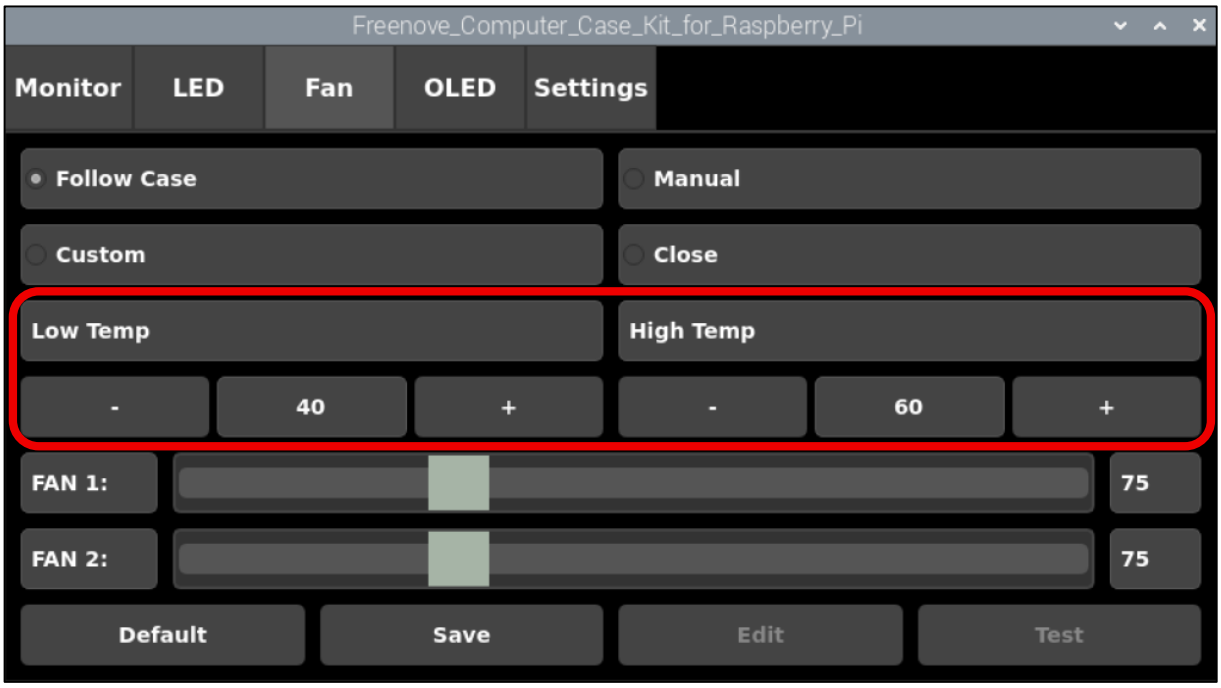
In Follow Case mode, the GPIO Adapter board samples the temperature sensor at regular intervals and regulates the fan PWM signal based on the readings.
The specific control logic governing this process is depicted in the following figure.



The diagram uses various colors to represent different temperature zones. All values shown are configurable defaults.

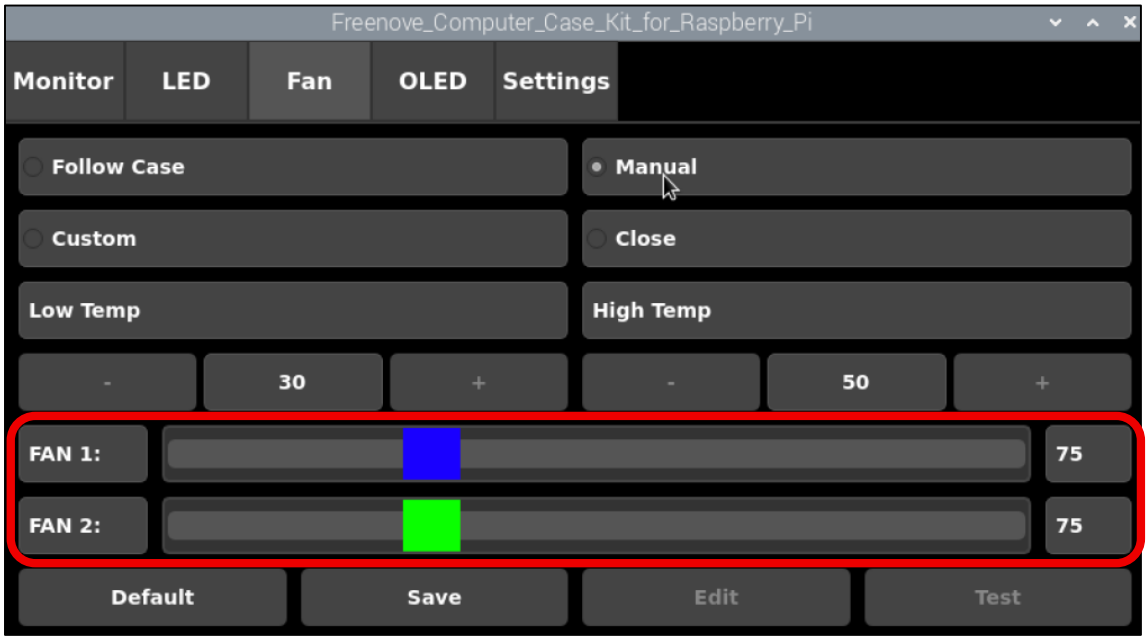
- **Heating Phase:**
  - When temperature ⩾ **Low Temp** (default: 30°C), fan PWM set value: 100.
  - When temperature ⩾ **High Temp** (default: 50°C), fan PWM set value: 175.
- **Cooling Phase:**
  - When temperature < **High Temp** (default: 50°C), fan PWM set value: 100.
  - **Low Temp - Schmitt** (default: 30 - 3 = 27°C) ⩽ When temperature < **Low Temp** (default: 30°C), fan PWM set value: 75.
  - When temperature < **Low Temp- Schmitt** (default: 30 - 3 = 27°C), the fan stops.

You can configure the following parameters via the software.



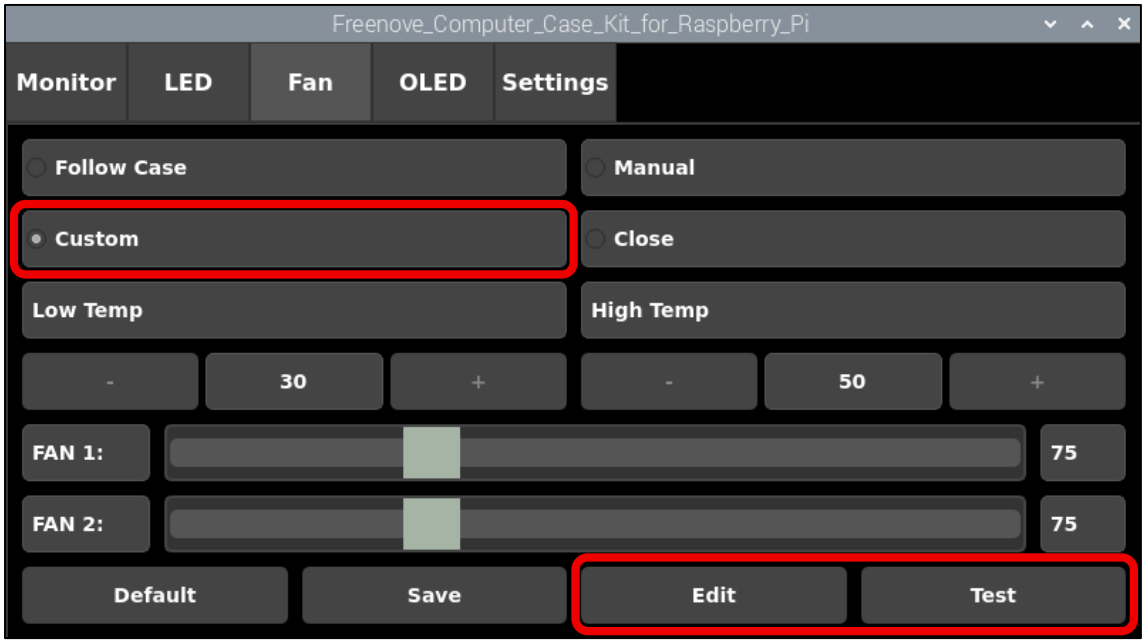
Manual

In Manual mode, you can adjust FAN1 and FAN2 PWM values via the respective sliders in manual mode.



Custom

If the predefined modes do not meet your needs, you can select the **Custom** mode for advanced configuration.



Click "**Edit**" will open the "task_fan.py" file. You can modify the code in the editor.
Click "**Test**" will create a temporary thread and run task_fan.py, allowing you to quickly debug and observe the fan effects.
You can also view the source code directly:
Freenove_Computer_Case_Kit_for_Raspberry_Pi/Code/task_fan.py

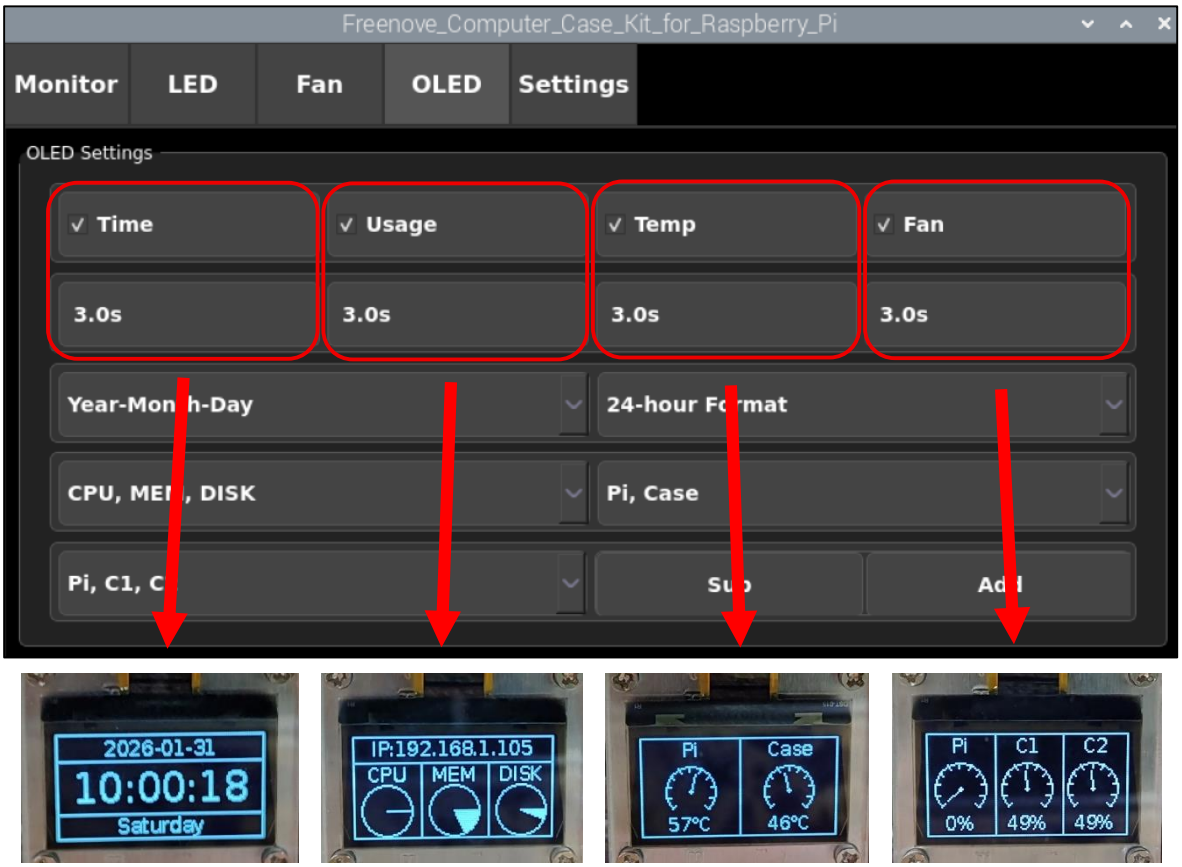
Important Notes:

1. Back up the source code before making any modifications.
2. Custom mode is an advanced feature. Modifying the code in task_fan.py requires a basic understanding of Python programming. If you are not familiar with programming, we do not recommend editing this file.
3. The “Test” button only runs the code in the current state. The code will not automatically run after rebooting the Raspberry Pi.
4. If the fans do no response after you modify the code and click “Test,” it indicates an error in your code preventing it from running properly. Please review your code.
5. If you want the Custom mode code to execute automatically on every startup, please refer to the instructions in the “3.2.5 Settings Interface” Section.

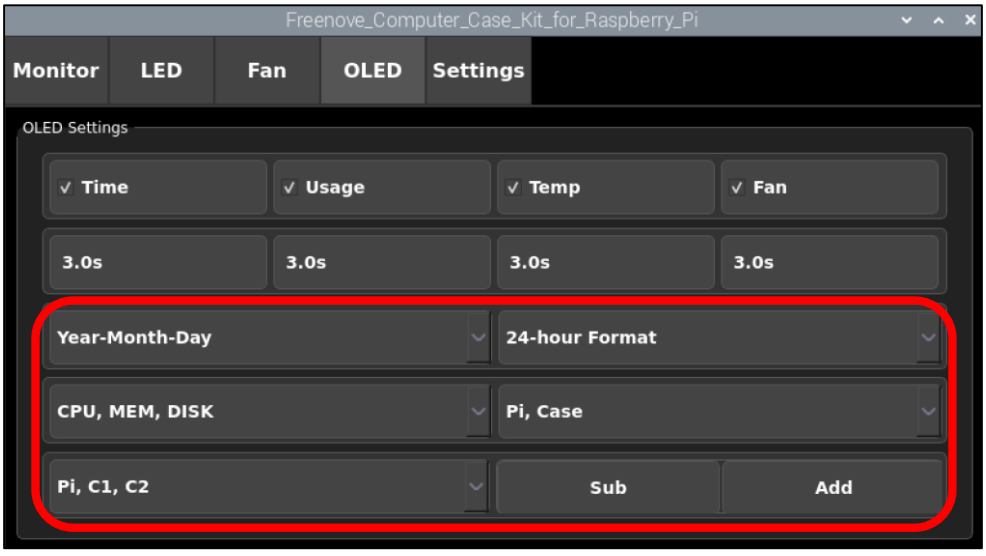
3.2.4 OLED Control Interface

This is the interface for controlling the OLED display content. You can choose from different display layouts according to your preference.

By default, the OLED cycles through four screens: Time, Usage, Temp, and Fan. If you uncheck any of these, that screen will be skipped during the cycle.

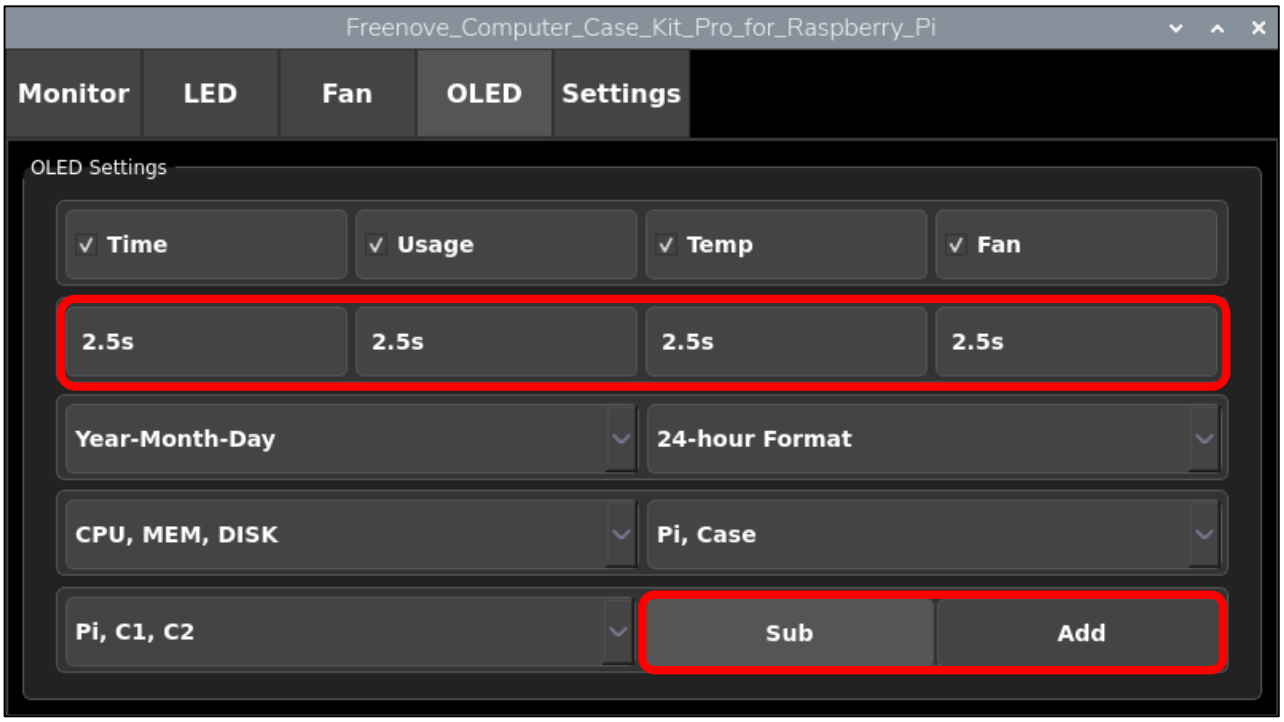


Below are six customizable options. Click the drop-down lists to select different time formats, data display orders, display durations, and more.

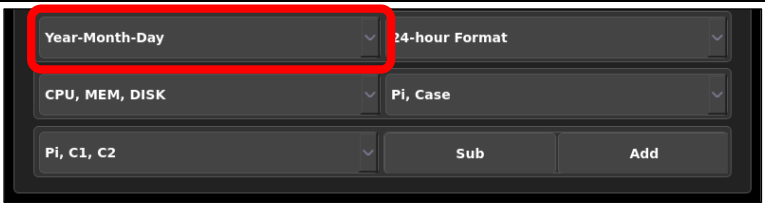
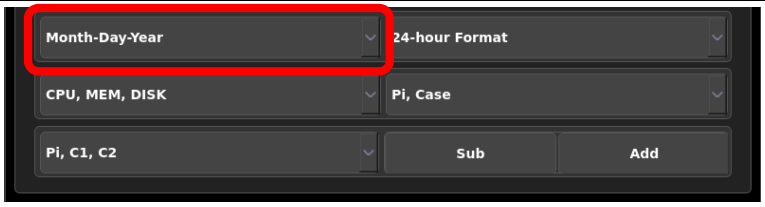
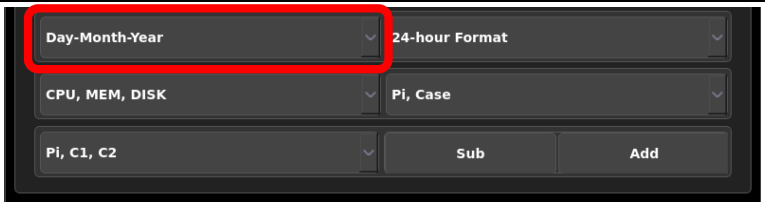


Adjusting Display Durations

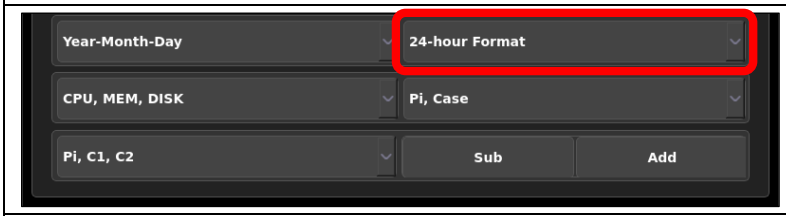
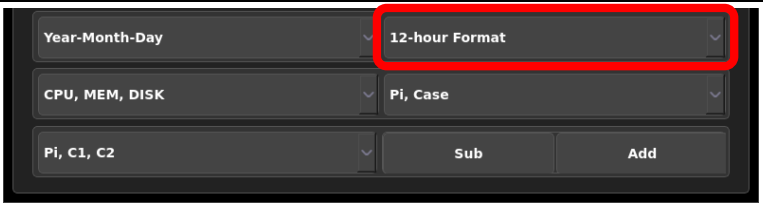
The duration for each display screen can be adjusted by clicking the "**Sub**" button to decrease by 0.5s or the "**Add**" button to increase by 0.5s.
To adjust the display time for a single interface, simply uncheck the other screens so that only the desired one remains active.



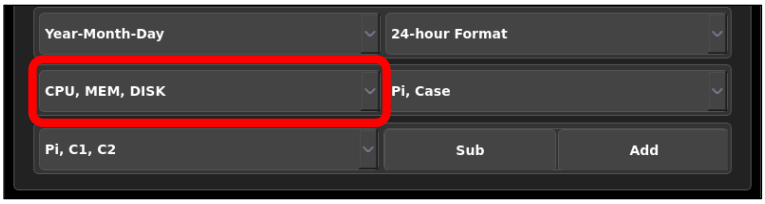
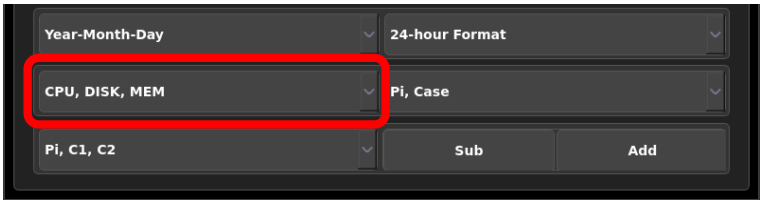
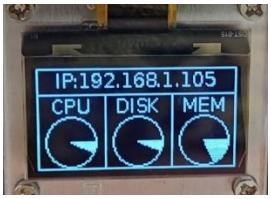
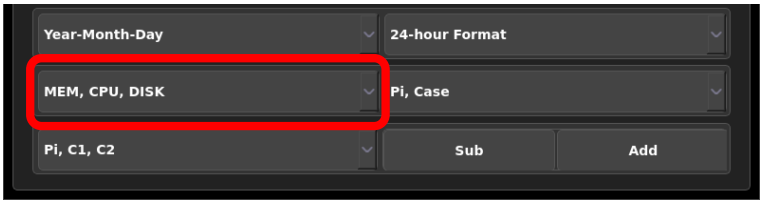
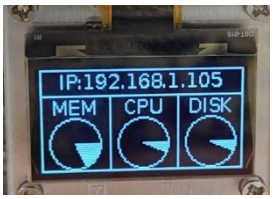
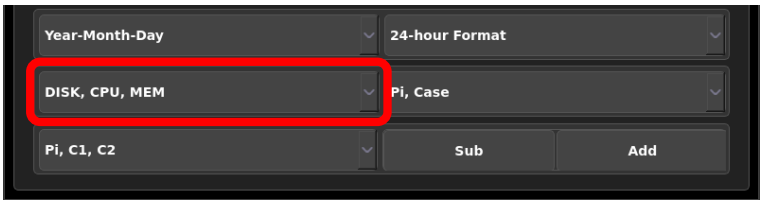
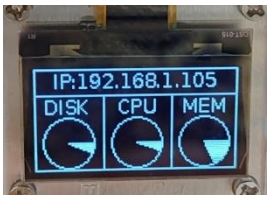
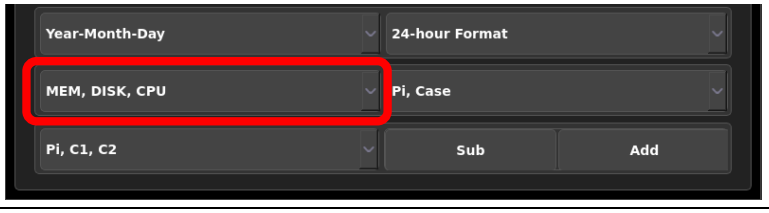
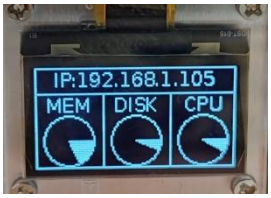
Modifying Date Format

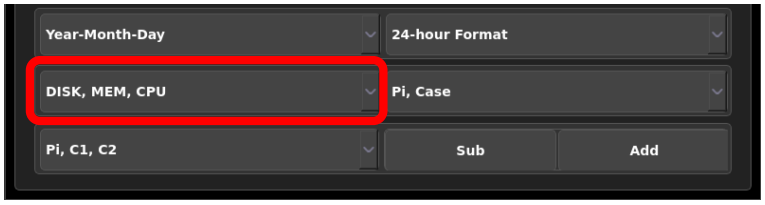
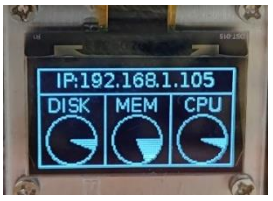
Options	OLED Display
	
	
	

Changing Time Format

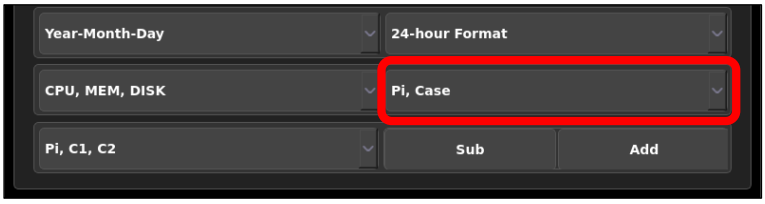
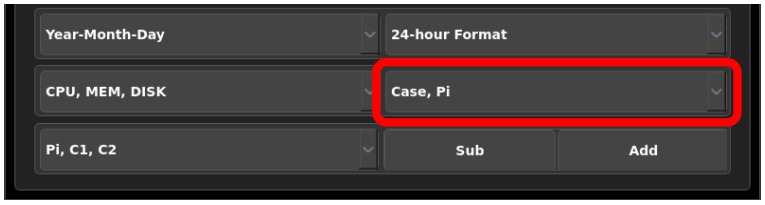
Options	OLED Display
	
	

Modifying the Display Order of CPU, MEM, and DISK Usage

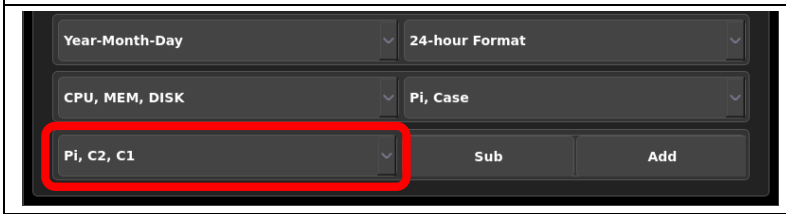
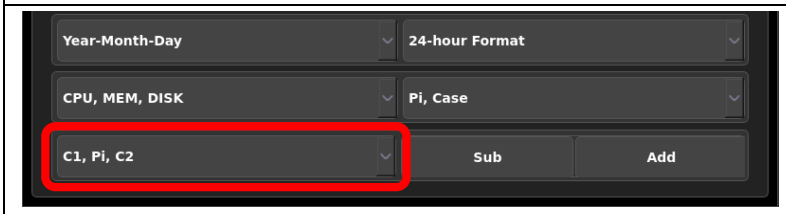
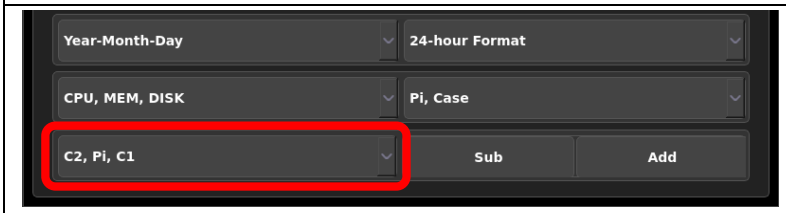
Options	OLED Display
	
	
	
	
	

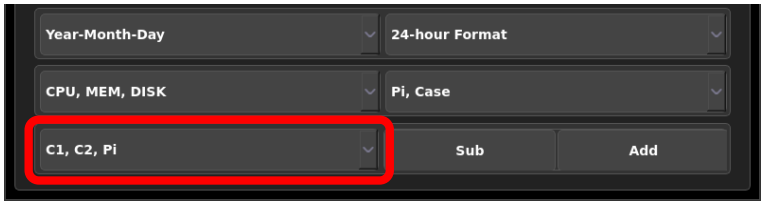
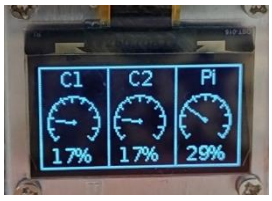
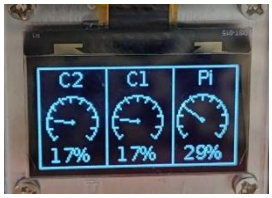
	

Changing the Display Order of Pi and Case Temperature

Options	OLED Display
	
	

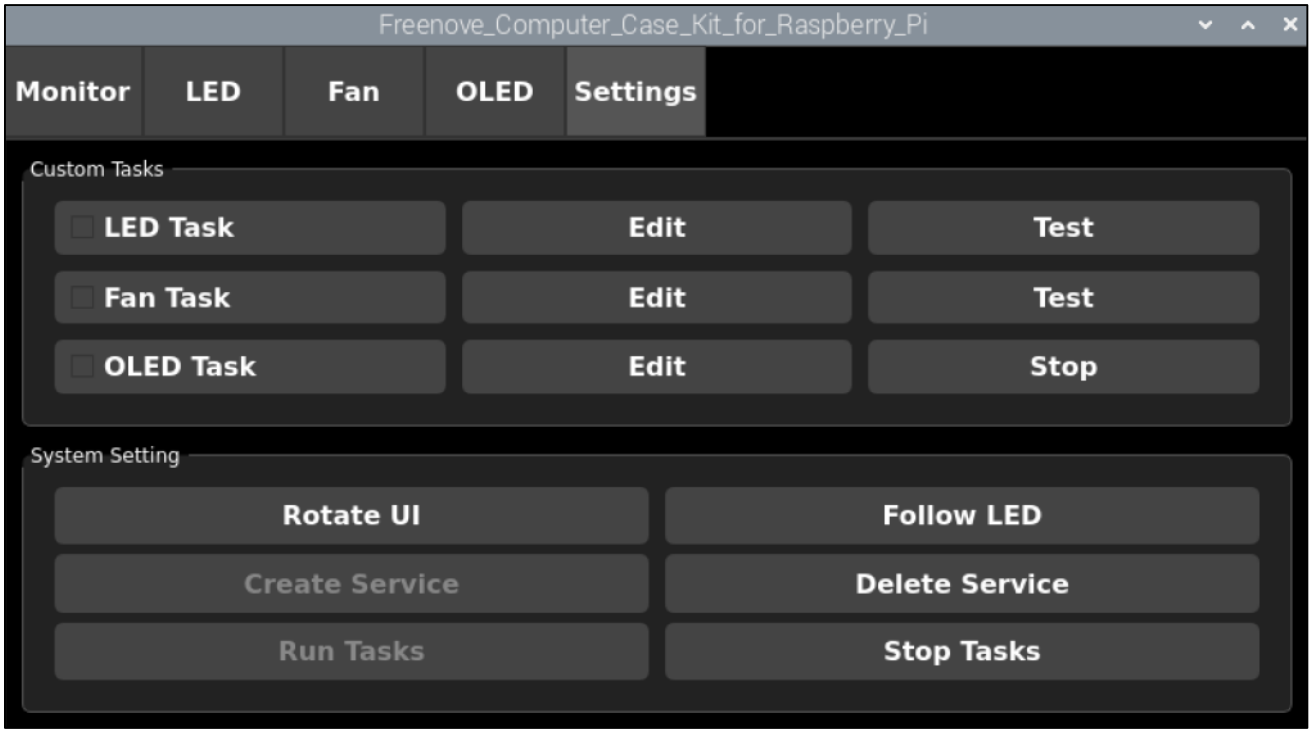
Adjusting the Display Order of Pi, C1, and C2 Fans PWM

Options	OLED Display
	
	
	
	

3.2.5 Settings Interface

This interface integrates advanced customization features, allowing you to edit code, debug, and create startup tasks for the case LED lights, fans, and OLED display.



Disable Terminal Display & Disable sudo Password

If you do not need to view the debugging information and want to close the command terminal window, follow the steps below.

Open the terminal on your Raspberry Pi.

Run the following command to set the correct permissions for the Freenove desktop software:

```
ls -al ~/Desktop/Freenove.desktop
sudo chmod 777 ~/Desktop/Freenove.desktop
```

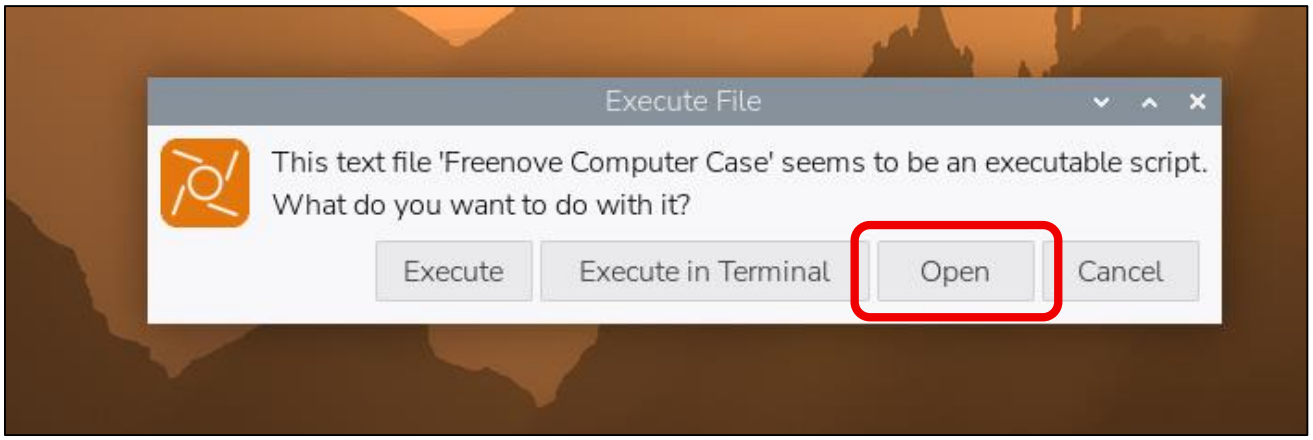
```
pi@raspberrypi:~ $ ls -al ~/Desktop/Freenove.desktop
-rwxr-xr-x 1 root root 340 Mar 10 10:09 /home/pi/Desktop/Freenove.desktop
pi@raspberrypi:~ $ sudo chmod 777 ~/Desktop/Freenove.desktop
pi@raspberrypi:~ $ ls -al ~/Desktop/Freenove.desktop
-rwxrwxrwx 1 root root 340 Mar 10 10:09 /home/pi/Desktop/Freenove.desktop
pi@raspberrypi:~ $
```

To revert to the default permissions, run the following command:

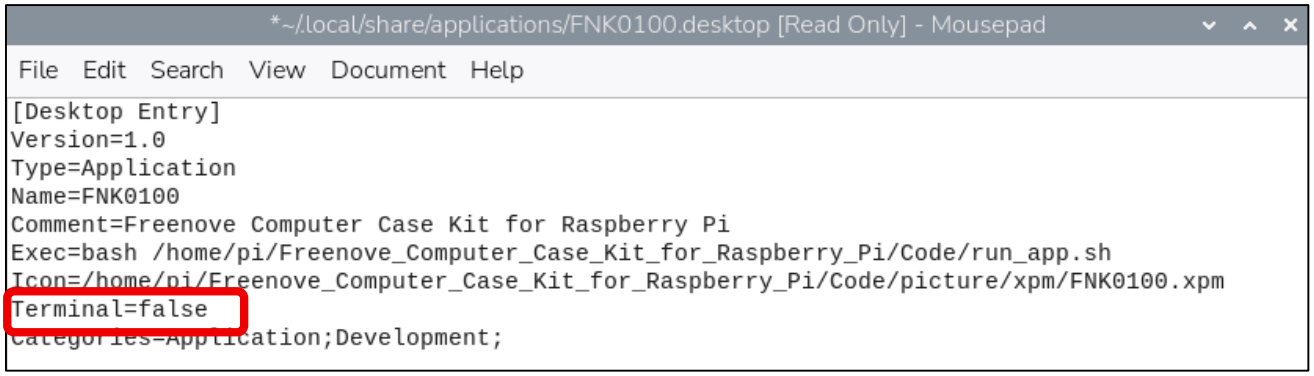
```
ls -al ~/Desktop/Freenove.desktop
sudo chmod 755 ~/Desktop/Freenove.desktop
```

```
pi@raspberrypi:~ $ ls -al ~/Desktop/Freenove.desktop
-rwxrwxrwx 1 root root 340 Mar 10 10:09 /home/pi/Desktop/Freenove.desktop
pi@raspberrypi:~ $ sudo chmod 755 ~/Desktop/Freenove.desktop
pi@raspberrypi:~ $ ls -al ~/Desktop/Freenove.desktop
-rwxr-xr-x 1 root root 340 Mar 10 10:09 /home/pi/Desktop/Freenove.desktop
pi@raspberrypi:~ $
```

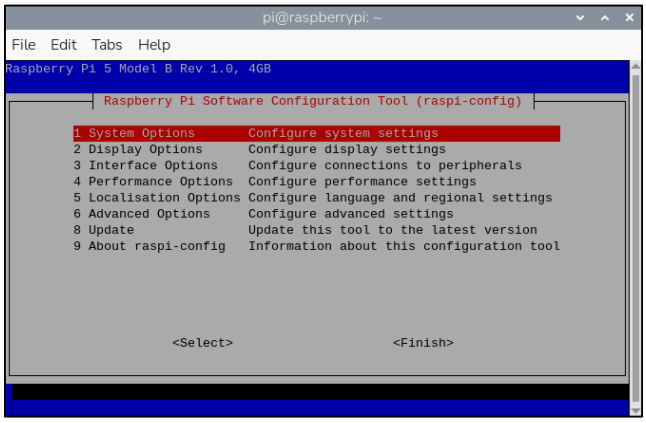
1. Close the current upper-computer software completely.
2. Double-click the Freenove software on the desktop to restart the application.
3. In the pop-up window, click **Open** to launch the software.

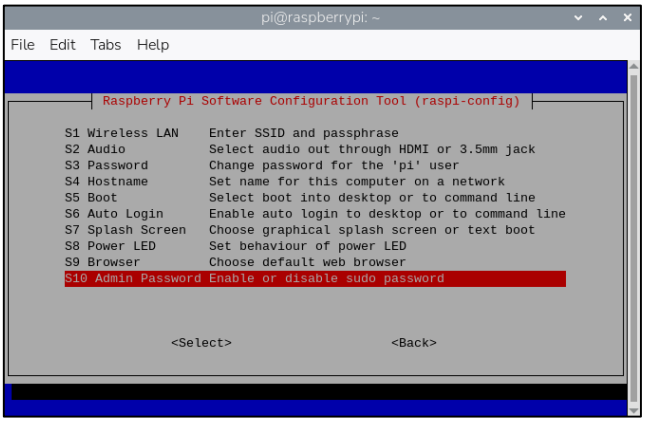


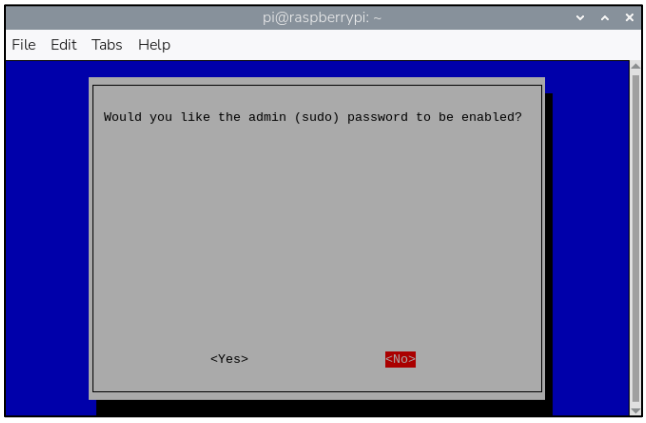
Change the parameter of **Terminal** from true to **false**, save and clost the file. Restart the software – the command terminal window will no longer open.

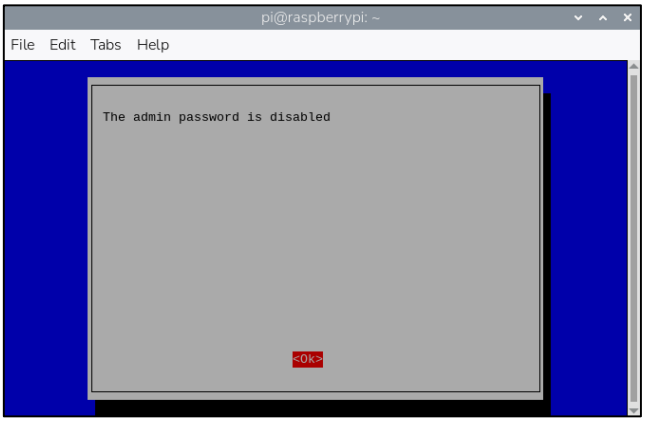


However, with the command terminal closed, you will have nowhere to enter your password when sudo is needed (because passwordless sudo is disabled by default in Trixie). To allow the software to run properly, you must enable passwordless sudo. To do this, open a terminal and run "**sudo raspi-config**", then select: "1 System Options" → "S10 Admin Password" → "No" → "OK".



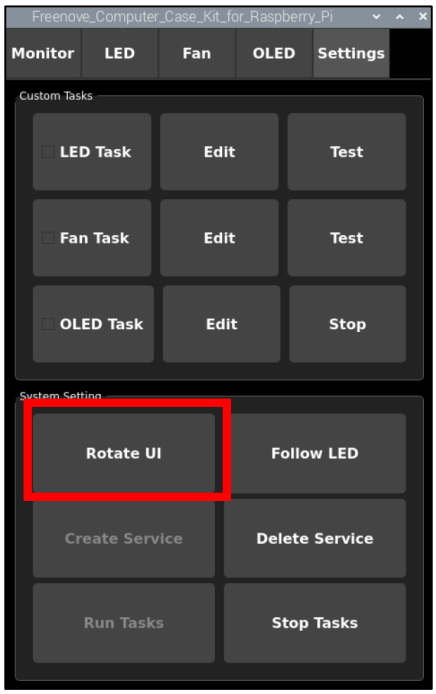


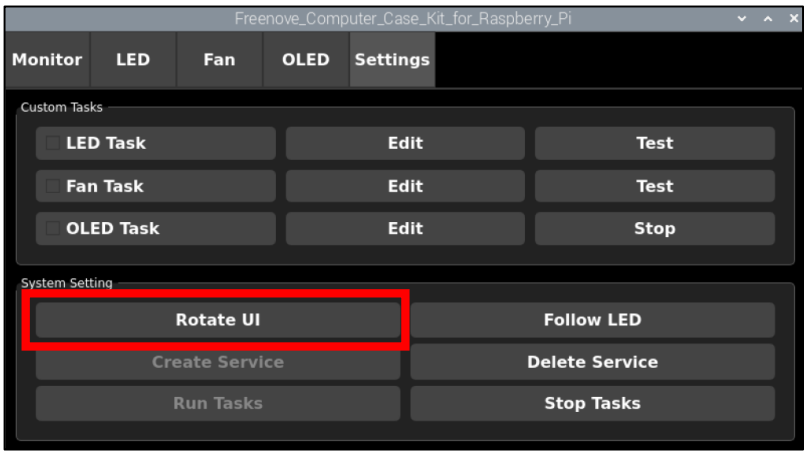




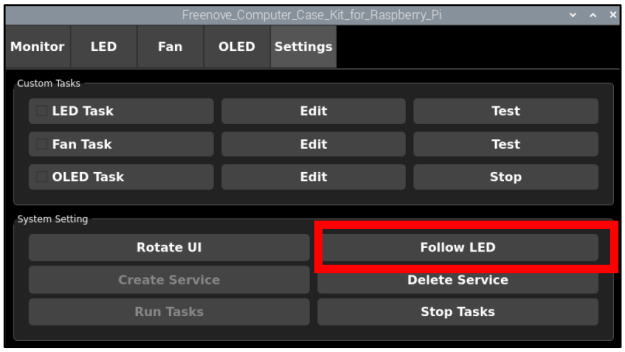
Interface Configuration

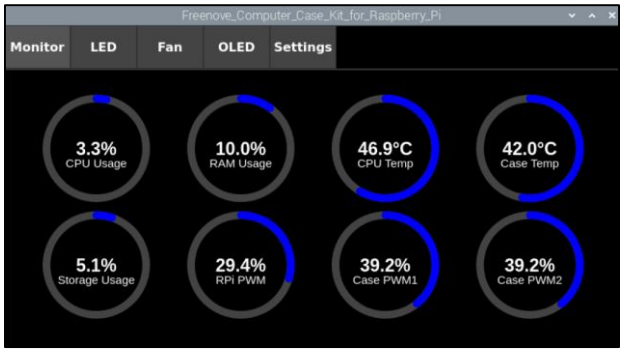
The "**Rotate UI**" button toggles the software interface instantly between landscape and portrait display modes.



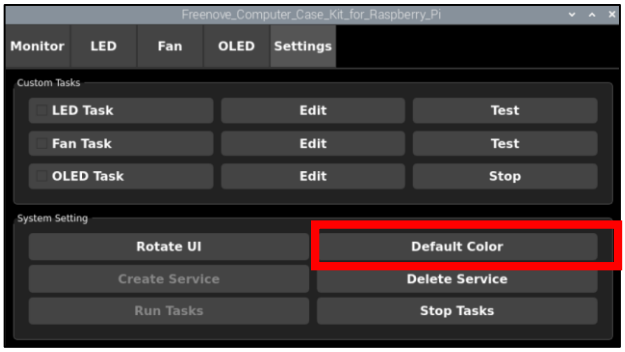


Clicking the "**Follow LED**" button activates synchronization between the LED effects and the Monitor interface. Once enabled, the color of the circular progress bar will mirror any real-time changes you make to the LED colors.





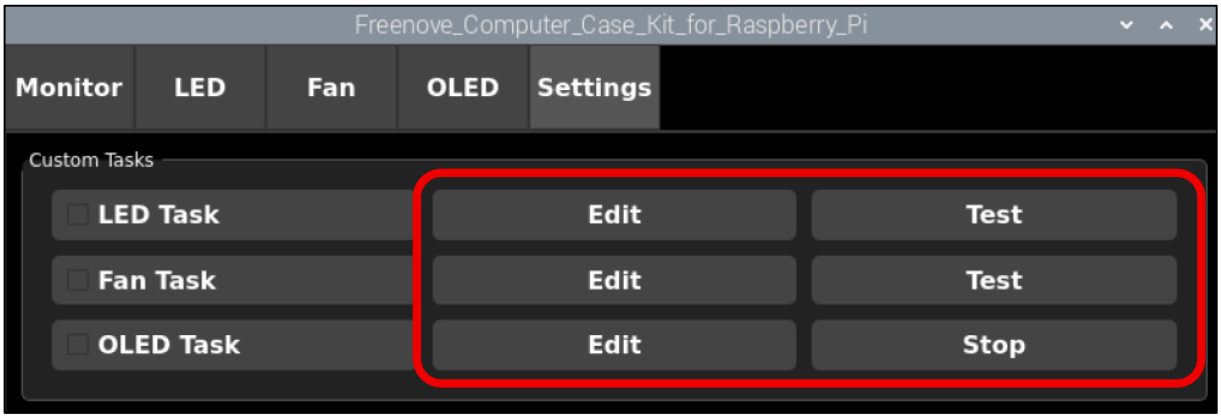
The "**Default Color**" button reverts the circular progress bar (on the Monitor interface) to its default color.



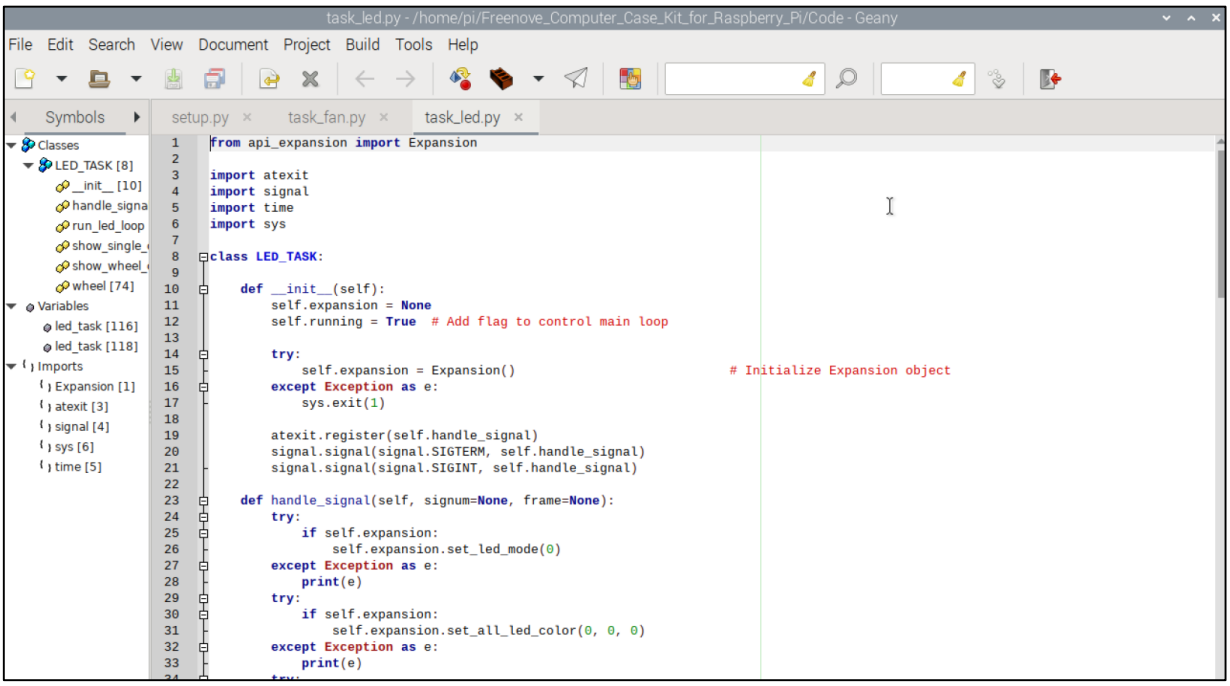


Code Editing and Testing

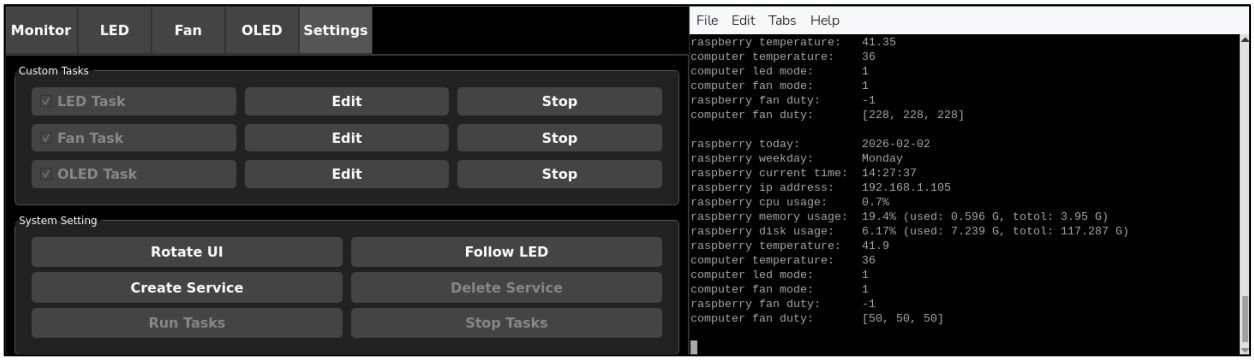
The Edit and Test buttons on the Settings interface allows for quick editing and testing of your code.



You can open and edit the task_led.py, task_fan.py, task_oled.py files by click the corresponding Edit button.

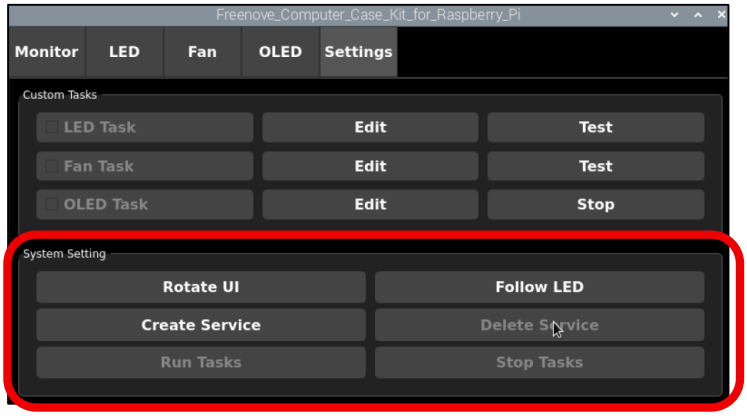


Similarly, you can click Test to quickly verify if the edited code performs as expected. Debug information will typically be printed in the terminal window.



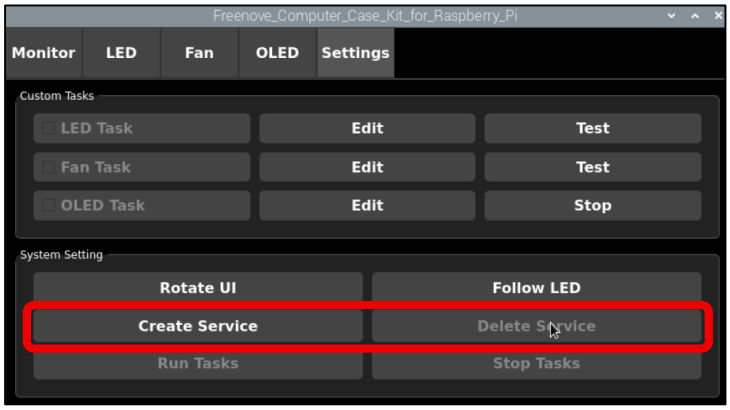
Background Task Management

This section is for comprehensive configuration. Here you can set up the software's interface features and configure background tasks for automatic operation of LEDs, fans, and the OLED display.



Click the "**Create Service**" button creates a background system service on your Raspberry Pi. This service will automatically run every time the case boots up, ensuring that your customized task_led.py, task_fan.py, and task_oled.py scripts are executed after startup.

This service is activated with every boot of the Raspberry Pi 5.

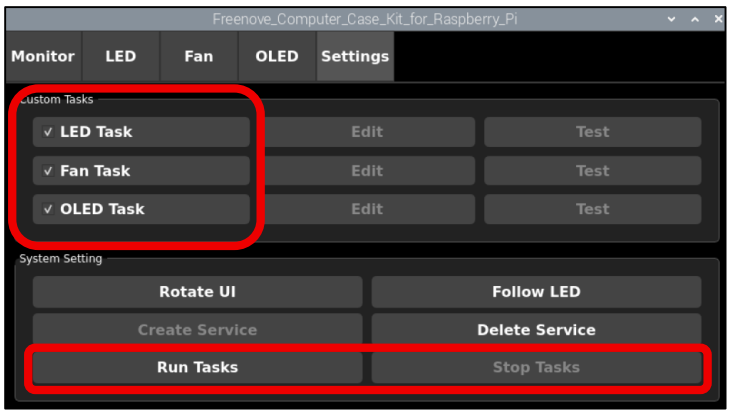


To prevent the background service from running on startup, click the **Delete Service button** to remove it.

Delete Service: Click to permanently remove the service. This prevents it from running automatically upon case startup.

Run Tasks: Click to temporarily start the service for immediate testing.

Stop Tasks: Click to temporarily halt the currently running service.

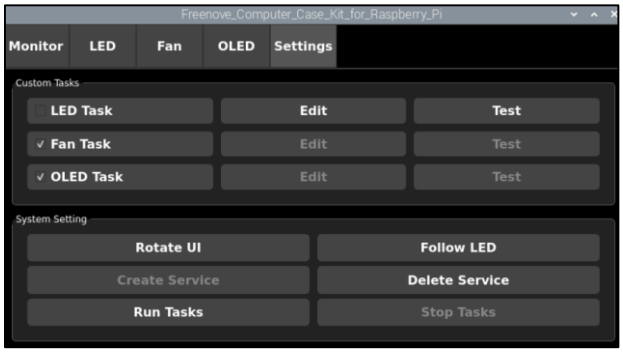


You can enable the execution of one or more custom tasks by checking their corresponding boxes

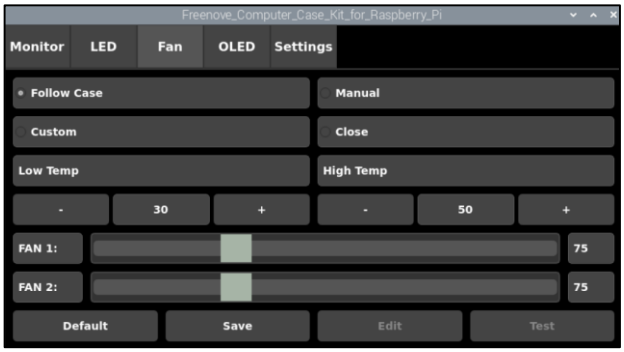
Please Note:
When switching from the Settings interface to the LED or Fan control interface, the corresponding **LED Task** or **Fan Task** will be automatically unchecked, pausing their background operation. You can re-enable them by checking the boxes again.

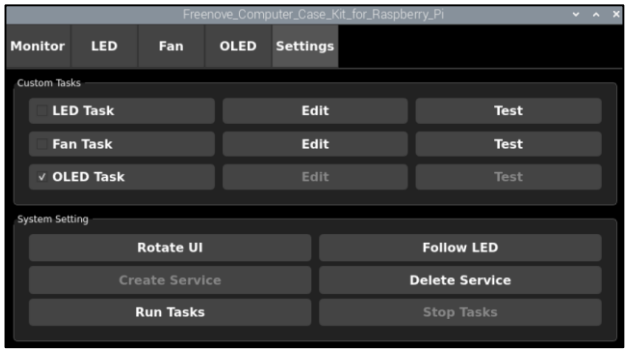
To stop the custom LED task and use a built-in mode instead, simply switch from the Settings interface to the LED interface and select one of the preset modes (Rainbow, Breathing, Follow, or Manual). This action will automatically uncheck the **LED Task**.





To stop the custom Fan Task and use a preset mode instead, simply switch from the Settings interface to the Fan interface and select one of the standard modes (Follow Case or Manual). This action will automatically uncheck the **Fan Task**.





Chapter 4 Speed Test & PCIe Gen3.0

The Raspberry Pi 5 includes a PCIe x1 slot that is certified for PCIe Gen 2.0, providing a theoretical maximum throughput of 5GT/sec, which roughly translates to 500MB/sec for read and write operations. Although this slot is not officially certified for PCIe Gen 3.0, it is possible to force the use of Gen 3.0 for potentially higher speeds.

In actual tests, it is found that most SSDs can work stably at PCIE2.0, but are slightly unstable at PCIE3.0, while others are just the opposite. Therefore, please choose PCIE2.0 or PCIE3.0 according to your actual situation.

The PCIe consortium states that the speed of PCIe Gen 3.0 x1 is up to 8GT/sec, which translates to approximately 985MB/sec; however, Raspberry Pi claims that their implementation can achieve a speed of 10GT/sec, equivalent to around 1231MB/sec.
https://en.wikipedia.org/wiki/PCI_Express#Comparison_table
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#pcie-gen-3-0

4.1 Disk Speed Test

This is an additional chapter for those who wish to test the read and write speeds of their SSD.

Open the terminal and enter the following command:

```
git clone https://github.com/TheRemote/PiBenchmarks
```

```
pi@raspberrypi:~ $ git clone https://github.com/TheRemote/PiBenchmarks
Cloning into 'PiBenchmarks'...
remote: Enumerating objects: 652, done.
remote: Counting objects: 100% (112/112), done.
remote: Compressing objects: 100% (80/80), done.
remote: Total 652 (delta 69), reused 65 (delta 31), pack-reused 540
Receiving objects: 100% (652/652), 119.41 KiB | 417.00 KiB/s, done.
Resolving deltas: 100% (386/386), done.
pi@raspberrypi:~ $
```

Enter the directory:

```
cd PiBenchmarks/
```

Grant executable permissions to the script:

```
chmod +x Storage.sh
```

```
pi@raspberrypi:~ $ cd PiBenchmarks/
pi@raspberrypi:~/PiBenchmarks $ ls
CODE_OF_CONDUCT.md  CONTRIBUTING.md  LICENSE  README.md  Storage.sh
pi@raspberrypi:~/PiBenchmarks $ chmod +x Storage.sh
pi@raspberrypi:~/PiBenchmarks $
```

Start the speed test. Please be aware that the first execution will involve downloading the required dependencies, so the process could take a relatively long time.

```
sudo ./Storage.sh ~/
```

```
pi@raspberrypi:~/PiBenchmarks $ sudo ./Storage.sh ~/
Trimming and syncing drives ...
/media/pi/bootfs: 433.1 MiB (454152192 bytes) trimmed on /dev/mmcblk0p1
/media/pi/rootfs: 2.5 GiB (2681069568 bytes) trimmed on /dev/mmcblk0p2
/boot/firmware: 436.4 MiB (457588736 bytes) trimmed on /dev/nvme0n1p1
/: 457.4 GiB (491171225600 bytes) trimmed on /dev/nvme0n1p2
Board information: Manufacturer: Raspberry Pi Foundation - Model: Raspberry Pi 5
 Model B Rev 1.0 - Architecture: aarch64 - OS: Debian GNU/Linux 12 (bookworm)
Fetching required components ...
Install lshw
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian-security bookworm-security InRelease
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease
Hit:4 http://archive.raspberrypi.com/debian bookworm InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
```

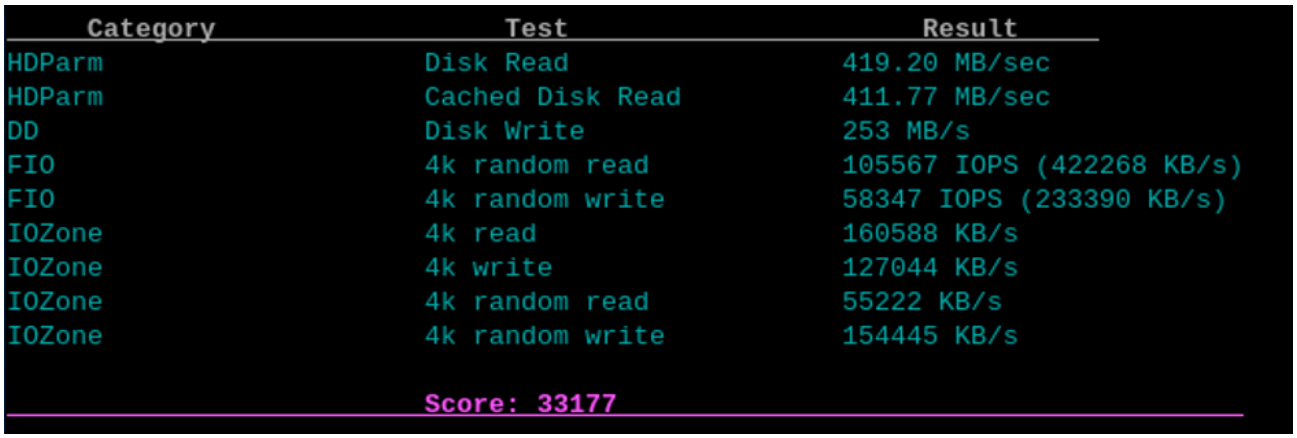
After the speed test is completed, follow the prompts to enter a description and a name for your SSD (you can use any arbitrary characters).

```
iozone test complete.
RandRead: 55044 - RandWrite: 169735 - Read: 194074 - Write: 159843
Enter a description of your storage and setup (Example: Kingston A400 SSD on Pi
4 using StarTech SATA to USB adapter)
Description: ssd1
(Optional) Enter alias to use on benchmark results.  Leave blank for completely
anonymous.
Alias (leave blank for Anonymous): ssd1
Result submitted successfully and will appear live on https://pibenchmarks.com w
ithin a couple of minutes.
```

Test result:

Category	Test	Result
HDParm	Disk Read	381.84 MB/sec
HDParm	Cached Disk Read	376.91 MB/sec
DD	Disk Write	312 MB/s
FIO	4k random read	93090 IOPS (372363 KB/s)
FIO	4k random write	70378 IOPS (281512 KB/s)
IOZone	4k read	194074 KB/s
IOZone	4k write	159843 KB/s
IOZone	4k random read	55044 KB/s
IOZone	4k random write	169735 KB/s
	Score: 36670	

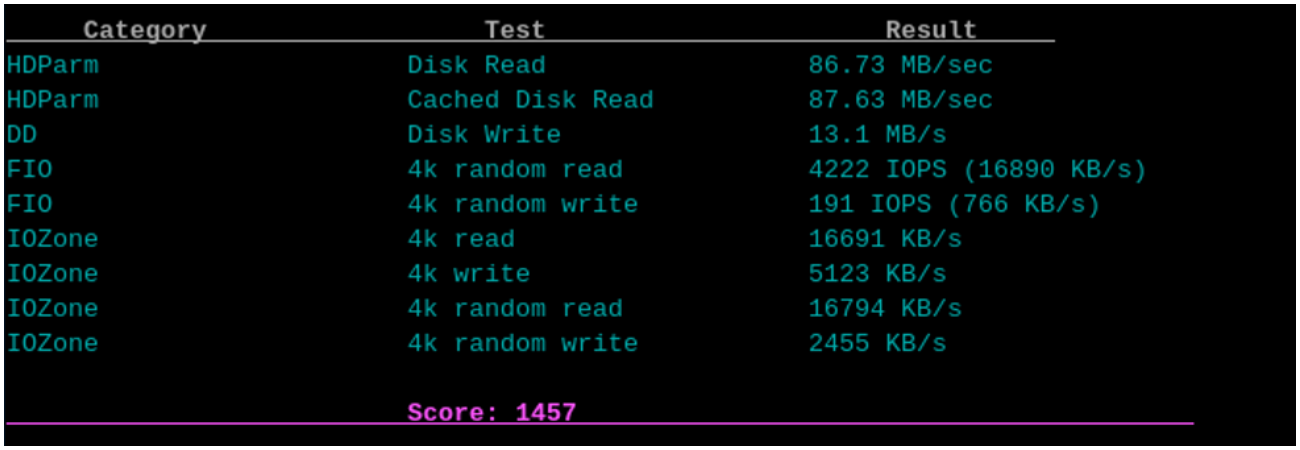
The performance varies among different SSDs, and each test may have certain error, which is normal. The following image shows the speed test results for another SSD:



Category	Test	Result
HDParm	Disk Read	419.20 MB/sec
HDParm	Cached Disk Read	411.77 MB/sec
DD	Disk Write	253 MB/s
FIO	4k random read	105567 IOPS (422268 KB/s)
FIO	4k random write	58347 IOPS (233390 KB/s)
IOZone	4k read	160588 KB/s
IOZone	4k write	127044 KB/s
IOZone	4k random read	55222 KB/s
IOZone	4k random write	154445 KB/s

Score: 33177

This is a speed test result for a TF (microSD) card, and it shows a significant difference in speed compared to an SSD.



Category	Test	Result
HDParm	Disk Read	86.73 MB/sec
HDParm	Cached Disk Read	87.63 MB/sec
DD	Disk Write	13.1 MB/s
FIO	4k random read	4222 IOPS (16890 KB/s)
FIO	4k random write	191 IOPS (766 KB/s)
IOZone	4k read	16691 KB/s
IOZone	4k write	5123 KB/s
IOZone	4k random read	16794 KB/s
IOZone	4k random write	2455 KB/s

Score: 1457

4.2 PCIe Gen3.0

In the Preface, it is mentioned that the Raspberry Pi's PCIe Gen 3.0 has not been officially certified. While it is functional, its performance is not as reliable as desired. This chapter is presented as an exploratory section for assessing the speed capabilities of SSDs when used with PCIe Gen 3.0. **For practical applications, it is advised to opt for PCIe Gen 2.0 to ensure greater stability and dependability.**

Enable PCIe Gen3.0

Add the line `dtparam=pciex1_gen=3` to /boot/firmware/config.txt to enable PCIe Gen3.0.
As shown below, enter the command to open the file.

```
sudo nano /boot/firmware/config.txt
```

Add the line `dtparam=pciex1_gen=3` to the end of the file, as shown below:

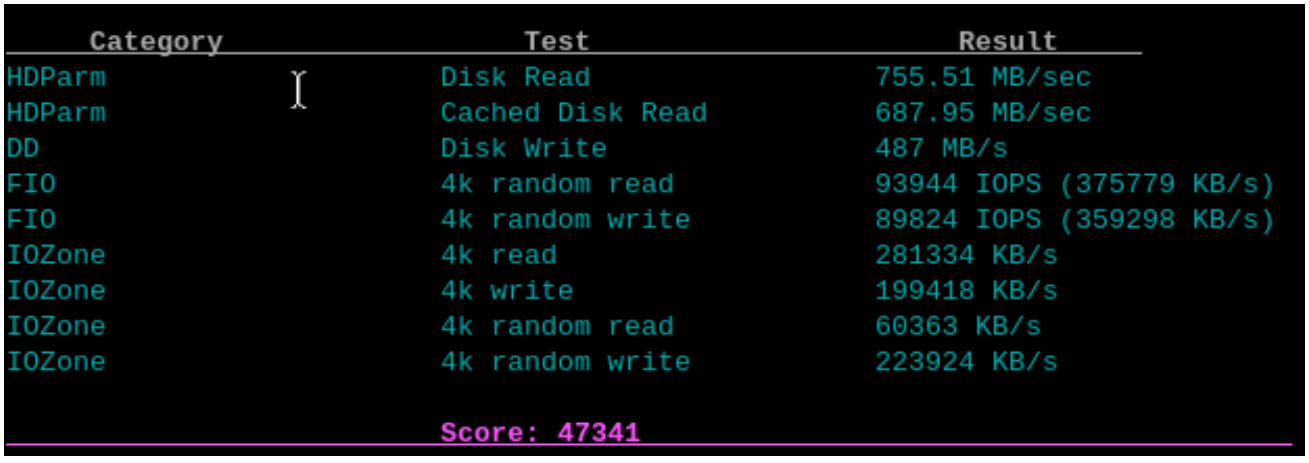


Press Ctrl-O to save the file, Enter to confirm, and Ctrl-X to exit.

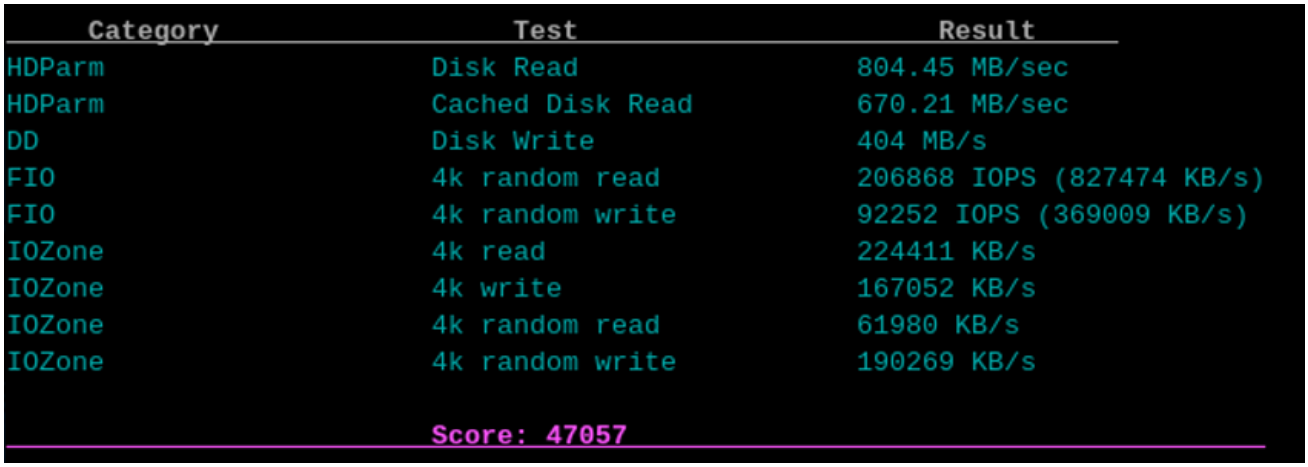
Reboot your Raspberry Pi.

```
sudo reboot
```

After rebooting, test the speed again.



The speed of another SSD.



Category	Test	Result
HDParm	Disk Read	804.45 MB/sec
HDParm	Cached Disk Read	670.21 MB/sec
DD	Disk Write	404 MB/s
FIO	4k random read	206868 IOPS (827474 KB/s)
FIO	4k random write	92252 IOPS (369009 KB/s)
IOZone	4k read	224411 KB/s
IOZone	4k write	167052 KB/s
IOZone	4k random read	61980 KB/s
IOZone	4k random write	190269 KB/s
	Score: 47057	

Disable PCIe Gen3.0

Delete the line added with the previous step to disable PCIe Gen3.0.
Delete the line `dtparam=pciex1_gen=3` in the boot/firmware/config.txt file.
After the line is removed, it will change to PCIe Gen2.0.

What's next?

Thank you again for choosing Freenove products.

THANK YOU for participating in this learning experience!

We have reached the end of this Tutorial. If you find errors, omissions, or you have suggestions and/or questions about the Tutorial or component contents of this Kit, please feel free to contact us:
support@freenove.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

If you want to learn more about Arduino, ESP32, Raspberry Pi Pico W, Raspberry Pi, Smart Cars, Robotics and other interesting products in science and technology, please continue to visit our website. We will continue to launch fun, cost effective, innovative and exciting products.
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